Reactive Flows and Cantera

A from-scratch tutorial for the ASEC Lab onboarding project

thermochemistry · equilibrium · kinetics · transport · 0D reactors
premixed flames · non-premixed flames · Cantera · project guide

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Chapter 1

How to use this tutorial

1.1 What this document is for

The ASEC Lab onboarding chapter *Reactive Flows in Cantera* starts from the fully reactive Navier–Stokes equations and then, very sensibly, refuses to solve them in three dimensions straight away. Instead it climbs a ladder:

$$\text{0D (homogeneous reactor)} \longrightarrow \text{1D (laminar flames)} \longrightarrow \text{3D (turbulent reacting flow).}$$

The onboarding project then asks you to work through hydrogen/air combustion in three parts:

Part 1: Equilibrium — the thermodynamic end state (adiabatic flame temperature) as a function of equivalence ratio ϕ, preheat temperature T_u and pressure p.

Part 2: Non-premixed flames — counterflow diffusion flames, mixture fraction Z, scalar dissipation rate χ, departure from equilibrium and extinction.

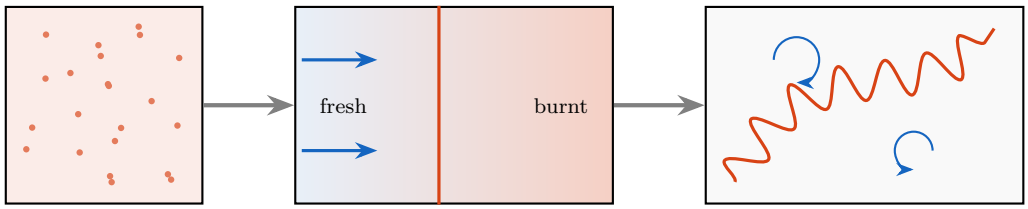
Part 3: Premixed flames — freely propagating flames, flame speed S_L, thickness δ_L, transport models (unity Lewis vs. mixture-averaged).

This tutorial assumes **nothing** beyond school-level calculus. Every equation that appears in the project is derived here, every derivation is followed by an *intuition* box that says what the mathematics means physically, and wherever a picture helps there is a schematic or a plot computed with Cantera. The data figures were generated with `latex/figures/scripts/make_figures.py`; you can (and should) re-run and modify that script.

Pitfall: the demonstrations are not the project

To keep the project yours, the worked numerical examples use **methane**/air or conditions that differ from the project tasks. The physics, methods and code patterns transfer directly to hydrogen; the numbers do not. When a *Project link* box gives a hint, it tells you what to look for and why, not the answer.

1.2 The ladder of complexity



0D reactor
$Y_k(t)$, $T(t)$
chemistry + thermo

1D flame
$Y_k(x)$, $T(x)$, $u(x)$
+ convection, diffusion

3D turbulent flame
$Y_k(\boldsymbol{x}, t)$, T, $\boldsymbol{u}$, p
+ turbulence, geometry

Figure 1.1: Each rung adds one physical ingredient. Chemistry and thermodynamics do not depend on the spatial dimension, so they are learnt first in 0D and reused unchanged in 1D and 3D.

The reason this ladder works is that the *chemical source term* and the *thermodynamic properties* are local: at a point in space they depend only on the local temperature, pressure and composition, never on gradients. Whatever you learn about them in a perfectly mixed box (0D) remains true inside a flame (1D) or a jet engine (3D).

1.3 Map of the chapters

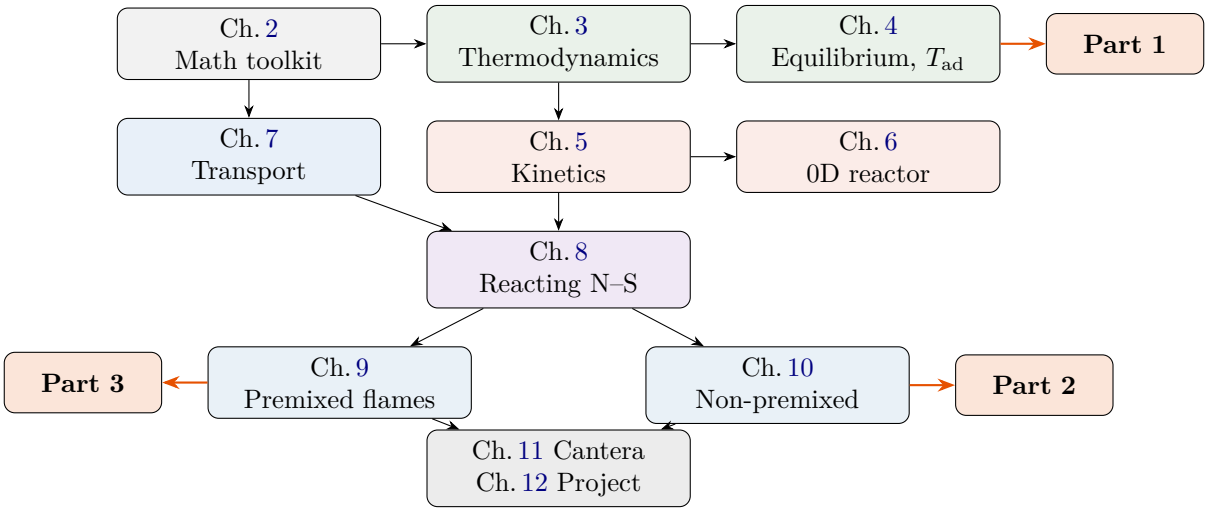


Figure 1.2: Dependencies between chapters and the project parts they prepare you for.

Suggested reading order.

- Before Part 1: Chapters 2–4 and Sections 11.2–11.4 of the Cantera chapter.
- Before Part 2: Chapters 5–8 and 10.
- Before Part 3: Chapter 9 and the grid-refinement material in Section 11.8.
- Chapter 12 is the step-by-step project companion; read the relevant part right before you start coding it.

1.4 Conventions

- SI units throughout. Cantera uses **kmol** rather than mol, so molar quantities are per kmol and $\mathcal{R} = 8314.46\,\mathrm{J/(kmol\,K)}$.
- Species are indexed by $k = 1, \ldots, K$, reactions by $i = 1, \ldots, I$ and elements by $j = 1, \ldots, J$.
- Bold symbols are vectors ($\boldsymbol{u}$, $\boldsymbol{x}$); ∇ is the gradient operator.
- Subscript u means *unburnt* (fresh) gas, b *burnt* gas, st stoichiometric, ad adiabatic.
- A full nomenclature is in Appendix A.

Boxes used in the text.

Physical intuition

What the mathematics means, in words and pictures.

Derivation

A step-by-step derivation where no step is skipped. It is always followed by a *Symbols and units* table listing every symbol, what it means and its SI unit (also after important standalone equations).

Symbols and units

x	every symbol used in the derivation just above, with what it means	its SI unit

Key result

A result you should remember.

Pitfall

A mistake that is easy to make, or a result Cantera will happily give you that is wrong.

Project link

How the section connects to a specific project task.

Chapter 2

Mathematical toolkit

This chapter collects every piece of mathematics used later. If a symbol in a later chapter looks unfamiliar, it is defined here.

2.1 Scalars, vectors, fields

A **scalar** is a single number (temperature T, density ρ). A **vector** has magnitude and direction; in 3D Cartesian coordinates $\boldsymbol{u} = (u, v, w)$. A **field** is a quantity defined at every point and time: $T(\boldsymbol{x}, t)$ is a scalar field, $\boldsymbol{u}(\boldsymbol{x}, t)$ a vector field.

The **dot product** $\boldsymbol{a} \cdot \boldsymbol{b} = a_x b_x + a_y b_y + a_z b_z = |\boldsymbol{a}||\boldsymbol{b}| \cos\theta$ measures how much two vectors point the same way. With a unit normal $\boldsymbol{n}$ to a surface, $\boldsymbol{u} \cdot \boldsymbol{n}$ is the velocity component *through* that surface.

Index notation. Writing u_i $(i = 1, 2, 3)$ for the components and using the *summation convention* (a repeated index is summed), $\boldsymbol{a} \cdot \boldsymbol{b} = a_i b_i$. We use it only in the momentum equation.

2.2 The gradient: which way is uphill

For a scalar field $T(x, y, z)$,

$$\nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right). \tag{2.2.1}$$

∇T points in the direction of steepest increase of T and its magnitude is the rate of increase per metre. The change of T over a small displacement $\mathrm{d}\boldsymbol{x}$ is $\mathrm{d}T = \nabla T \cdot \mathrm{d}\boldsymbol{x}$.

Physical intuition

Heat flows from hot to cold, i.e. *down* the gradient. That is why Fourier's law (Chapter 7) is $\boldsymbol{q} = -\lambda \nabla T$, with a minus sign. Species diffuse down their concentration gradient in the same way. In a flame, the gradients are enormous: temperature rises by $\sim 2000\,\mathrm{K}$ over $\sim 0.5\,\mathrm{mm}$, i.e. $|\nabla T| \sim 4 \times 10^6\,\mathrm{K/m}$.

2.3 The divergence: net outflow per unit volume

Let $\boldsymbol{F}$ be a *flux*: an amount of something crossing a unit area per unit time (e.g. heat flux in $\mathrm{W/m^2}$, mass flux $\rho\boldsymbol{u}$ in $\mathrm{kg/(m^2\,s)}$).

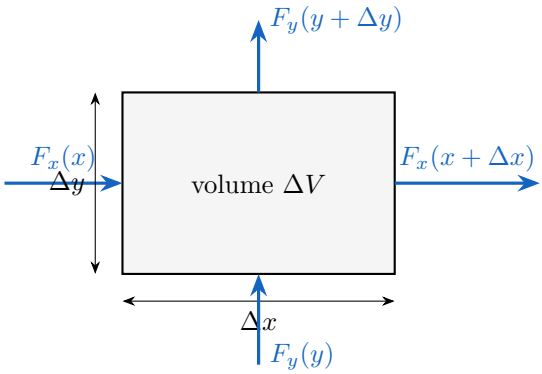


Figure 2.1: A small box. What leaves through the right face minus what enters through the left face, plus the same in y and z, is the net outflow.

Derivation: divergence from a box

Net outflow in x per unit time through faces of area $\Delta y \Delta z$:

$$\big[F_x(x+\Delta x) - F_x(x)\big]\Delta y\,\Delta z \approx \frac{\partial F_x}{\partial x}\,\Delta x\,\Delta y\,\Delta z = \frac{\partial F_x}{\partial x}\,\Delta V,$$

using a first-order Taylor expansion $F_x(x+\Delta x) \approx F_x(x) + \Delta x\,\partial F_x/\partial x$. Adding the y and z contributions and dividing by ΔV:

$$\frac{\text{net outflow}}{\Delta V} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} \equiv \nabla\cdot\boldsymbol{F}. \tag{2.3.1}$$

Symbols and units

$\boldsymbol{F} = (F_x, F_y, F_z)$	flux vector: amount of some quantity crossing unit area per unit time	e.g. $\mathrm{W/m^2}$ (heat), $\mathrm{kg/(m^2\,s)}$ (mass)
$\Delta x, \Delta y, \Delta z$	side lengths of the small box	m
ΔV	volume of the box, $\Delta x \Delta y \Delta z$	$\mathrm{m^3}$
$\partial F_x/\partial x$	how fast F_x changes with x	unit of F per m
$\nabla\cdot\boldsymbol{F}$	divergence: net outflow per unit volume (per unit time)	unit of F per m

Key result: Gauss divergence theorem

For any volume V with closed surface S and outward normal $\boldsymbol{n}$,

$$\oint_S \boldsymbol{F}\cdot\boldsymbol{n}\,\mathrm{d}S = \int_V \nabla\cdot\boldsymbol{F}\,\mathrm{d}V. \tag{2.3.2}$$

This is just "add up the boxes": interior faces cancel because what leaves one box enters its neighbour. It converts surface fluxes (easy to reason about physically) into volume integrals (easy to turn into differential equations).

The **Laplacian** $\nabla^2 T = \nabla\cdot(\nabla T) = \partial^2 T/\partial x^2 + \dots$ measures how much T at a point differs from

the average of its neighbours. Diffusion $\partial T/\partial t = \alpha \nabla^2 T$ therefore smooths things out: a local hot spot (negative curvature) cools.

2.4 The universal conservation law

Every equation in this tutorial has the same structure. Take any quantity with amount per unit volume $\rho\varphi$ (e.g. $\varphi = 1$ for mass, $\varphi = Y_k$ for species k, $\varphi = \boldsymbol{u}$ for momentum, $\varphi = h$ for enthalpy). For a fixed control volume V:

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\varphi \,\mathrm{d}V}_{\text{rate of change inside}} = \underbrace{-\oint_S \rho\varphi\, \boldsymbol{u}\cdot\boldsymbol{n}\,\mathrm{d}S}_{\text{carried in by flow (convection)}} \underbrace{-\oint_S \boldsymbol{J}_\varphi \cdot \boldsymbol{n}\,\mathrm{d}S}_{\text{carried in by molecular motion (diffusion)}} + \underbrace{\int_V S_\varphi \,\mathrm{d}V}_{\text{created inside (source)}} .$$

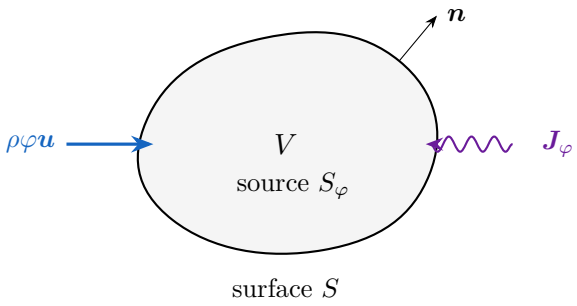


Figure 2.2: Bookkeeping on a control volume: convection, diffusion, source.

Apply Gauss's theorem (2.3.2) to the two surface integrals; since V is arbitrary the integrands must balance pointwise:

$$\boxed{\frac{\partial(\rho\varphi)}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}\varphi) = -\nabla\cdot\boldsymbol{J}_\varphi + S_\varphi} \tag{2.4.1}$$

Symbols and units

φ	transported quantity per kg (1 for mass, Y_k, $\boldsymbol{u}$, h, ...)	unit of φ
$\rho\varphi$	amount of that quantity per unit volume	unit of φ × kg/m^3
$\boldsymbol{u}$	flow velocity	m/s
$\boldsymbol{J}_\varphi$	diffusive flux of φ	unit of $\rho\varphi$ × m/s
S_φ	source: amount created per unit volume and time	unit of $\rho\varphi$ per s
V, S, $\boldsymbol{n}$	control volume, its surface, outward unit normal	m^3, m^2, –

Key result

Reading any transport equation: **unsteady + convection = diffusion + source**. Chapter 8 fills in $\boldsymbol{J}_\varphi$ and S_φ for mass, species, momentum and energy. A 0D reactor keeps only unsteady = source. A steady 1D flame keeps convection = diffusion + source in one direction.

2.5 The material derivative

Imagine a thermometer drifting with the flow. The rate of change it records is the **material derivative**

$$\frac{\mathrm{D}\varphi}{\mathrm{D}t} = \frac{\partial \varphi}{\partial t} + \boldsymbol{u} \cdot \nabla \varphi. \tag{2.5.1}$$

The first term is change at a fixed point, the second is change because you moved to a place where φ is different. Combining (2.4.1) with mass conservation $\partial\rho/\partial t + \nabla\cdot(\rho\boldsymbol{u}) = 0$ gives the useful identity

$$\frac{\partial(\rho\varphi)}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}\varphi) = \rho\frac{\mathrm{D}\varphi}{\mathrm{D}t}. \tag{2.5.2}$$

Symbols and units

$\mathrm{D}\varphi/\mathrm{D}t$	rate of change seen while moving with the fluid	unit of φ per s
$\partial\varphi/\partial t$	rate of change at a fixed point	unit of φ per s
$\boldsymbol{u}\cdot\nabla\varphi$	change due to moving to where φ differs	unit of φ per s

(Expand the product rule: $\varphi[\partial_t\rho + \nabla\cdot(\rho\boldsymbol{u})] + \rho[\partial_t\varphi + \boldsymbol{u}\cdot\nabla\varphi]$; the first bracket is zero.)

2.6 Ordinary differential equations and stiffness

A 0D reactor is a system of ODEs $\mathrm{d}\boldsymbol{y}/\mathrm{d}t = \boldsymbol{f}(\boldsymbol{y})$. The simplest is exponential decay $\mathrm{d}y/\mathrm{d}t = -y/\tau$ with solution $y = y_0 e^{-t/\tau}$; τ is the **time scale**.

Chemistry contains time scales from 10^{-10} s (radical recombination) to 10^{-1} s (NO formation). A system with widely separated time scales is **stiff**.

Worked example: why explicit methods fail

Explicit Euler, $y_{n+1} = y_n + \Delta t f(y_n)$, applied to $\mathrm{d}y/\mathrm{d}t = -y/\tau$ gives $y_{n+1} = (1-\Delta t/\tau)y_n$. If $\Delta t > 2\tau$ then $|1-\Delta t/\tau| > 1$ and the numerical solution explodes, even though the true solution has long since decayed to zero. So the *fastest* time scale ($\tau \sim 10^{-10}$ s) dictates the step, while you want to integrate over the *slowest* (10^{-1} s): 10^9 steps.
Implicit (backward) Euler, $y_{n+1} = y_n + \Delta t f(y_{n+1})$, gives $y_{n+1} = y_n/(1+\Delta t/\tau)$, stable for any Δt. Cantera uses the implicit BDF integrator CVODES for reactors and a damped Newton method for 1D flames; you never pick the time step yourself, but you should know why the solver may become slow.

2.7 Newton's method

To solve $\boldsymbol{F}(\boldsymbol{y}) = \boldsymbol{0}$ (a steady flame discretised on N points with $K+3$ unknowns per point), linearise around a guess: $\boldsymbol{F}(\boldsymbol{y}+\delta\boldsymbol{y}) \approx \boldsymbol{F}(\boldsymbol{y}) + \mathsf{J}\,\delta\boldsymbol{y} = \boldsymbol{0}$, with Jacobian $\mathsf{J}_{ab} = \partial F_a/\partial y_b$. Solve $\mathsf{J}\delta\boldsymbol{y} = -\boldsymbol{F}$ and update. Convergence is very fast *if* the starting guess is close; otherwise it diverges.

Symbols and units

$\boldsymbol{y}$	vector of all unknowns (T, Y_k, u, …at every grid point)	mixed
$\boldsymbol{F}(\boldsymbol{y})$	residuals of the discretised equations (zero at the solution)	mixed
J	Jacobian matrix $\partial F_a/\partial y_b$	mixed
$\delta\boldsymbol{y}$	Newton correction	as $\boldsymbol{y}$

Physical intuition

This is why Cantera's 1D solver alternates Newton iterations with a few pseudo-time steps (which are robust but slow) and why, for parameter sweeps, you should start each new flame from the previous converged one: it is a guess that is already close.

2.8 Constrained minimisation: Lagrange multipliers

Equilibrium is "minimise G subject to conservation of atoms". To minimise $f(\boldsymbol{x})$ subject to constraints $g_j(\boldsymbol{x}) = 0$, form $\mathcal{L} = f - \sum_j \lambda_j g_j$ and set $\partial\mathcal{L}/\partial x_k = 0$:

$$\frac{\partial f}{\partial x_k} = \sum_j \lambda_j \frac{\partial g_j}{\partial x_k}. \tag{2.8.1}$$

Symbols and units

$f(\boldsymbol{x})$	function to minimise (later: Gibbs energy G)	e.g. J
x_k	variables (later: amounts n_k)	e.g. kmol
$g_j(\boldsymbol{x}) = 0$	constraints (later: element conservation)	e.g. kmol
λ_j	Lagrange multipliers	unit of f / unit of g_j

Physical intuition

At a constrained minimum you cannot go further downhill without leaving the allowed surface, so the downhill direction $-\nabla f$ must be perpendicular to that surface, i.e. a combination of the constraint normals ∇g_j. The multipliers λ_j are the "prices" of the constraints; for equilibrium they turn out to be the chemical potentials of the elements.

2.9 Exponentials, logarithms and the Arrhenius plot

Reaction rates behave like $k = Ae^{-E/(\mathcal{R}T)}$. Taking logarithms,

$$\ln k = \ln A - \frac{E}{\mathcal{R}}\frac{1}{T}, \tag{2.9.1}$$

Symbols and units

k	rate constant	depends on order
A	pre-exponential factor	same as k
E	activation energy	J/kmol
$\mathcal{R}$, T	universal gas constant, temperature	8314.46 J/(kmol K), K

a straight line in $(1/T, \ln k)$ with slope $-E/\mathcal{R}$. That is why kinetics plots use $1000/T$ on the x-axis and a log y-axis (Fig. 5.3). Curvature in such a plot signals that more than one process is competing.

Physical intuition: how violent the exponential is

For a typical $E/\mathcal{R} = 20\,000\,\mathrm{K}$, going from 1000 K to 1100 K multiplies k by $e^{20000(1/1000-1/1100)} = e^{1.82} \approx 6$. A 10% temperature change gives a 600% rate change. This extreme sensitivity is what makes flames thin, ignition sudden and extinction abrupt.

2.10 The error function

$$\operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-s^2} \mathrm{d}s, \qquad \operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta). \tag{2.10.1}$$

Symbols and units

η	argument (later x/ℓ)	– (dimensionless)
s	integration variable	– (dimensionless)
c, D, t	concentration, diffusivity, time in the example	kmol/m^3, m^2/s, s

erf rises smoothly from -1 to 1; it is the shape of a diffusion front, e.g. the solution of $\partial_t c = D\partial_{xx} c$ with a step initial condition is $c = \frac{1}{2}\operatorname{erfc}(x/\sqrt{4Dt})$. We will meet it as the shape of the mixture fraction in a counterflow flame (Section 10.5). Its inverse is written erf^{-1} (`scipy.special.erfinv`).

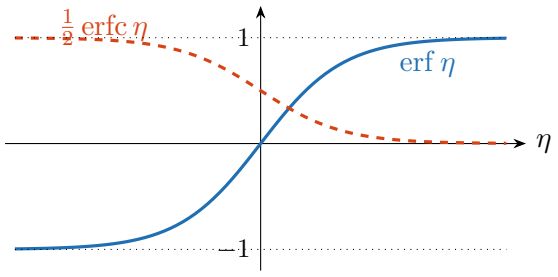


Figure 2.3: Shapes of erf and $\frac{1}{2}$erfc (sketched). $\frac{1}{2}\operatorname{erfc}$ goes from 1 to 0: exactly what a mixture fraction does between a fuel stream and an air stream.

Chapter 3

Thermodynamics of reacting ideal-gas mixtures

Thermodynamics answers two questions: *how much energy* is released when a mixture reacts, and *where it ends up* if left alone for long enough. It says nothing about how fast; that is kinetics (Chapter 5).

3.1 System, state and properties

A **system** is the piece of matter we look at. Its **state** is fully fixed by a small number of **state variables**. For a gas mixture of K species, the *state postulate* says that two thermodynamic variables plus the composition are enough:

$$\text{state} = (T,\ p \text{ (or } \rho),\ \text{composition } Y_1, \ldots, Y_K). \tag{3.1.1}$$

Everything else (h, s, c_p, viscosity, reaction rates…) is a function of these. In Cantera this is literally the statement `gas.TPY = T, p, Y`: after it, every property is available.

Extensive properties scale with the amount of matter (volume V, enthalpy H); **intensive** ones do not (T, p). Dividing an extensive property by mass gives a *specific* property (lower case, per kg: $h = H/m$); dividing by moles gives a *molar* property (with a hat or bar, per kmol: $\hat{h} = H/n$).

3.2 The ideal gas

Derivation: ideal gas law from molecules hitting a wall

Consider N molecules of mass m in a box of volume V. A molecule with velocity component c_x that hits the wall at $x = L$ bounces back with $-c_x$: momentum change $2mc_x$. It returns after time $2L/c_x$, so its average force on the wall is $2mc_x/(2L/c_x) = mc_x^2/L$. Summing over all molecules and dividing by area A:

$$p = \frac{\sum mc_x^2}{LA} = \frac{Nm\overline{c_x^2}}{V} = \frac{Nm\overline{c^2}}{3V},$$

because $\overline{c^2} = \overline{c_x^2} + \overline{c_y^2} + \overline{c_z^2} = 3\overline{c_x^2}$ by symmetry. Temperature is *defined* microscopically through the mean kinetic energy, $\frac{1}{2}m\overline{c^2} = \frac{3}{2}k_BT$, so $pV = Nk_BT$. With $n = N/N_A$ moles

and $\mathcal{R} = N_A k_B$:

$$pV = n\mathcal{R}T \qquad \Longleftrightarrow \qquad p = \rho\frac{\mathcal{R}}{W}T, \tag{3.2.1}$$

where W is the molar mass [kg/kmol] and $\rho = nW/V$.

Symbols and units

N	number of molecules in the box	– (dimensionless)
m	mass of one molecule	kg
V, L, A	box volume, box length, wall area	m^3, m, m^2
c_x, c	molecular velocity component normal to the wall, molecular speed	m/s
$\overline{c^2}$	mean of the squared molecular speed	$\mathrm{m}^2/\mathrm{s}^2$
p	pressure (force per unit wall area)	Pa
k_B	Boltzmann constant, 1.380649×10^{-23}	J/K
T	temperature	K
n	amount of substance	kmol
N_A	Avogadro's number, 6.022×10^{26}	1/kmol
$\mathcal{R} = N_A k_B$	universal gas constant	8314.46 J/(kmol K)
W	molar mass	kg/kmol
ρ	density	kg/m^3

Physical intuition

Pressure is the momentum flux of molecules hammering the wall; temperature is how hard each one hits. "Ideal" means the molecules do not attract each other and take up no volume. At flame conditions (high T, moderate p) this is an excellent approximation. Note: the typical molecular speed $\sqrt{3k_BT/m}$ is $\sim 500\,\mathrm{m/s}$ for N_2 at room temperature and $\sim 1900\,\mathrm{m/s}$ for H_2. Light molecules are fast, which is why hydrogen diffuses so quickly (Chapter 7).

3.3 Describing a mixture

Let species k have n_k kmol, mass $m_k = n_k W_k$, in a volume V. Three ways to describe the composition:

$$\text{mole fraction} \quad X_k = \frac{n_k}{\sum_j n_j}, \qquad \sum_k X_k = 1, \tag{3.3.1}$$

$$\text{mass fraction} \quad Y_k = \frac{m_k}{\sum_j m_j}, \qquad \sum_k Y_k = 1, \tag{3.3.2}$$

$$\text{molar concentration} \quad C_k = [X_k] = \frac{n_k}{V} \ \ [\mathrm{kmol}/\mathrm{m}^3]. \tag{3.3.3}$$

Derivation: conversions

The **mean molar mass** is total mass over total moles:

$$W = \frac{\sum_j n_j W_j}{\sum_j n_j} = \sum_j X_j W_j \qquad \text{and equivalently} \qquad \frac{1}{W} = \sum_j \frac{Y_j}{W_j}. \tag{3.3.4}$$

Then

$$Y_k = \frac{n_k W_k}{\sum_j n_j W_j} = X_k \frac{W_k}{W}, \qquad X_k = Y_k \frac{W}{W_k}, \qquad C_k = \frac{X_k\, p}{\mathcal{R}T} = \frac{\rho Y_k}{W_k}. \tag{3.3.5}$$

The last one uses $n/V = p/(\mathcal{R}T)$ from (3.2.1).

Symbols and units

n_k	amount of species k	kmol
$m_k = n_k W_k$	mass of species k	kg
X_k, Y_k	mole fraction, mass fraction of species k	– (dimensionless)
W_k, W	molar mass of species k, mean molar mass of the mixture	kg/kmol
C_k	molar concentration of species k	kmol/m^3
p, T, ρ	pressure, temperature, density	Pa, K, kg/m^3
$\mathcal{R}$	universal gas constant	8314.46 J/(kmol K)

Worked example: stoichiometric H_2/air

Air is modelled as $O_2 : N_2 = 1 : 3.76$ by moles. Burning one H_2 needs $\frac{1}{2}O_2$, which comes with $\frac{1}{2} \cdot 3.76 = 1.88$ N_2. Total moles $= 1 + 0.5 + 1.88 = 3.38$:

$$X_{H_2} = 0.296, \quad X_{O_2} = 0.148, \quad X_{N_2} = 0.556.$$

Mean molar mass $W = 0.296(2.016) + 0.148(32.00) + 0.556(28.01) = 20.9$ kg/kmol, so

$$Y_{H_2} = 0.296 \cdot 2.016/20.9 = 0.0285.$$

Hydrogen is 30% of the molecules but under 3% of the mass. Whether a formula uses X or Y matters enormously for H_2.

Partial pressure. Each species in an ideal mixture behaves as if alone in the volume: $p_k = n_k \mathcal{R}T/V = X_k p$ (Dalton's law).

3.4 First law: energy, work, enthalpy

For a closed system, $\mathrm{d}U = \delta Q - \delta W$: internal energy rises by heat added minus work done by the system. With only expansion work, $\delta W = p\,\mathrm{d}V$.

Derivation: why combustion uses enthalpy

At constant pressure, $\delta Q = \mathrm{d}U + p\,\mathrm{d}V = \mathrm{d}(U + pV)$. Define the **enthalpy**

$$H \equiv U + pV, \qquad h = u + p/\rho. \tag{3.4.1}$$

Then at constant p, heat added equals the enthalpy change. Most combustion (open flames, gas turbines, Cantera's `FreeFlame`) happens at essentially constant pressure, so h is the natural energy variable.

In a flowing system there is a second reason: pushing a parcel of fluid of volume V into a control volume against pressure p requires flow work pV. The energy carried by a parcel is therefore $U + pV = H$ (plus kinetic energy).

Symbols and units

U, u	internal energy (total, per kg)	J, J/kg
H, h	enthalpy $U + pV$ (total, per kg)	J, J/kg
Q, δQ	heat added to the system (finite, infinitesimal)	J
W, δW	work done by the system (here not a molar mass)	J
p, V	pressure, volume	Pa, m^3
p/ρ	flow work per kg	J/kg

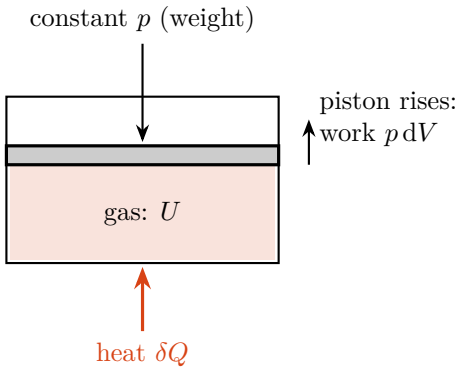


Figure 3.1: At constant pressure part of the heat added goes into lifting the piston. Enthalpy accounts for both: $\delta Q = \mathrm{d}H$.

Heat capacities. $c_p = (\partial h/\partial T)_p$ and $c_v = (\partial u/\partial T)_v$. For an ideal gas h and u depend on T only, and since $h = u + \mathcal{R}T/W$,

$$c_p - c_v = \frac{\mathcal{R}}{W}, \qquad \gamma = c_p/c_v. \tag{3.4.2}$$

Symbols and units

c_p, c_v	specific heat capacity at constant pressure, volume	J/(kg K)
γ	ratio of heat capacities	– (dimensionless)
$\mathcal{R}/W$	specific gas constant	J/(kg K)

Physical intuition: why c_p grows with temperature

Energy stored in a molecule is shared among its "modes": translation (always $\frac{3}{2}\mathcal{R}T$), rotation ($\mathcal{R}T$ for linear molecules) and vibration. Vibrations are quantised and only switch on when $k_B T$ is comparable to their energy spacing, i.e. at high temperature. So a cold N_2 has $\hat{c}_p \approx \frac{7}{2}\mathcal{R}$ and a hot one approaches $\frac{9}{2}\mathcal{R}$. Triatomic molecules (H_2O, CO_2) have more vibrational modes and a larger, faster-rising c_p. See Fig. 3.2: this rise with T is a major reason flames are cooler than a constant-c_p estimate predicts.

3.5 Chemical energy: enthalpy of formation

To compare energies of *different* molecules we need a common zero. The convention:

- Reference state: $T^\circ = 298.15\,\mathrm{K}$, $p^\circ = 1\,\mathrm{bar}$ (Cantera uses 1 atm = 101 325 Pa as standard pressure for gases; the difference is negligible here).

- Elements in their natural form at T° (H_2, O_2, N_2, graphite C) have zero enthalpy.
- The **standard enthalpy of formation** $\Delta\hat{h}^\circ_{f,k}$ of species k is the enthalpy change when one kmol of k is made from its elements at T°.

The enthalpy of species k at any temperature is then

$$\hat{h}_k(T) = \underbrace{\Delta\hat{h}^\circ_{f,k}}_{\text{chemical}} + \underbrace{\int_{T^\circ}^{T} \hat{c}_{p,k}(T')\,\mathrm{d}T'}_{\text{sensible}}. \tag{3.5.1}$$

Symbols and units

$\hat{h}_k(T)$	molar enthalpy of species k	J/kmol
$\Delta\hat{h}^\circ_{f,k}$	standard enthalpy of formation (chemical part)	J/kmol
$\hat{c}_{p,k}$	molar heat capacity of k	J/(kmol K)
T°, T'	reference temperature 298.15, integration variable	K

Table 3.1: Standard enthalpies of formation (gas phase).

species	H_2	O_2	N_2	H	OH	H_2O	CO_2
$\Delta\hat{h}^\circ_f$ [MJ/kmol]	0	0	0	+218.0	+37.3	−241.8	−393.5

Physical intuition

A positive Δh°_f means the molecule is "uphill" from its elements: radicals like H and OH store energy because a bond was broken to make them. Products like H_2O are downhill: energy was released when they formed. Combustion converts uphill-ish reactants into deep-downhill products; the height difference is the heat release.

Worked example: heat of reaction

For $H_2 + \frac{1}{2}O_2 \to H_2O$ at T°:

$$\Delta\hat{h}^\circ_R = \sum_{\text{prod}} \nu_k \Delta\hat{h}^\circ_{f,k} - \sum_{\text{reac}} \nu_k \Delta\hat{h}^\circ_{f,k} = -241.8 - 0 - 0 = -241.8\,\text{MJ/kmol of } H_2.$$

Per kg of H_2: $241.8/2.016 = 120\,\text{MJ/kg}$, the *lower heating value*. Methane gives 50 MJ/kg and gasoline about 44 MJ/kg. Per kg, hydrogen is the most energetic fuel there is.

3.6 Second law, entropy and Gibbs energy

The second law states that for an isolated system entropy never decreases: $\mathrm{d}S \geq 0$, with equality only for reversible processes. For heat exchanged reversibly, $\mathrm{d}S = \delta Q_{\text{rev}}/T$.

Derivation: entropy of an ideal gas species

From $T\mathrm{d}s = \mathrm{d}h - v\,\mathrm{d}p$ (combine first law with $\delta Q_{\text{rev}} = T\mathrm{d}s$ and $h = u + pv$) and $v = \mathcal{R}T/(Wp)$, per kmol:

$$\mathrm{d}\hat{s} = \hat{c}_p \frac{\mathrm{d}T}{T} - \mathcal{R}\frac{\mathrm{d}p}{p} \quad\Longrightarrow\quad \hat{s}_k(T, p_k) = \hat{s}^\circ_k(T) - \mathcal{R}\ln\frac{p_k}{p^\circ},$$

where $\hat{s}_k^\circ(T)$ is the entropy at standard pressure. In a mixture, species k is at its partial pressure $p_k = X_k p$, so

$$\hat{s}_k = \hat{s}_k^\circ(T) - \mathcal{R}\ln\frac{p}{p^\circ} - \mathcal{R}\ln X_k. \tag{3.6.1}$$

Symbols and units

$\hat{s}$, $\hat{s}_k$	molar entropy (mixture, species k)	J/(kmol K)
$\hat{s}_k^\circ(T)$	standard-state molar entropy of k (at p°)	J/(kmol K)
$\hat{c}_p$	molar heat capacity at constant pressure	J/(kmol K)
$v = 1/\rho$	specific volume	m^3/kg
$p_k = X_k p$	partial pressure of species k	Pa
p°	standard pressure (Cantera: 101 325)	Pa
X_k	mole fraction of k	– (dimensionless)
T, $\mathcal{R}$	temperature, universal gas constant	K, 8314.46 J/(kmol K)

Physical intuition: the mixing term $-\mathcal{R}\ln X_k$

Since $X_k < 1$, $-\mathcal{R}\ln X_k > 0$: spreading a species out in a mixture raises its entropy. As $X_k \to 0$ this term goes to $+\infty$. This is why at equilibrium *every* species allowed by the mechanism is present at least in traces: the first few molecules of any species are entropically "free".

Gibbs free energy. $G \equiv H - TS$, per kmol of species k (its **chemical potential**):

$$\mu_k = \hat{g}_k = \hat{h}_k - T\hat{s}_k = \hat{g}_k^\circ(T) + \mathcal{R}T\ln\frac{p}{p^\circ} + \mathcal{R}T\ln X_k, \qquad \hat{g}_k^\circ = \hat{h}_k^\circ - T\hat{s}_k^\circ. \tag{3.6.2}$$

Symbols and units

$\mu_k = \hat{g}_k$	chemical potential (molar Gibbs energy of k in the mixture)	J/kmol
$\hat{g}_k^\circ$	standard-state molar Gibbs energy	J/kmol
$\hat{h}_k^\circ$, $\hat{s}_k^\circ$	standard molar enthalpy, entropy	J/kmol, J/(kmol K)
p, p°, X_k	pressure, standard pressure, mole fraction	Pa, Pa, –

Chapter 4 shows that at fixed T and p, a mixture evolves until $G = \sum_k n_k\mu_k$ is minimal.

3.7 NASA polynomials: how property data are stored

Cantera and CHEMKIN store $\hat{c}_{p,k}(T)$ for every species as a fourth-order polynomial, on two temperature ranges (typically 300–1000 and 1000–3500 K), each with seven coefficients $a_1, \ldots, a_7$:

$$\frac{\hat{c}_p^\circ}{\mathcal{R}} = a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4. \tag{3.7.1}$$

Derivation: enthalpy and entropy from c_p

Integrate $\mathrm{d}\hat{h}^\circ = \hat{c}_p^\circ \mathrm{d}T$ and divide by $\mathcal{R}T$:

$$\frac{\hat{h}^\circ}{\mathcal{R}T} = a_1 + \frac{a_2}{2}T + \frac{a_3}{3}T^2 + \frac{a_4}{4}T^3 + \frac{a_5}{5}T^4 + \frac{a_6}{T}. \tag{3.7.2}$$

Integrate $\mathrm{d}\hat{s}^\circ = \hat{c}_p^\circ \mathrm{d}T/T$:

$$\frac{\hat{s}^\circ}{\mathcal{R}} = a_1 \ln T + a_2 T + \frac{a_3}{2}T^2 + \frac{a_4}{3}T^3 + \frac{a_5}{4}T^4 + a_7. \tag{3.7.3}$$

a_6 and a_7 are the integration constants. a_6 is chosen so that $\hat{h}^\circ(298.15) = \Delta\hat{h}_f^\circ$: **it carries the chemical energy**. a_7 fixes the absolute entropy (third law).

Symbols and units

$\hat{c}_p^\circ$, $\hat{h}^\circ$, $\hat{s}^\circ$	standard-state molar heat capacity, enthalpy, entropy	J/(kmol K), J/kmol, J/(kmol K)
a_1	constant part of $\hat{c}_p^\circ/\mathcal{R}$	– (dimensionless)
a_2, a_3, a_4, a_5	temperature coefficients of $\hat{c}_p^\circ/\mathcal{R}$	1/K, 1/K^2, 1/K^3, K^{-4}
a_6	enthalpy integration constant (carries $\Delta\hat{h}_f^\circ$), equal to $\hat{h}^\circ/\mathcal{R}$ offset	K
a_7	entropy integration constant	– (dimensionless)
$\Delta\hat{h}_f^\circ$	standard enthalpy of formation	J/kmol
T	temperature	K

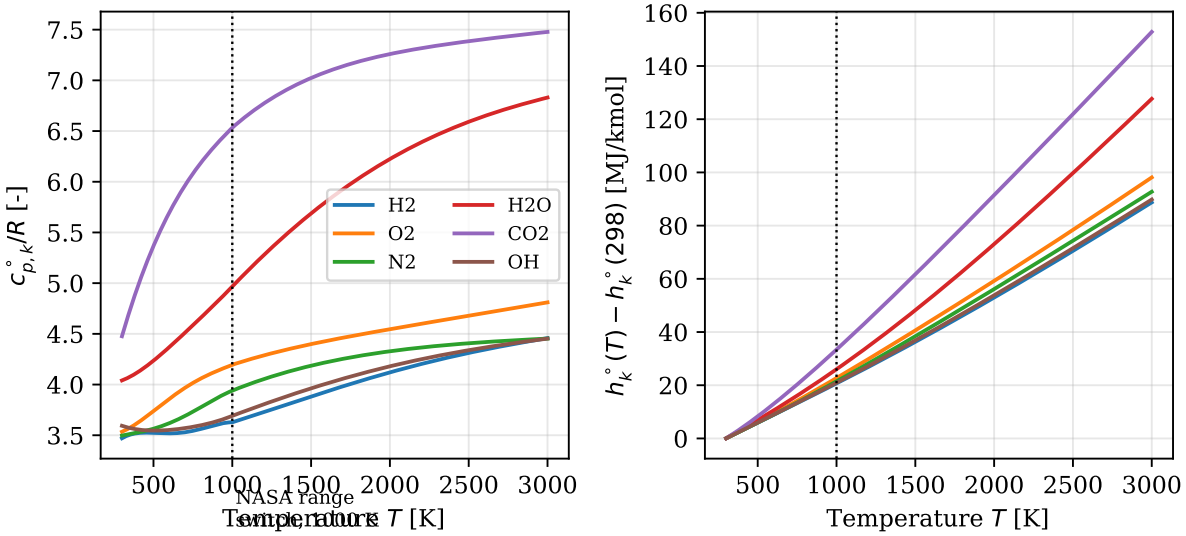


Figure 3.2: Left: c_p/R from the NASA polynomials of GRI-Mech 3.0. Diatomics start near 3.5 (translation + rotation) and grow as vibration switches on. H_2O and CO_2 grow more. Right: sensible enthalpy. The dotted line at 1000 K is where the polynomial ranges join; any kink there is a fitting artefact, not physics.

3.8 Mixture properties

Ideal mixtures have no interaction energy, so mixture properties are simple averages of species properties:

$$h = \sum_k Y_k h_k(T), \qquad c_p = \sum_k Y_k c_{p,k}(T) \qquad \text{(per kg: mass-weighted)}, \tag{3.8.1}$$

$$\hat{h} = \sum_k X_k \hat{h}_k(T) \qquad \text{(per kmol: mole-weighted)}, \tag{3.8.2}$$

$$s = \sum_k Y_k s_k(T, p, X_k) \qquad \text{(including the mixing term (3.6.1))}. \tag{3.8.3}$$

Check your understanding

1. Why can the mixture enthalpy h be negative? (Hint: H_2O at room temperature.)
2. A burnt-gas mixture has h equal to that of the fresh gas, yet is at 2300 K. Where did the "chemical" part of the enthalpy go?
3. In (3.5.1), which term changes when you preheat the reactants, and which when you change the fuel?

Chapter 4

Chemical equilibrium and the adiabatic flame temperature

Wait long enough and any reacting mixture stops changing. That end state is **chemical equilibrium**. It sets the maximum temperature a flame can reach and the composition it relaxes towards, and it is the yardstick for every simulation in the project.

4.1 Why equilibrium minimises the Gibbs energy

Derivation: the direction of spontaneous change at fixed T, p

The second law for a system exchanging heat δQ with surroundings at temperature T is Clausius's inequality $\mathrm{d}S \geq \delta Q/T$. At constant pressure $\delta Q = \mathrm{d}H$, so

$$\mathrm{d}H - T\mathrm{d}S \leq 0.$$

At constant T this is $\mathrm{d}(H - TS) = \mathrm{d}G \leq 0$. Any spontaneous change at fixed T and p *lowers* G, and the system stops when G is minimal:

$$\text{equilibrium at fixed } (T, p): \qquad G = \sum_k n_k \mu_k = \text{minimum}. \tag{4.1.1}$$

Symbols and units

S	entropy of the system	J/K
δQ	heat received from the surroundings	J
H, $G = H - TS$	enthalpy, Gibbs free energy of the system	J
T, p	temperature, pressure (held fixed)	K, Pa
n_k	amount of species k	kmol
μ_k	chemical potential of k (molar Gibbs energy in the mixture)	J/kmol

Physical intuition: a tug of war

$G = H - TS$. The H term wants strongly bonded, low-energy molecules (H_2O). The $-TS$ term wants many free fragments (H, O, OH), because more particles means more entropy. At low T, H wins and the products are almost pure H_2O. At high T the entropy

term weighs more, and a noticeable fraction of the water *dissociates*. Flame temperatures sit right where this tug of war starts to matter.

4.2 One reaction: extent of reaction and the equilibrium constant

Take one reaction $\sum_k \nu'_k \mathcal{M}_k \rightleftharpoons \sum_k \nu''_k \mathcal{M}_k$ with net stoichiometric coefficients $\nu_k = \nu''_k - \nu'_k$ (positive for products). If it advances by $\mathrm{d}\xi$ (the **extent of reaction**), then $\mathrm{d}n_k = \nu_k \mathrm{d}\xi$ and

$$\mathrm{d}G = \sum_k \mu_k \mathrm{d}n_k = \Big(\sum_k \nu_k \mu_k\Big)\mathrm{d}\xi.$$

At the minimum $\mathrm{d}G/\mathrm{d}\xi = 0$:

$$\sum_k \nu_k \mu_k = 0. \tag{4.2.1}$$

Derivation: equilibrium constant

Insert $\mu_k = \hat{g}^\circ_k(T) + \mathcal{R}T \ln(X_k p/p^\circ)$ from (3.6.2):

$$\underbrace{\sum_k \nu_k \hat{g}^\circ_k}_{\Delta G^\circ(T)} + \mathcal{R}T \sum_k \nu_k \ln \frac{X_k p}{p^\circ} = 0 \quad \Longrightarrow \quad \prod_k \left(\frac{X_k p}{p^\circ}\right)^{\nu_k} = \exp\left(-\frac{\Delta G^\circ}{\mathcal{R}T}\right) \equiv K_p(T).$$

Separating the pressure, with $\Delta\nu = \sum_k \nu_k$ (change in number of moles):

$$\prod_k X_k^{\nu_k} = K_p(T) \left(\frac{p}{p^\circ}\right)^{-\Delta\nu}. \tag{4.2.2}$$

Symbols and units

ν'_k, ν''_k, ν_k	stoichiometric coefficients of k as reactant, as product, net $\nu''_k - \nu'_k$	– (dimensionless)
ξ	extent of reaction	kmol
μ_k	chemical potential	J/kmol
$\hat{g}^\circ_k(T)$	standard-state molar Gibbs energy of k	J/kmol
$\Delta G^\circ = \sum_k \nu_k \hat{g}^\circ_k$	standard Gibbs energy change per kmol of reaction	J/kmol
$K_p(T)$	equilibrium constant (pressure-based)	– (dimensionless)
$\Delta\nu = \sum_k \nu_k$	change in number of moles per unit reaction	– (dimensionless)
X_k, p, p°	mole fraction, pressure, standard pressure	–, Pa, Pa

Key result

K_p depends **only on temperature** and is computed from tabulated $\hat{g}^\circ_k$ (NASA polynomials). The pressure enters the composition only through $(p/p^\circ)^{-\Delta\nu}$. The same K_p links forward and reverse reaction rates in Chapter 5, so thermodynamics and kinetics stay consistent.

Worked example: dissociation of water vapour at 3000 K

$H_2O \rightleftharpoons H_2 + \frac{1}{2}O_2$, $\Delta\nu = +\frac{1}{2}$. From the NASA data, $K_p(3000\,\mathrm{K}) = 0.0448$. Start with 1 kmol H_2O; after dissociating ξ kmol we have $1-\xi$ H_2O, ξ H_2, $\xi/2$ O_2, total $1+\xi/2$. Then

$$\frac{X_{H_2} X_{O_2}^{1/2}}{X_{H_2O}} = \frac{\xi\,(\xi/2)^{1/2}}{(1-\xi)(1+\xi/2)^{1/2}} = 0.0448 \left(\frac{p}{p^\circ}\right)^{-1/2}.$$

At 1 atm, trial and error gives $\xi \approx 0.146$: about 15% of the water is dissociated. At 10 atm the right-hand side drops by $\sqrt{10}$ and $\xi \approx 0.07$. Fig. 4.1 shows the same result as the minimum of $G(\xi)$.

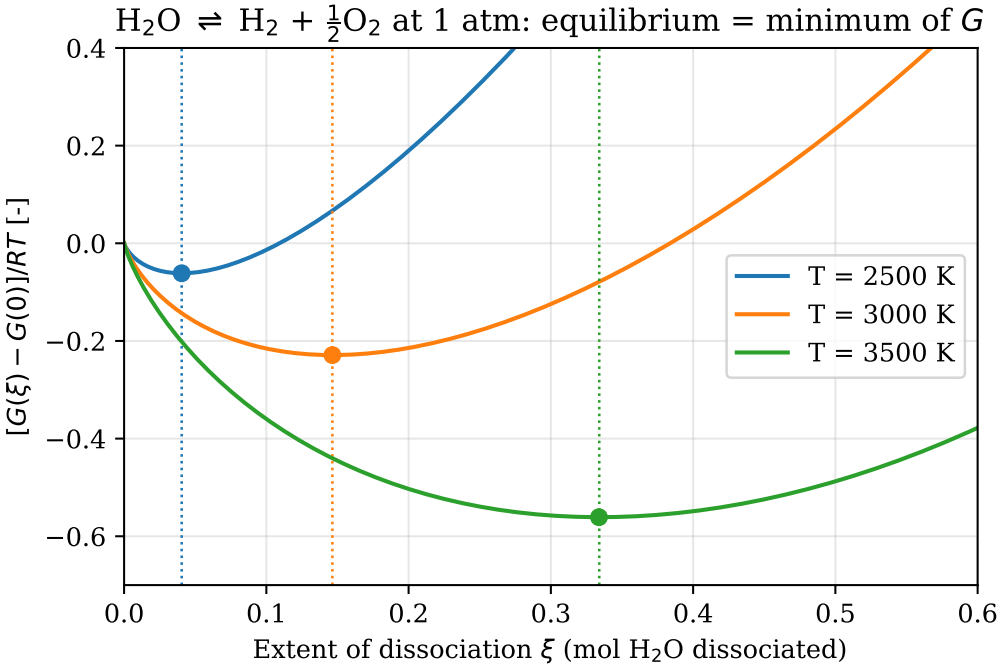


Figure 4.1: $G(\xi)$ for 1 kmol of H_2O allowed to dissociate into H_2 and O_2 at 1 atm, computed from the NASA $\hat{g}_k^\circ$ with the mixing term. Dots: minima. Dotted vertical lines: Cantera's `equilibrate("TP")` restricted to the same three species, which lands exactly on the minima. Hotter means deeper dissociation.

4.3 Le Chatelier's principle, derived

Pressure. From (4.2.2), if a reaction increases the number of moles ($\Delta\nu > 0$, as for every dissociation), raising p makes the right-hand side smaller, so the product side (fragments) shrinks. *Higher pressure suppresses dissociation.*

Temperature. Differentiating $\ln K_p = -\Delta G^\circ/\mathcal{R}T$ and using $\partial(\hat{g}^\circ/T)/\partial T = -\hat{h}^\circ/T^2$ gives the **van 't Hoff equation**

$$\frac{\mathrm{d}\ln K_p}{\mathrm{d}T} = \frac{\Delta H^\circ}{\mathcal{R}T^2}. \tag{4.3.1}$$

Symbols and units

K_p	equilibrium constant	– (dimensionless)
ΔH°	standard enthalpy of reaction per kmol of reaction	J/kmol
$\mathcal{R}$, T	gas constant, temperature	8314.46 J/(kmol K), K

Dissociation is endothermic ($\Delta H^\circ > 0$: breaking bonds costs energy), so K_p grows with T. *Higher temperature promotes dissociation.*

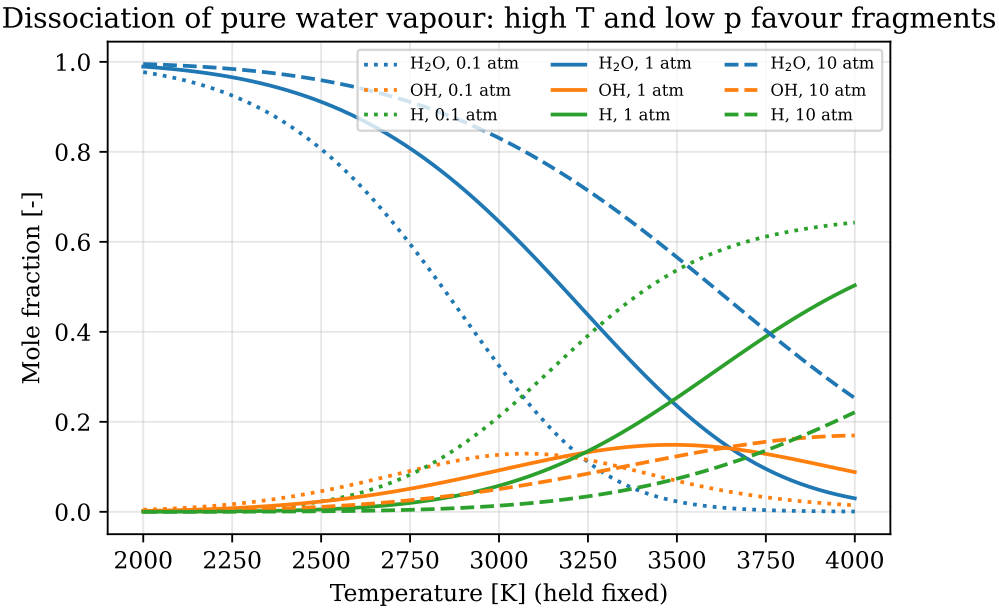


Figure 4.2: Pure water vapour held at fixed T and p (`equilibrate("TP")`). Fragments (OH, H) grow steeply with T and are suppressed by pressure, exactly as (4.2.2) and van 't Hoff predict.

Physical intuition

The system "pushes back" against what you do to it. Squeeze it (raise p) and it reduces the number of molecules to relieve the pressure. Heat it and it runs the reaction that absorbs heat (bond breaking). In a flame, dissociation acts as a thermostat: the hotter it gets, the more energy is parked in broken bonds instead of temperature.

4.4 Many species: element-constrained minimisation

Real products contain dozens of species linked by many reactions. Writing one K_p per reaction is clumsy. Instead, minimise $G = \sum_k n_k \mu_k$ directly, subject only to conservation of each element j: $\sum_k a_{jk} n_k = b_j$, where a_{jk} is the number of atoms of element j in species k and b_j is fixed by the initial mixture.

Derivation: the element potential method

With Lagrange multipliers λ_j (Section 2.8), $\partial/\partial n_k$ of $G - \sum_j \lambda_j(\sum_k a_{jk} n_k - b_j)$ gives

$$\mu_k = \sum_j \lambda_j a_{jk} \qquad \Longrightarrow \qquad X_k = \frac{p^\circ}{p} \exp\left(\frac{-\hat{g}_k^\circ(T) + \sum_j \lambda_j a_{jk}}{\mathcal{R}T}\right). \tag{4.4.1}$$

So every species' amount follows from just J unknowns λ_j (one per element: 3 for H/O/N), found together with $\sum_k X_k = 1$ and the element balances. Each λ_j is the chemical potential of one atom of element j. This is what Cantera's default equilibrium solver does, and it is why adding species to a mechanism does not make equilibrium much harder.

Symbols and units

a_{jk}	number of atoms of element j in one molecule of species k	– (dimensionless)
b_j	total amount of element j in the system (fixed)	kmol
λ_j	Lagrange multiplier = chemical potential of element j	J/kmol
J	number of elements (3 for H/O/N)	– (dimensionless)
$\hat{g}_k^\circ$, X_k, p, p°	as in the previous derivation	J/kmol, –, Pa

4.5 Which quantity is held fixed?

Minimum G applies at fixed (T, p). Other pairs give other extremum principles:

Table 4.1: Cantera's `equilibrate(XY)` options.

`XY`	held fixed	extremum	physical situation
`"TP"`	T, p	min G	isothermal reactor, furnace with perfect temperature control
`"HP"`	h, p	max S	**adiabatic constant-pressure combustion** (flames, gas turbines)
`"UV"`	u, v	max S	adiabatic closed bomb (constant-volume explosion)
`"SP"`	s, p	min H	isentropic compression or expansion of an equilibrium mixture

Pitfall

`equilibrate("TP")` after `gas.TPX = 300, p, mix` gives the equilibrium *at 300 K*: essentially all H_2O, very cold. To get a flame temperature you must use `"HP"`. Mixing these up is the most common Part 1 bug.

4.6 The equivalence ratio

Derivation: stoichiometric fuel/air ratio

For a hydrocarbon C_xH_y burning completely in air ($O_2 + 3.76\ N_2$):

$$C_xH_y + a\,(O_2 + 3.76\,N_2) \rightarrow x\,CO_2 + \tfrac{y}{2}\,H_2O + 3.76a\,N_2,$$

where the oxygen atom balance gives $2a = 2x + y/2$, i.e. $a = x + y/4$. For CH_4, $a = 2$; for H_2 ($x = 0, y = 2$), $a = \frac{1}{2}$.

Symbols and units

x, y	number of C and H atoms in the fuel C_xH_y	– (dimensionless)
$a = x + y/4$	kmol of O_2 needed per kmol of fuel	– (dimensionless)
3.76	kmol of N_2 accompanying each kmol of O_2 in air	– (dimensionless)

The **equivalence ratio** compares the actual fuel/oxidiser ratio with the stoichiometric one:

$$\phi = \frac{(n_F/n_{O_2})}{(n_F/n_{O_2})_{\mathrm{st}}} = \frac{(Y_F/Y_{O_2})}{(Y_F/Y_{O_2})_{\mathrm{st}}}. \tag{4.6.1}$$

Symbols and units

ϕ	equivalence ratio	– (dimensionless)
n_F, n_{O_2}	amounts of fuel and oxygen	kmol
Y_F, Y_{O_2}	mass fractions of fuel and oxygen	– (dimensionless)
$(\cdot)_{\mathrm{st}}$	value at stoichiometric conditions	–

It is the same whether you use moles or masses, because the molar-mass factor cancels. $\phi < 1$ is **lean** (excess air), $\phi > 1$ is **rich** (excess fuel). For H_2/air at equivalence ratio ϕ, the reactant mixture is $\phi\,H_2 + \frac{1}{2}(O_2 + 3.76N_2)$.

```
gas.set_equivalence_ratio(phi, fuel="H2", oxidizer="O2:1, N2:3.76")
gas.TP = 300.0, ct.one_atm      # set T and p *after* composition
```

4.7 The adiabatic flame temperature

If reactants at T_u burn at constant pressure with no heat loss, the first law says h is unchanged. The product temperature that satisfies this is the **adiabatic flame temperature**:

$$h_{\mathrm{products}}(T_{\mathrm{ad}}) = h_{\mathrm{reactants}}(T_u), \qquad p = \mathrm{const.} \tag{4.7.1}$$

Symbols and units

$h_{\mathrm{reactants}}$, h_{products}	specific enthalpy of the mixture before and after combustion	J/kg
T_u	unburnt (reactant) temperature	K
T_{ad}	adiabatic flame temperature	K

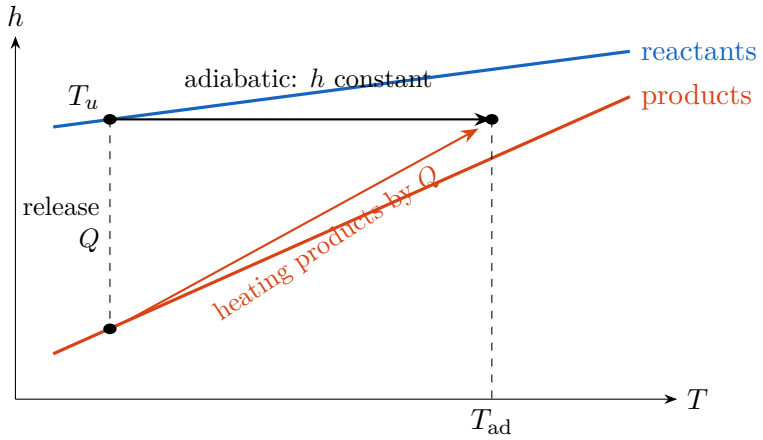


Figure 4.3: Enthalpy–temperature diagram. The products' curve lies lower by the heat of reaction. To keep h fixed, the products must be hot enough that their sensible enthalpy makes up the difference. The slope of each line is c_p: a steeper product line (larger c_p) gives a lower T_{ad}.

Worked example: hand estimate for H_2/air, $\phi = 1$

Per kmol of H_2: products are 1 H_2O + 1.88 N_2, and the heat released is 241.8 MJ. Using sensible enthalpies from the NASA fits (MJ/kmol):

T [K]	Δh_{H_2O}	$1.88\,\Delta h_{N_2}$	total
2400	94.3	132.8	227.1
2500	99.7	139.7	239.4

Interpolating to 241.8 gives $T_{\mathrm{ad}} \approx 2520$ K for *complete* combustion. Cantera gives:

- 2429 K if only H_2, O_2, N_2, H_2O are allowed, so water can partly dissociate back to $H_2 + O_2$;
- 2388 K with the full `h2o2.yaml` (radicals H, O, OH, ...).

Dissociation costs about 130 K. The project's sanity value of $\approx$ 2382 K is the full-equilibrium number.

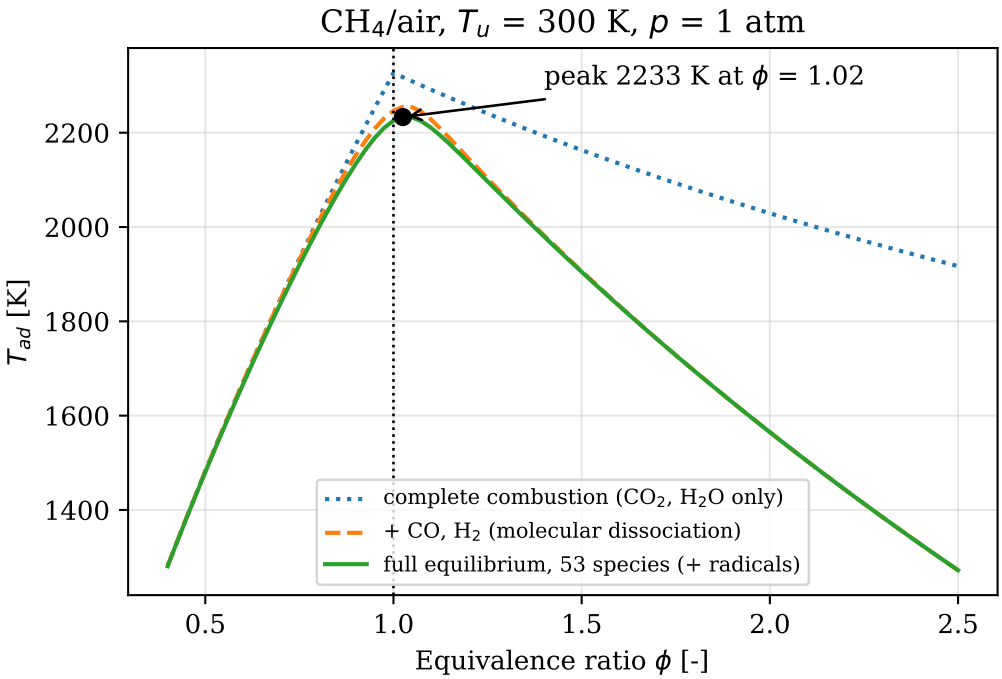


Figure 4.4: $T_{\mathrm{ad}}(\phi)$ for CH_4/air at three levels of chemical realism. Complete combustion peaks at $\phi = 1$; on its rich side the excess CH_4 simply rides along as a hot diluent. Allowing CO and H_2 (molecular dissociation and, on the rich side, partial oxidation) already accounts for most of the drop. Radicals remove a little more. The loss is about 100 K near stoichiometric, where it is hottest and dissociation strongest, and the peak moves slightly rich.

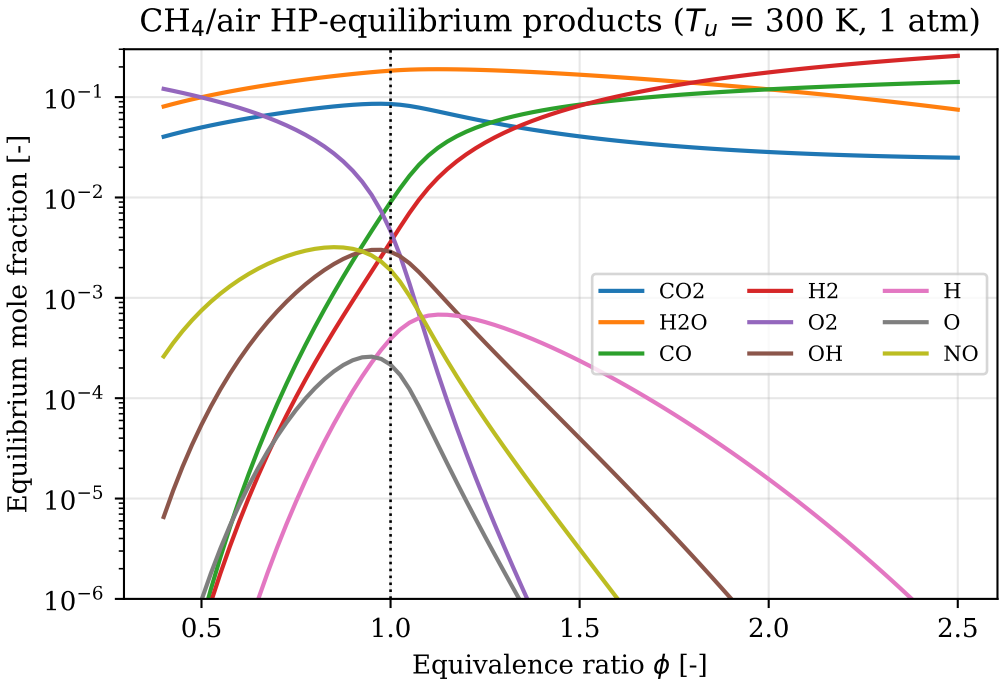


Figure 4.5: Equilibrium composition behind the curve in Fig. 4.4. Lean: excess O_2 survives. Rich: CO and H_2 replace CO_2 and H_2O. Radicals (OH, H, O) peak near stoichiometric where T is highest.

4.7.1 Why the curve has the shape it has

Derivation: a simple model of $T_{\mathrm{ad}}(\phi)$

Approximate the products' heat capacity per kmol of mixture as constant, $\bar{c}_p$, and let Q be the heat released per kmol of fuel burnt. Per kmol of air-borne O_2 we have ϕ/a kmol of fuel, where a is the stoichiometric O_2 per fuel.

- **Lean** ($\phi < 1$): all the fuel burns. Heat $\propto \phi$, while the total moles are dominated by the air and are nearly constant:

$$T_{\mathrm{ad}} - T_u \approx \frac{Q\,\phi/a}{n_{\mathrm{mix}}\bar{c}_p} \propto \phi.$$

- **Rich** ($\phi > 1$): the *oxygen* limits. Heat released is fixed at Q/a, but every extra kmol of fuel adds heat capacity:

$$T_{\mathrm{ad}} - T_u \approx \frac{Q/a}{(n_{\mathrm{mix,st}} + (\phi-1)/a)\,\bar{c}_p}.$$

Lean: the numerator falls linearly. Rich: the numerator is fixed and the denominator grows slowly. So T_{ad} falls off faster on the lean side. How much slower the rich side falls depends on how much heat capacity each extra kmol of fuel brings, which depends on the fuel.

Symbols and units

Q	heat released per kmol of fuel burnt	J/kmol
a	stoichiometric kmol O_2 per kmol fuel	– (dimensionless)
ϕ	equivalence ratio	– (dimensionless)
n_{mix}, $n_{\mathrm{mix,st}}$	kmol of product mixture per kmol of O_2 (general, stoichiometric)	– (dimensionless)
$\bar{c}_p$	average molar heat capacity of the products	J/(kmol K)
T_{ad}, T_u	adiabatic flame temperature, unburnt temperature	K

Physical intuition: why the peak is slightly rich

Two effects, both absent in the complete-combustion curve of Fig. 4.4:

1. At $\phi = 1$ dissociation leaves some O_2, O and OH in the products. A little extra fuel mops these up and releases more heat.
2. On the rich side the triatomic products (CO_2, H_2O, high c_p) are partly replaced by diatomic ones (CO, H_2, lower c_p). Less heat capacity means a hotter product for the same heat.

4.7.2 Preheating

Derivation: sensitivity to the unburnt temperature

Write (4.7.1) with average heat capacities of reactants $\bar{c}_{p,u}$ and products $\bar{c}_{p,b}$ (per kg), relative to the reference T°:

$$\bar{c}_{p,u}(T_u - T^\circ) + Q_m = \bar{c}_{p,b}(T_{\mathrm{ad}} - T^\circ) \quad \Longrightarrow \quad T_{\mathrm{ad}} = T^\circ + \frac{Q_m + \bar{c}_{p,u}(T_u - T^\circ)}{\bar{c}_{p,b}},$$

with Q_m the heat release per kg of mixture. Hence

$$\frac{\partial T_{\mathrm{ad}}}{\partial T_u} \approx \frac{\bar{c}_{p,u}}{\bar{c}_{p,b}}. \tag{4.7.2}$$

Since the hot, triatomic-rich products have the larger c_p, this ratio is below one. With dissociation it is lower still: the extra enthalpy pushes T_{ad} up, which increases dissociation, which absorbs part of that enthalpy.

Symbols and units

$\bar{c}_{p,u}$, $\bar{c}_{p,b}$	average specific heat capacity of reactants, of products	J/(kg K)
Q_m	heat released per kg of mixture	J/kg
T°	reference temperature, 298.15	K
T_u, T_{ad}	unburnt temperature, adiabatic flame temperature	K
$\partial T_{\mathrm{ad}}/\partial T_u$	gain in flame temperature per kelvin of preheat	– (dimensionless)

4.7.3 Pressure

Pressure does not enter (4.7.1) for an ideal gas: h does not depend on p. T_{ad} still depends on p, but *only through dissociation*. By Le Chatelier, higher p suppresses dissociation, so less energy is parked in broken bonds and T_{ad} rises. Without dissociation (the complete-combustion curve) T_{ad} would be exactly pressure-independent.

Project link: Part 1

Every task in Part 1 is an `equilibrate("HP")` loop over one or two parameters, using `h2o2.yaml`. The derivations above tell you what to expect and how to explain it:

- Task 1 (ϕ-sweep): peak location (dissociation and c_p argument), lean/rich asymmetry (the simple model; think about how much heat capacity each extra kmol of H_2 adds). For the flammability question, recall that kinetics needs $T \gtrsim 1000\,\mathrm{K}$ to sustain chain branching (Chapter 5).
- Task 2 (T_u-sweep): compare your slope with $\bar{c}_{p,u}/\bar{c}_{p,b}$, which you can compute with Cantera.
- Task 3 (p-sweep): Le Chatelier on OH, H, O. For NO_x, thermal NO formation ($N_2 + O \rightarrow NO + N$) depends on O atoms and very strongly on T.
- Task 4 (maps): which derivative is larger, $\partial T_{\mathrm{ad}}/\partial T_u$ or $\partial T_{\mathrm{ad}}/\partial \ln p$?

Check your understanding

1. Why does `equilibrate("HP")` conserve elements but not species?
2. Would T_{ad} change with pressure in a world without dissociation? Why?
3. Pure O_2 instead of air as oxidiser: is T_{ad} higher or lower, and does dissociation matter more or less?

Chapter 5

Chemical kinetics

Thermodynamics tells us the destination; kinetics tells us the speed and the route. Flames exist only because the route is fast at high temperature and almost blocked at room temperature. A stoichiometric H_2/air mixture can sit in a balloon for years, yet burns in microseconds once a spark has made some radicals.

5.1 Global versus elementary reactions

The overall reaction $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ never happens as written: three molecules would have to collide in exactly the right way and rearrange four bonds at once. What happens instead is a network of **elementary reactions**, each a single collision event, mostly involving highly reactive fragments called **radicals** (species with an unpaired electron: H, O, OH, HO_2).

Physical intuition

A global reaction is like a train timetable that lists only "London → Edinburgh". Elementary reactions are the individual stations and connections. If you want to know what happens when a station closes (extinction), when the train sets off (ignition) or how fast it travels (flame speed), you need the full route.

5.2 Why reactions need energy: collision theory

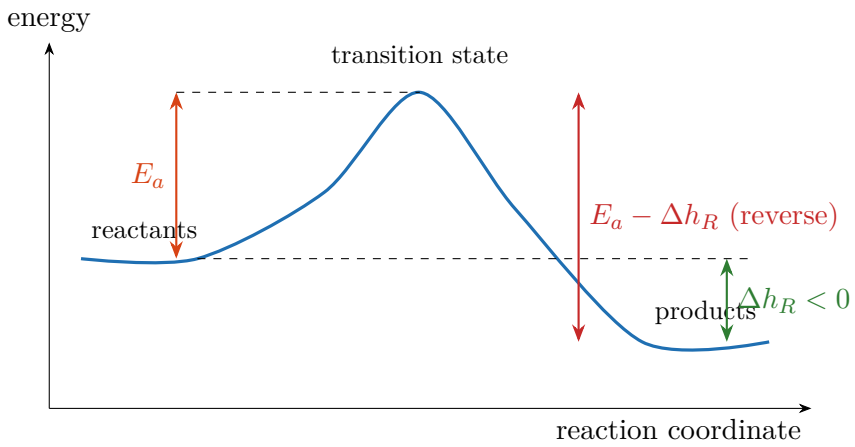


Figure 5.1: Energy along the reaction path. Even an exothermic reaction must first climb a barrier E_a. The reverse reaction must climb $E_a + |\Delta h_R|$, which is why exothermic reactions are effectively one-way at low temperature.

Derivation: Arrhenius law from collisions

Consider $\mathrm{A} + \mathrm{B} \to$ products. Treat molecules as hard spheres with collision cross-section $\sigma_{AB} = \pi(r_A + r_B)^2$. A molecule A moving at relative speed $\bar{c}_{rel}$ sweeps a volume $\sigma_{AB}\bar{c}_{rel}$ per second and hits every B in it. Kinetic theory gives $\bar{c}_{rel} = \sqrt{8k_BT/(\pi m_{AB})}$, with reduced mass $m_{AB} = m_A m_B/(m_A + m_B)$. Collisions per unit volume per second:

$$Z_{AB} = \sigma_{AB}\,\bar{c}_{rel}\,n_A n_B.$$

Only collisions whose energy along the line of centres exceeds E_a react. From the Maxwell–Boltzmann distribution that fraction is $e^{-E_a/\mathcal{R}T}$. Add a "steric factor" $P \le 1$ for the orientation. The reaction rate in kmol/m^3/s is then $k\,[\mathrm{A}][\mathrm{B}]$ with

$$k(T) = \underbrace{P\,\sigma_{AB}\sqrt{\frac{8k_BT}{\pi m_{AB}}}\,N_A}_{\propto T^{1/2}}\; e^{-E_a/\mathcal{R}T}. \tag{5.2.1}$$

Generalising the $T^{1/2}$ to T^{β} (to absorb effects the hard-sphere model misses) gives the **modified Arrhenius** form used in every mechanism file:

$$\boxed{k(T) = A\,T^{\beta}\exp\!\left(-\frac{E_a}{\mathcal{R}T}\right)} \tag{5.2.2}$$

Symbols and units

r_A, r_B	molecular radii	m
σ_{AB}	collision cross-section $\pi(r_A + r_B)^2$	m^2
$\bar{c}_{rel}$	mean relative speed of A and B	m/s
m_{AB}	reduced mass $m_A m_B/(m_A + m_B)$	kg
n_A, n_B	number densities	$1/\mathrm{m}^3$
Z_{AB}	collision rate per unit volume	$1/(\mathrm{m}^3\,\mathrm{s})$
E_a	activation energy (barrier height)	J/kmol
P	steric (orientation) factor	– (dimensionless)
k_B, N_A	Boltzmann constant, Avogadro's number	J/K, 1/kmol
$k(T)$	rate constant; for a bimolecular reaction	$\mathrm{m}^3/(\mathrm{kmol}\,\mathrm{s})$
A	pre-exponential factor (order n reaction)	$(\mathrm{m}^3/\mathrm{kmol})^{n-1}1/\mathrm{s}\,\mathrm{K}^{-\beta}$
β	temperature exponent	– (dimensionless)
[A], [B]	molar concentrations	$\mathrm{kmol}/\mathrm{m}^3$

Physical intuition

AT^β counts how often molecules meet in the right orientation; it varies slowly with T. $e^{-E_a/\mathcal{R}T}$ is the fraction that meet hard enough; it varies violently. Radical–radical recombinations have $E_a \approx 0$: every encounter works, and they are fast at any T. Reactions that break a strong bond (like $\mathrm{H{+}O_2} \to \mathrm{O{+}OH}$, $E_a \approx 70\,\mathrm{kJ/mol}$) need high temperatures.

Units of A. For a reaction of overall order n (number of reactant molecules, including third bodies), k has units $(\mathrm{m}^3/\mathrm{kmol})^{n-1}\,1/\mathrm{s}$. Mechanism files often use cm, mol, cal. Cantera converts on load, but be careful when copying numbers by hand.

5.3 Law of mass action and production rates

Write reaction i as $\sum_k \nu'_{ki}\mathcal{M}_k \rightleftharpoons \sum_k \nu''_{ki}\mathcal{M}_k$. The collision argument generalises: the rate is proportional to the product of reactant concentrations, each raised to its stoichiometric coefficient. The **rate of progress** is

$$q_i = k_{f,i}\prod_k C_k^{\nu'_{ki}} - k_{r,i}\prod_k C_k^{\nu''_{ki}} \qquad [\mathrm{kmol}/(\mathrm{m}^3\,\mathrm{s})], \tag{5.3.1}$$

and the **net molar production rate** of species k sums over all reactions:

$$\dot{\omega}_k = \sum_{i=1}^{I} \nu_{ki}\, q_i, \qquad \nu_{ki} = \nu''_{ki} - \nu'_{ki}. \tag{5.3.2}$$

Symbols and units

q_i	net rate of progress of reaction i	$\mathrm{kmol}/(\mathrm{m}^3\,\mathrm{s})$
$k_{f,i}$, $k_{r,i}$	forward, reverse rate constants	depend on order
C_k	molar concentration	$\mathrm{kmol}/\mathrm{m}^3$
ν'_{ki}, ν''_{ki}, ν_{ki}	reactant, product, net stoichiometric coefficients	– (dimensionless)
$\dot{\omega}_k$	net molar production rate of k	$\mathrm{kmol}/(\mathrm{m}^3\,\mathrm{s})$
$W_k\dot{\omega}_k$	net mass production rate of k	$\mathrm{kg}/(\mathrm{m}^3\,\mathrm{s})$
I, K	number of reactions, of species	– (dimensionless)

In matrix form $\dot{\boldsymbol{\omega}} = \mathsf{N}\boldsymbol{q}$, with N the $K \times I$ stoichiometric matrix. The mass production rate is $W_k\dot{\omega}_k$ [kg/(m^3 s)]. Mass is conserved in every reaction, so $\sum_k W_k\dot{\omega}_k = 0$ exactly.

```
gas.forward_rate_constants     # k_f,i
gas.net_rates_of_progress      # q_i            [kmol/m^3/s]
gas.net_production_rates       # omega_dot_k    [kmol/m^3/s]
gas.heat_release_rate          # -sum_k h_k omega_dot_k   [W/m^3]
```

5.4 Reverse rates come from thermodynamics

Derivation: detailed balance

At equilibrium every elementary reaction is individually balanced: $q_i = 0$. Therefore

$$\frac{k_{f,i}}{k_{r,i}} = \prod_k C_{k,\mathrm{eq}}^{\nu_{ki}} \equiv K_{c,i}(T).$$

Using $C_k = X_k p/(\mathcal{R}T)$ and (4.2.2):

$$K_{c,i} = K_{p,i}\left(\frac{p^\circ}{\mathcal{R}T}\right)^{\Delta\nu_i}, \qquad K_{p,i} = \exp\!\left(-\frac{\Delta G_i^\circ}{\mathcal{R}T}\right). \tag{5.4.1}$$

Mechanisms therefore list only (A, β, E_a) for the forward direction. Cantera computes $k_{r,i} = k_{f,i}/K_{c,i}$ from the NASA polynomials. This guarantees that a kinetic simulation run for long enough ends exactly at the thermodynamic equilibrium of Chapter 4.

Symbols and units

$k_{f,i}$, $k_{r,i}$	forward, reverse rate constants of reaction i	depend on reaction order
$C_{k,\mathrm{eq}}$	equilibrium concentration of species k	kmol/m^3
$K_{c,i}$	concentration-based equilibrium constant	(kmol/m^3)$^{\Delta\nu_i}$
$K_{p,i}$	pressure-based equilibrium constant	– (dimensionless)
$\Delta\nu_i$	net change in moles in reaction i	– (dimensionless)
ΔG_i°	standard Gibbs energy change of reaction i	J/kmol
p°, $\mathcal{R}$, T	standard pressure, gas constant, temperature	Pa, J/(kmol K), K

5.5 Third bodies and pressure dependence

Recombination such as H + OH → H_2O releases a lot of energy into the new bond. If nothing removes it, the molecule flies apart again. A **third body** M (any molecule) must collide at the same time and carry the energy away:

$$\mathrm{H} + \mathrm{OH} + \mathrm{M} \rightarrow \mathrm{H_2O} + \mathrm{M}, \qquad q = k\,[\mathrm{H}][\mathrm{OH}][\mathrm{M}], \qquad [\mathrm{M}] = \sum_k \epsilon_k C_k.$$

Symbols and units

[M]	effective third-body concentration	kmol/m^3
ϵ_k	collision efficiency of species k	– (dimensionless)
k	termolecular rate constant	m^6/kmol^2/s

The **efficiencies** ϵ_k capture that some molecules absorb energy better: H_2O ($\epsilon \approx 12$) is far better than Ar ($\epsilon \approx 0.7$).

Fall-off (Lindemann). For reactions like $2OH(+M) \rightleftharpoons H_2O_2(+M)$ the collision with M is only needed at low pressure. At high pressure the energised complex is always stabilised and the rate stops depending on [M]:

$$k = \frac{k_0[\mathrm{M}]\,k_\infty}{k_\infty + k_0[\mathrm{M}]} = k_\infty \frac{P_r}{1+P_r}, \qquad P_r = \frac{k_0[\mathrm{M}]}{k_\infty}, \tag{5.5.1}$$

Symbols and units

k_0	low-pressure limit rate constant	$\mathrm{m^6/kmol^2/s}$
k_∞	high-pressure limit rate constant	$\mathrm{m^3/(kmol\,s)}$
P_r	reduced pressure	– (dimensionless)
F	Troe broadening factor	– (dimensionless)

so $k \approx k_0[\mathrm{M}]$ at low p and $k \approx k_\infty$ at high p. Troe's form multiplies this by a broadening factor F. In the YAML file these appear as `type: falloff` with `low-P-rate-constant`, `high-P-rate-constant` and `Troe:` entries.

5.6 The hydrogen–oxygen chain

The heart of all hydrogen combustion (and of the H/O sub-mechanism of every hydrocarbon) is:

	reaction	role
R1	$H_2 + O_2 \rightarrow HO_2 + H$	**initiation**: makes the first radicals (slow, high E_a)
R2	$H + O_2 \rightleftharpoons O + OH$	**chain branching**: 1 radical $\rightarrow$ 2
R3	$O + H_2 \rightleftharpoons H + OH$	**chain branching**: 1 radical $\rightarrow$ 2
R4	$OH + H_2 \rightleftharpoons H_2O + H$	**chain propagation**: 1 radical $\rightarrow$ 1, makes H_2O
R5	$H + O_2 + M \rightleftharpoons HO_2 + M$	**termination** (at low T, high p): reactive H $\rightarrow$ sluggish HO_2
R6	$H + OH + M \rightleftharpoons H_2O + M$	**termination / recombination**: releases most of the heat

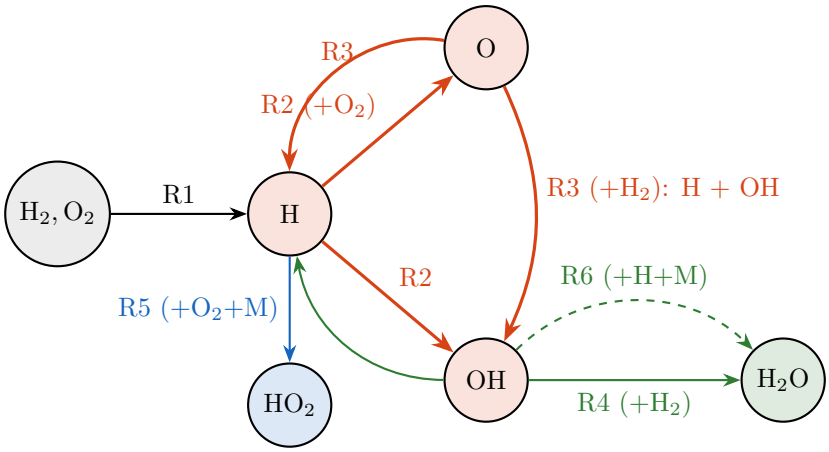


Figure 5.2: The H_2/O_2 chain. Red arrows multiply radicals (branching), green convert radicals into product while keeping the chain alive (R4 also hands the H atom back, curved green arrow), blue removes the active H atom into the much less reactive HO_2.

Derivation: net effect of one branching cycle

Add R2 + R3 + 2×R4:

$$\mathrm{H} + \mathrm{O_2} + 3\,\mathrm{H_2} \;\rightarrow\; 2\,\mathrm{H_2O} + 3\,\mathrm{H}.$$

One H atom goes in, **three** come out. Each cycle multiplies the radical pool, and the radical population grows exponentially. This is chain branching.

Symbols and units

H, O	hydrogen and oxygen atoms (radicals)	–
OH	hydroxyl radical	–
HO_2	hydroperoxyl radical (much less reactive than H)	–
R2, R3, R4	reactions of the table above (branching, branching, propagation)	–

Derivation: branching versus termination: the crossover temperature

Let n be the H-atom concentration and focus on its two fates: R2 (rate $k_2[\mathrm{O_2}]n$), which via the cycle above returns net +2 radicals, and R5 (rate $k_5[\mathrm{M}][\mathrm{O_2}]n$), which removes it. Then

$$\frac{\mathrm{d}n}{\mathrm{d}t} \approx \dot{r}_{\mathrm{init}} + (2k_2 - k_5[\mathrm{M}])[\mathrm{O_2}]\,n. \tag{5.6.1}$$

If $2k_2 > k_5[\mathrm{M}]$ the coefficient is positive and n grows like $e^{t/\tau}$: **explosion**. If it is negative, n settles at a small steady value: **no explosion**. The boundary $2k_2(T) = k_5(T)[\mathrm{M}]$ is the **crossover temperature**. k_2 has a large activation energy and k_5 has none, so branching wins at high T. $[\mathrm{M}] = p/\mathcal{R}T$, so termination gains with pressure.

Symbols and units

n	concentration of H atoms	kmol/m^3
$\dot{r}_{\text{init}}$	rate at which initiation creates radicals	kmol/(m^3 s)
k_2	rate constant of R2 (bimolecular)	m^3/(kmol s)
k_5	rate constant of R5 (termolecular)	m^6/kmol^2/s
[M], [O_2]	third-body and oxygen concentrations; [M] $\approx p/\mathcal{R}T$	kmol/m^3
$(2k_2 - k_5[\text{M}])[O_2]$	net radical growth rate $= 1/\tau$	1/s

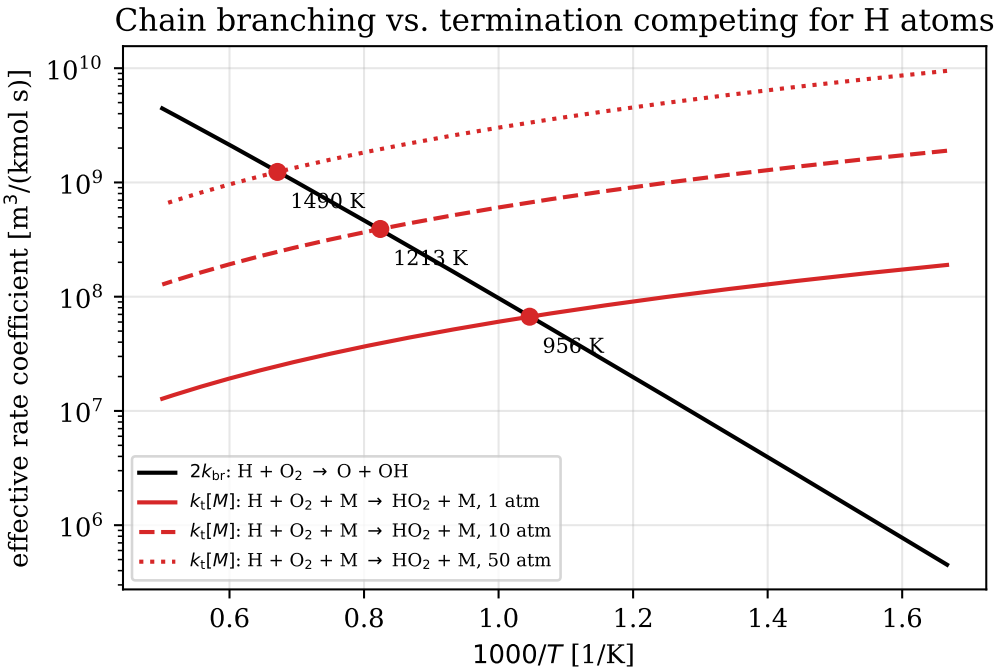


Figure 5.3: Arrhenius plot computed from `h2o2.yaml`. The branching line (black) is steep (large E_a). The termination lines (red) are almost flat and move up with pressure. Their intersections are the crossover temperatures: about 950 K at 1 atm, rising with p. Below crossover, hydrogen chemistry is "slow"; above it, "explosive".

Physical intuition

This single competition explains a surprising amount:

- Why mixtures need $T \gtrsim 1000$ K to burn vigorously, which matters for flammability limits (Project Part 1, Task 1).
- Why ignition delay has a kink as a function of temperature (Fig. 6.2).
- Why hydrogen flame speeds *decrease* with pressure: higher [M] favours R5 over R2 (Project Part 3, Task 5).
- Why a flame can be extinguished: if strain or heat loss lowers the reaction-zone temperature towards crossover, branching collapses (Project Part 2).

5.7 Heat release

Most of the heat is *not* released in the branching steps (R2 is endothermic!) but in the recombination and propagation steps that form H_2O (R4, R6). The local volumetric **heat release rate** is

$$\dot{q} = -\sum_k \hat{h}_k(T)\,\dot{\omega}_k \qquad [\mathrm{W/m^3}], \tag{5.7.1}$$

Symbols and units

$\dot{q}$	volumetric heat release rate	$\mathrm{W/m^3}$
$\hat{h}_k$	molar enthalpy of k (including formation)	J/kmol
$\dot{\omega}_k$	net molar production rate	$\mathrm{kmol/(m^3\,s)}$

because destroying a species of enthalpy $\hat{h}_k$ and making another of lower enthalpy frees the difference as sensible heat.

5.8 Stiffness of chemistry

Each species has a characteristic time scale $\tau_k \sim C_k/|\dot{\omega}_k|$. Radicals live for nanoseconds; NO forms over milliseconds to seconds. Formally the time scales are the inverse eigenvalues of the chemical Jacobian $\partial\dot{\boldsymbol{\omega}}/\partial\boldsymbol{C}$, and they span 8–10 orders of magnitude. This is the stiffness of Section 2.6. It is why Cantera uses implicit integrators and why you never want to write your own explicit chemistry loop.

5.9 Mechanism files

A mechanism is a list of species (with thermo and transport data) plus a list of reactions (with Arrhenius parameters). `h2o2.yaml` has 10 species and 29 reactions; GRI-Mech 3.0 (`gri30.yaml`) has 53 species and 325 reactions for natural gas. The anatomy of the YAML format is shown in Section 11.3.

Pitfall

A mechanism is a *model*, validated over a range of T, p, ϕ against experiments (ignition delays, flame speeds). Outside that range it can be wrong. GRI-Mech 3.0 was designed for natural gas, not hydrogen. For H_2 use a dedicated H/O mechanism (`h2o2.yaml`, or literature mechanisms by Li et al., Burke et al., Ó Conaire et al.).

5.10 Global one-step reactions (for scaling arguments)

For back-of-envelope reasoning we sometimes pretend the whole chain is one step:

$$\dot{\omega}_F = -A\,[F]^a[O]^b\,e^{-E/\mathcal{R}T}, \qquad n = a + b \text{ (overall order)}. \tag{5.10.1}$$

Symbols and units

$[F]$, $[O]$	fuel, oxidiser concentrations	kmol/m^3
a, b, n	reaction orders in fuel, oxidiser, overall	– (dimensionless)
A	pre-factor	(m^3/kmol)$^{n-1}$1/s
E	global activation energy	J/kmol

Since $[F] = \rho Y_F / W_F \propto p$, the mass reaction rate scales like p^n at fixed T and composition. We will use this in Chapter 9 to predict how flame speed depends on pressure. The *effective* order of H_2/air chemistry is less than 2 near crossover, precisely because the termination step R5 gains with pressure.

Check your understanding

1. R2 is endothermic. Why is it nevertheless the most important reaction in hydrogen combustion?
2. Why does the recombination R6 become faster at high pressure, and why does that move heat release closer to the reaction zone?
3. Using Fig. 5.3, estimate the crossover temperature at 10 atm. What does it imply for ignition at engine-like pressures?

Chapter 6

The homogeneous (0D) reactor

The simplest reacting system is a perfectly mixed box: no gradients, so no convection and no diffusion. Only chemistry and thermodynamics are left. It is the first rung of the ladder (Fig. 1.1) and the place to see ignition with your own eyes.

6.1 Governing equations

Derivation: constant-pressure adiabatic reactor

Species. Mass of species k in the reactor, mY_k, changes only by chemistry. The total mass m is constant (closed system):

$$m\frac{\mathrm{d}Y_k}{\mathrm{d}t} = V\,W_k\dot{\omega}_k \quad\Longrightarrow\quad \frac{\mathrm{d}Y_k}{\mathrm{d}t} = \frac{W_k\dot{\omega}_k}{\rho}. \tag{6.1.1}$$

Energy. Adiabatic and at constant p, so $h = \sum_k Y_k h_k(T)$ is constant:

$$0 = \frac{\mathrm{d}h}{\mathrm{d}t} = \sum_k Y_k\frac{\mathrm{d}h_k}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}t} + \sum_k h_k\frac{\mathrm{d}Y_k}{\mathrm{d}t} = c_p\frac{\mathrm{d}T}{\mathrm{d}t} + \sum_k h_k\frac{W_k\dot{\omega}_k}{\rho},$$

using $\mathrm{d}h_k/\mathrm{d}T = c_{p,k}$ and $\sum_k Y_k c_{p,k} = c_p$. Therefore

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -\frac{1}{\rho c_p}\sum_k \hat{h}_k\dot{\omega}_k = \frac{\dot{q}}{\rho c_p}. \tag{6.1.2}$$

($h_k W_k = \hat{h}_k$ converts per-kg to per-kmol.) The density follows from the ideal gas law at fixed p; the volume expands as the gas heats.

Symbols and units

m, V	mass and volume of the reactor contents	kg, m^3
Y_k	mass fraction of species k	– (dimensionless)
W_k	molar mass of k	kg/kmol
$\dot{\omega}_k$	net molar production rate of k	kmol/(m^3 s)
ρ	density	kg/m^3
h, h_k	specific enthalpy of the mixture, of species k	J/kg
$\hat{h}_k = h_k W_k$	molar enthalpy of k	J/kmol
c_p, $c_{p,k}$	specific heat capacity of mixture, of species k	J/(kg K)
$\dot{q}$	heat release rate	W/m^3
T, t	temperature, time	K, s

Physical intuition

Read (6.1.2) as: temperature rises at the rate heat is released divided by the heat capacity of a cubic metre of gas. It is (2.4.1) with only the unsteady term and the source term. In a flame (Chapter 9) the same source appears, now competing with convection and diffusion.

Constant volume. For a closed bomb, replace h by u and c_p by c_v: $\mathrm{d}T/\mathrm{d}t = -\sum_k \hat{u}_k \dot{\omega}_k/(\rho c_v)$. Pressure then rises during combustion. In Cantera this is `IdealGasReactor` rather than `IdealGasConstPressureRe`

6.2 Anatomy of ignition

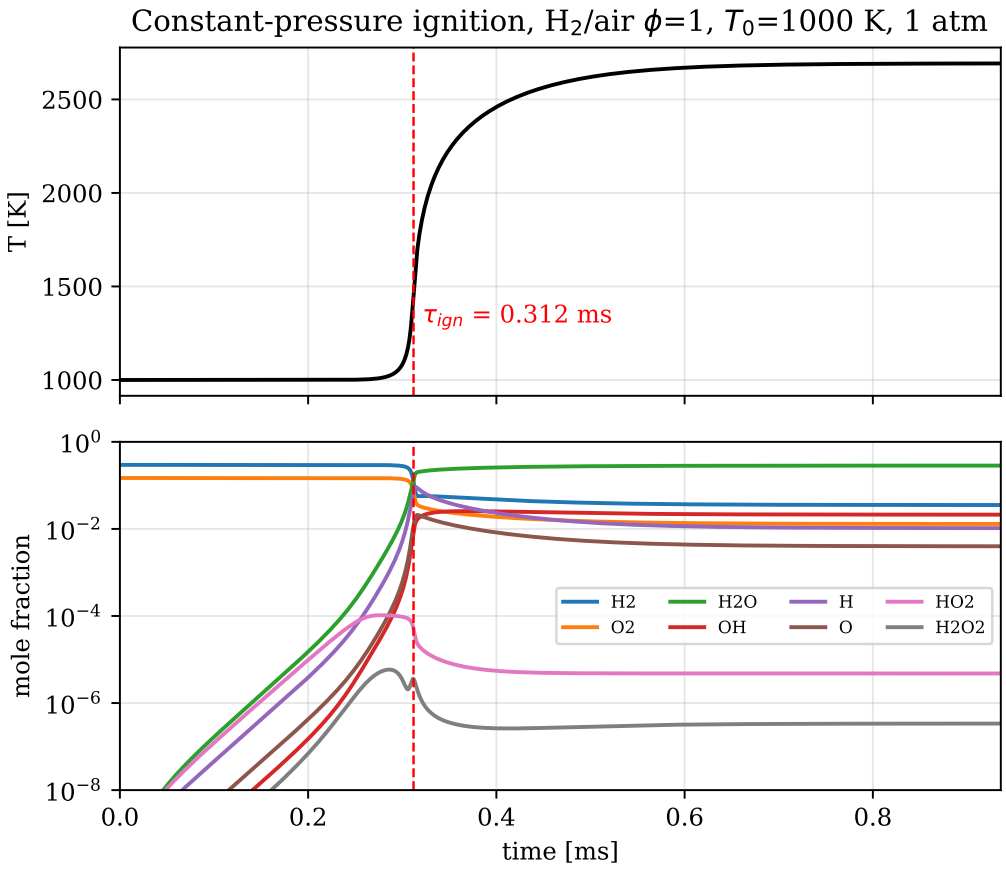


Figure 6.1: Constant-pressure ignition of stoichiometric H_2/air at $T_0 = 1000\,\mathrm{K}$, 1 atm (`h2o2.yaml`). Top: temperature. Bottom: species on a log scale. The ignition delay τ_{ign} is the time of maximum $\mathrm{d}T/\mathrm{d}t$.

Three phases are visible:

1. **Induction.** T is flat, but radicals (H, O, OH) grow *exponentially*: straight lines on the log plot, as predicted by the branching analysis of Section 5.6. Very little heat is released because the radical concentrations are still tiny.
2. **Ignition.** Once radicals reach $\sim 10^{-3}$, fuel consumption and recombination (heat release) become significant. Higher T accelerates branching even more. This feedback makes T jump by $\sim 1500\,\mathrm{K}$ in microseconds.
3. **Equilibration.** Radicals recombine slowly (three-body reactions) and T creeps towards T_{ad}, the same value `equilibrate("HP")` gives.

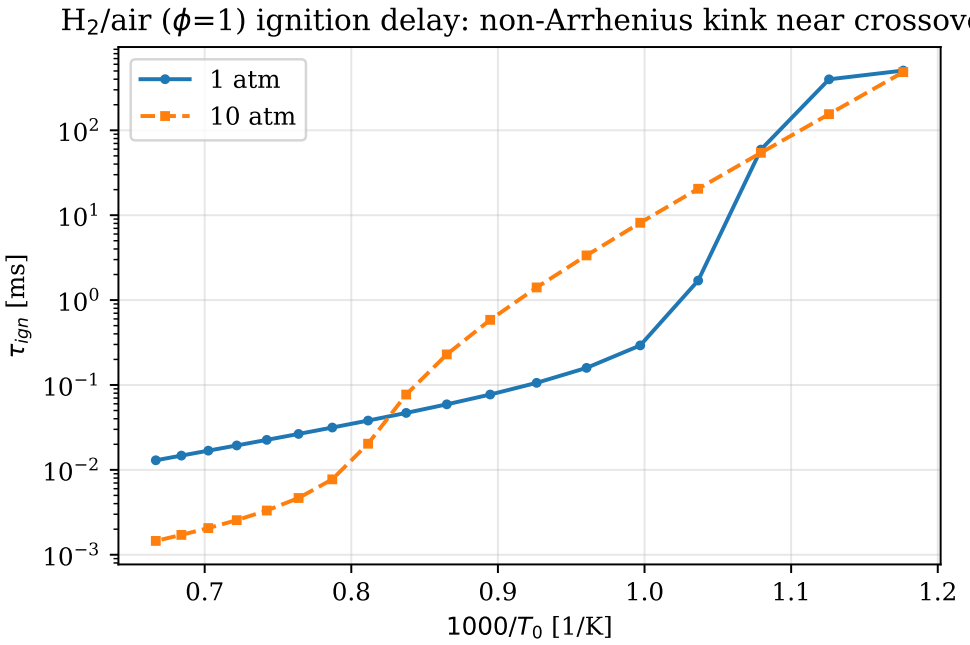


Figure 6.2: Ignition delay vs. $1000/T_0$. At high T the curves are nearly straight (Arrhenius). Near the crossover temperature (Fig. 5.3) the slope changes sharply: below it, R5 steals H atoms and ignition must proceed through slower HO_2/H_2O_2 chemistry. At 10 atm the crossover moves to higher T, which is why at intermediate temperatures high pressure ignites *later*.

6.3 Damköhler number and frozen chemistry

Compare a flow time τ_{flow} (residence time in a combustor, time for strain to remove heat in a flame) with a chemical time τ_{chem} (e.g. τ_{ign}):

$$\text{Da} = \frac{\tau_{\text{flow}}}{\tau_{\text{chem}}}. \tag{6.3.1}$$

Symbols and units

τ_{flow}	flow (residence or mixing) time	s
τ_{chem}	chemical time (e.g. ignition delay)	s
Da	Damköhler number	– (dimensionless)

- Da $\gg 1$: chemistry finishes; the state is close to equilibrium.
- Da $\ll 1$: chemistry has no time; composition is *frozen*. This is how CO and unburnt hydrocarbons escape from engines.
- Da ~ 1: chemistry and flow compete, as in extinction, ignition and pollutant formation.

The Damköhler number will reappear as the parameter of the S-curve in non-premixed flames (Section 10.7).

6.4 In Cantera

```python
import cantera as ct, numpy as np
gas = ct.Solution("h2o2.yaml")
gas.set_equivalence_ratio(1.0, "H2", "O2:1, N2:3.76")
gas.TP = 1000.0, ct.one_atm

reactor = ct.IdealGasConstPressureReactor(gas, clone=True)  # clone: own copy of gas
net = ct.ReactorNet([reactor])
states = ct.SolutionArray(gas, extra=["t"])
while net.time < 2e-3:
    net.step()                                      # adaptive implicit step (CVODES)
    states.append(reactor.phase.state, t=net.time)

tau = states.t[np.argmax(np.gradient(states.T, states.t))]
```

Pitfall: Cantera 3.2 API changes

`reactor.thermo` is deprecated in favour of `reactor.phase`. The `clone` argument will default to `True` after 3.2. Pass it explicitly to avoid warnings and surprises: with `clone=False` the reactor and your `gas` object share one state, so reading `gas.T` during integration gives the reactor temperature.

Check your understanding

1. Why does the 0D reactor's final temperature equal `equilibrate("HP")` exactly? (Hint: Section 5.4.)
2. In Fig. 6.1, which species peaks *before* ignition, and why? (Hint: R5 at 1000 K.)
3. What is the Damköhler number of a gas turbine combustor with residence time 5 ms burning a mixture whose τ_{ign} is 0.1 ms?

Chapter 7

Molecular transport

As soon as there are gradients, molecules carry momentum, energy and mass from one place to another simply by moving randomly. In a flame these fluxes are what bring heat and radicals *upstream* into the fresh gas and make the flame propagate.

7.1 Random walks: where diffusion comes from

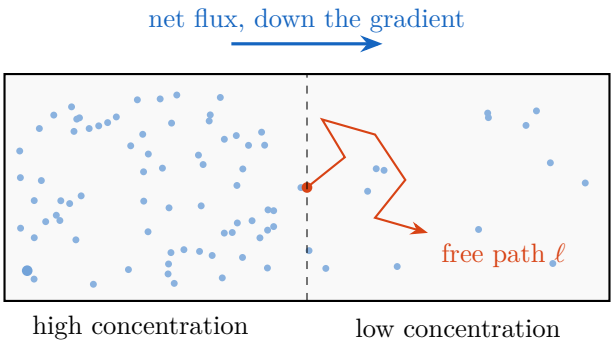


Figure 7.1: Each molecule moves randomly. More molecules on the left means more random crossings of the dashed plane from left to right than from right to left. The net flux is down the gradient, even though no single molecule "knows" about it.

Derivation: diffusion coefficient from kinetic theory

Molecules move at mean speed $\bar{c} = \sqrt{8k_BT/(\pi m)}$ and travel a **mean free path** $\ell = 1/(\sqrt{2}\,n\sigma)$ between collisions (n is number density, σ the collision cross-section). Consider the plane $x = 0$ with concentration $C(x)$. Molecules crossing it last collided about one free path away, so roughly $\frac{1}{6}\bar{c}\,C(-\ell)$ cross per unit area left-to-right and $\frac{1}{6}\bar{c}\,C(+\ell)$ right-to-left (one sixth of directions point along $+x$). Net flux:

$$J = \tfrac{1}{6}\bar{c}\,[C(-\ell) - C(\ell)] \approx -\tfrac{1}{3}\bar{c}\,\ell\,\frac{\mathrm{d}C}{\mathrm{d}x} \quad \Longrightarrow \quad D \approx \tfrac{1}{3}\bar{c}\,\ell.$$

Exactly the same argument for momentum and for energy gives the kinematic viscosity $\nu \approx \frac{1}{3}\bar{c}\ell$ and the thermal diffusivity $\alpha \approx \frac{1}{3}\bar{c}\ell$ (up to factors of order one).

Symbols and units

$\bar{c}$	mean molecular speed $\sqrt{8k_BT/\pi m}$	m/s
m	mass of one molecule	kg
ℓ	mean free path	m
n	number density	$1/\mathrm{m}^3$
σ	collision cross-section	m^2
$C(x)$	concentration of the diffusing species	$\mathrm{kmol}/\mathrm{m}^3$
J	net molar flux through the plane	$\mathrm{kmol}/(\mathrm{m}^2\,\mathrm{s})$
D, ν, α	mass diffusivity, kinematic viscosity, thermal diffusivity	m^2/s
$\mu = \rho\nu$	dynamic viscosity	$\mathrm{Pa\,s}$
$\lambda = \rho c_p \alpha$	thermal conductivity	$\mathrm{W}/(\mathrm{m\,K})$

Key result: scalings

With $\bar{c} \propto \sqrt{T/m}$ and $\ell \propto 1/n \propto T/p$:

$$D,\ \alpha,\ \nu\ \propto\ \frac{T^{3/2}}{p\sqrt{m}}, \qquad \mu = \rho\nu \propto \sqrt{T}\ \ \text{(independent of } p\text{)}, \qquad \lambda = \rho c_p \alpha \propto \sqrt{T}. \quad (7.1.1)$$

Real molecules attract and repel softly (Lennard-Jones potential), which changes the exponents to roughly $D \propto T^{1.7}/p$ and $\mu, \lambda \propto T^{0.7}$ (Fig. 7.2). Three consequences matter for flames:

1. All three diffusivities are of the same order, so Pr and Sc are of order one.
2. Diffusivities scale as $1/p$, so flames get thinner at high pressure.
3. Light molecules (H_2, H) diffuse much faster: $D \propto 1/\sqrt{m}$ (more precisely, on the reduced mass of the colliding pair).

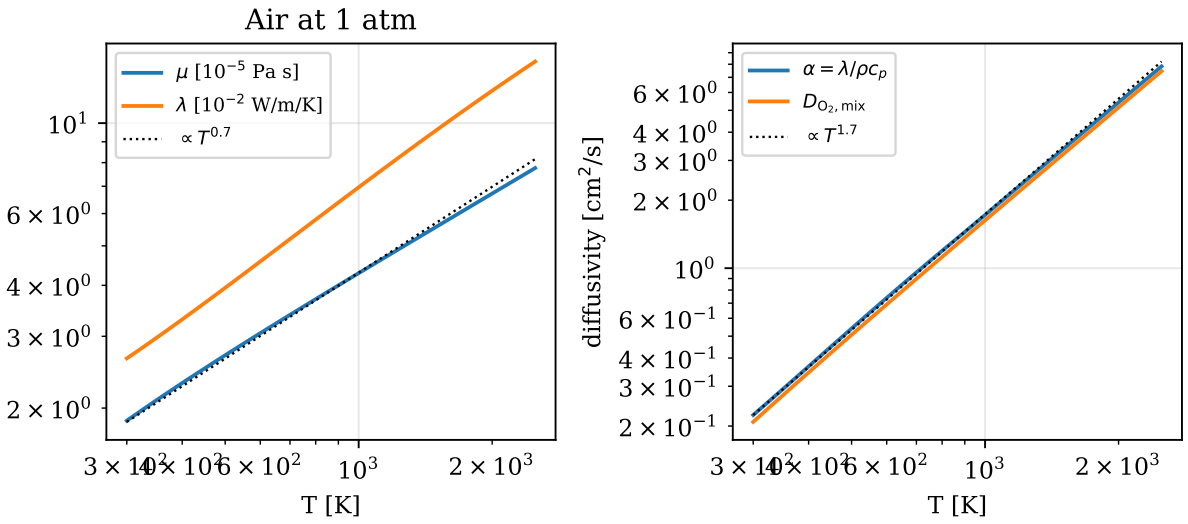


Figure 7.2: Transport properties of air computed by Cantera. Diffusivities grow by a factor of $\sim$ 20 between room temperature and flame temperature. This is why the preheat zone of a flame is much wider on its hot side.

7.2 The three flux laws

$$\text{Newton (momentum):} \quad \tau_{ij} = \mu\Big(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\Big) - \tfrac{2}{3}\mu\,\delta_{ij}\,\nabla\cdot\boldsymbol{u}, \tag{7.2.1}$$

$$\text{Fourier (heat):} \quad \boldsymbol{q} = -\lambda\nabla T + \sum_k h_k \boldsymbol{j}_k \quad (+\text{Dufour, negligible}), \tag{7.2.2}$$

$$\text{Fick (species):} \quad \boldsymbol{j}_k = \rho Y_k \boldsymbol{V}_k \approx -\rho D_k \nabla Y_k. \tag{7.2.3}$$

Symbols and units

τ_{ij}	viscous stress component	Pa
μ	dynamic viscosity	Pa s
u_i, x_i	velocity component, coordinate	m/s, m
δ_{ij}	Kronecker delta (1 if $i = j$, else 0)	– (dimensionless)
$\boldsymbol{q}$	heat flux	$\mathrm{W/m^2}$
λ	thermal conductivity	$\mathrm{W/(m\,K)}$
h_k	specific enthalpy of k	J/kg
$\boldsymbol{j}_k$, $\boldsymbol{V}_k$	diffusive mass flux, diffusion velocity of k	$\mathrm{kg/(m^2\,s)}$, m/s
D_k	diffusivity of k	$\mathrm{m^2/s}$

$\boldsymbol{V}_k$ is the **diffusion velocity** of species k relative to the mass-averaged velocity $\boldsymbol{u}$, and $\boldsymbol{j}_k$ the diffusive mass flux. The $\sum h_k \boldsymbol{j}_k$ term in $\boldsymbol{q}$ is enthalpy carried by species diffusing: a hot H atom diffusing upstream brings its enthalpy along.

Derivation: the mass-conservation constraint

$\boldsymbol{u}$ is defined as the mass-averaged velocity, $\rho\boldsymbol{u} = \sum_k \rho Y_k \boldsymbol{u}_k$. Hence the diffusive fluxes must add up to zero:

$$\sum_k \boldsymbol{j}_k = \sum_k \rho Y_k \boldsymbol{V}_k = 0. \tag{7.2.4}$$

Fick's law with arbitrary D_k violates this. Practical codes, including Cantera, add a **correction velocity** $\boldsymbol{V}_c = -\sum_k Y_k \boldsymbol{V}_k^{\mathrm{Fick}}$ to every species so that the sum is zero.

Symbols and units

$\boldsymbol{u}$	mass-averaged velocity	m/s
$\boldsymbol{u}_k$	velocity of species k	m/s
$\boldsymbol{V}_k = \boldsymbol{u}_k - \boldsymbol{u}$	diffusion velocity of k	m/s
$\boldsymbol{V}_c$	correction velocity added to every species	m/s
$\boldsymbol{j}_k = \rho Y_k \boldsymbol{V}_k$	diffusive mass flux of k	$\mathrm{kg/(m^2\,s)}$

7.3 Mixture-averaged and multicomponent diffusion

Kinetic theory gives binary coefficients $\mathcal{D}_{jk}$ for each pair of species. In a mixture, species k collides with everything. Two models:

- **Mixture-averaged** (Cantera `"mixture-averaged"`, the default): k diffuses into an effective "rest of the mixture" with

$$D_{km} = \frac{1 - Y_k}{\sum_{j\neq k} X_j/\mathcal{D}_{jk}}, \qquad \boldsymbol{j}_k = -\rho\frac{W_k}{W} D_{km}\nabla X_k + \rho Y_k \boldsymbol{V}_c. \tag{7.3.1}$$

Symbols and units

D_{km}	mixture-averaged diffusivity of k	m^2/s
$\mathcal{D}_{jk}$	binary diffusivity of the pair j–k	m^2/s
X_j, Y_k	mole, mass fractions	– (dimensionless)
W_k, W	molar mass of k, of the mixture	kg/kmol
$\boldsymbol{V}_c$	correction velocity	m/s

Cheap and accurate to a few percent for most flames.

- **Multicomponent** (`"multicomponent"`): solves the full Stefan–Maxwell relations $\nabla X_k = \sum_j \frac{X_k X_j}{\mathcal{D}_{kj}}(\boldsymbol{V}_j - \boldsymbol{V}_k) + \ldots$, plus **thermal diffusion** (Soret effect: light species are pushed towards hot regions). More expensive. It matters for H_2 and H, especially in lean H_2 flames.

```
flame.transport_model = "mixture-averaged"      # default
flame.transport_model = "multicomponent"; flame.soret_enabled = True
flame.transport_model = "unity-Lewis-number"    # D_k = alpha for every k
```

Mixture viscosity and conductivity come from species values with mixing rules (Wilke's formula for μ, an average of arithmetic and harmonic means for λ). Cantera does this for you from the Lennard-Jones parameters stored for each species in the YAML file.

7.4 Dimensionless numbers

Making the governing equations dimensionless with a length L, velocity U and the property values produces ratios that say which process dominates:

number	definition	compares	typical (flame)
Reynolds	$\mathrm{Re} = UL/\nu$	inertia / viscous diffusion	laminar: $\lesssim 10^3$
Prandtl	$\mathrm{Pr} = \nu/\alpha$	momentum / heat diffusion	≈ 0.7
Schmidt	$\mathrm{Sc}_k = \nu/D_k$	momentum / mass diffusion	0.2 (H_2) – 1.5
Lewis	$\mathrm{Le}_k = \alpha/D_k$	**heat / mass diffusion**	0.3 (H_2) – 2

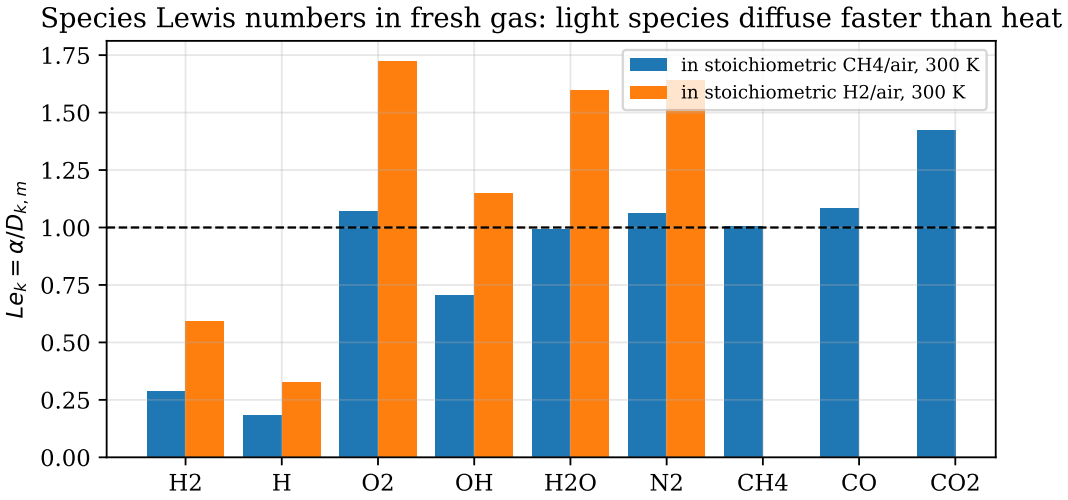


Figure 7.3: Lewis numbers in fresh stoichiometric gas. H_2 and especially H diffuse much faster than heat (Le < 1). CH_4, O_2 and N_2 are close to 1.

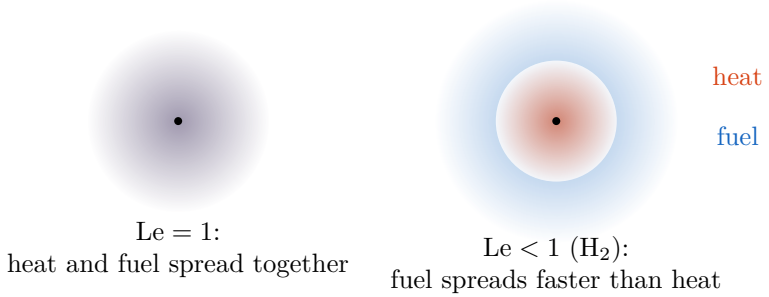


Figure 7.4: Lewis number as a race between heat and species. With Le < 1 a reacting region draws in fuel faster than it loses heat, so it becomes richer and hotter locally. This is **preferential diffusion**.

7.5 The unity-Lewis-number simplification

Derivation: enthalpy becomes a conserved scalar when all $\mathrm{Le}_k = 1$

Take the heat flux with Fick's law, $\boldsymbol{q} = -\lambda\nabla T - \sum_k h_k \rho D_k \nabla Y_k$. Since $h = \sum_k Y_k h_k(T)$,

$$\nabla h = c_p \nabla T + \sum_k h_k \nabla Y_k \quad \Longrightarrow \quad \lambda \nabla T = \frac{\lambda}{c_p}\nabla h - \frac{\lambda}{c_p}\sum_k h_k \nabla Y_k.$$

Substituting,

$$\boldsymbol{q} = -\rho\alpha\nabla h + \sum_k h_k\, \rho\,(\alpha - D_k)\nabla Y_k. \tag{7.5.1}$$

If $D_k = \alpha$ for every species ($\mathrm{Le}_k = 1$), the sum vanishes and $\boldsymbol{q} = -\rho\alpha\nabla h$. The enthalpy equation of Chapter 8 (low Mach number, no radiation) becomes

$$\rho\frac{\mathrm{D}h}{\mathrm{D}t} = \nabla\cdot(\rho\alpha\nabla h),$$

which has no source term: h is transported like a passive dye. By the same argument, every *elemental* mass fraction (and therefore the mixture fraction, Chapter 10) is conserved,

because chemistry rearranges atoms but never creates them.

Symbols and units

$\boldsymbol{q}$	heat flux	$\mathrm{W/m^2}$
λ	thermal conductivity	$\mathrm{W/(m\,K)}$
h, h_k	specific enthalpy of mixture, of species k	J/kg
c_p	specific heat capacity	$\mathrm{J/(kg\,K)}$
$\alpha = \lambda/\rho c_p$	thermal diffusivity	$\mathrm{m^2/s}$
D_k	diffusivity of species k	$\mathrm{m^2/s}$
$\mathrm{Le}_k = \alpha/D_k$	Lewis number of species k	– (dimensionless)
∇T, ∇Y_k	temperature and mass-fraction gradients	K/m, 1/m

Key result

With unity Lewis numbers, enthalpy and element mass fractions are linear functions of the mixture fraction in a two-stream problem. That makes the equilibrium temperature $T_{\mathrm{eq}}(Z)$ meaningful (Part 2) and keeps the mixture fraction constant through a premixed flame (Part 3). With real (mixture-averaged) transport and $\mathrm{Le}_{\mathrm{H_2}} \approx 0.3$, both statements break. The size of the breakage is exactly what Part 3, Task 3 asks you to measure.

Check your understanding

1. Doubling the pressure: what happens to D, μ and to a flame's thickness?
2. Why is Le_{H} even smaller than $\mathrm{Le}_{\mathrm{H_2}}$?
3. Under $\mathrm{Le} = 1$, can the local enthalpy inside a flame differ from the fresh-gas enthalpy? And under mixture-averaged transport?

Chapter 8

The reacting Navier–Stokes equations

The onboarding introduction lists four conservation laws. Here each one is built from the generic balance (2.4.1), so you can see where every term comes from. We then simplify them for the low-speed flames Cantera solves.

8.1 Mass

With $\varphi = 1$, no diffusion of total mass and no source:

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho \boldsymbol{u}) = 0. \tag{8.1.1}$$

Symbols and units		
ρ	density	kg/m^3
$\boldsymbol{u}$	velocity	m/s
t	time	s

8.2 Species

With $\varphi = Y_k$, diffusive flux $\boldsymbol{j}_k = \rho Y_k \boldsymbol{V}_k$ and chemical source $W_k \dot{\omega}_k$:

$$\frac{\partial (\rho Y_k)}{\partial t} + \nabla\cdot(\rho \boldsymbol{u} Y_k) = -\nabla\cdot(\rho Y_k \boldsymbol{V}_k) + W_k \dot{\omega}_k, \qquad k = 1 \dots K. \tag{8.2.1}$$

Symbols and units		
Y_k	mass fraction of species k	– (dimensionless)
$\boldsymbol{V}_k$	diffusion velocity of k	m/s
$\rho Y_k \boldsymbol{V}_k$	diffusive mass flux of k	kg/(m^2 s)
$W_k \dot{\omega}_k$	chemical mass production rate of k	kg/(m^3 s)

Summing over all k, and using $\sum Y_k = 1$, $\sum \rho Y_k \boldsymbol{V}_k = 0$ and $\sum W_k \dot{\omega}_k = 0$, gives back (8.1.1). So only $K - 1$ species equations are independent. Cantera solves all K and enforces the constraints numerically.

Physical intuition

Each species is a dye that is carried by the flow, smeared by diffusion at its own rate, and created or destroyed by chemistry. The equations are coupled only through $\dot{\omega}_k(T, \boldsymbol{Y})$ and through the flow. With 10 species (`h2o2.yaml`) that is 10 equations; with GRI-Mech 3.0, 53.

8.3 Momentum

Derivation: Newton's second law for a control volume

Momentum per unit volume is $\rho\boldsymbol{u}$. The forces are pressure on the surface ($-p\boldsymbol{n}\,\mathrm{d}S$), viscous stress on the surface ($\boldsymbol{\tau}\cdot\boldsymbol{n}\,\mathrm{d}S$) and gravity in the volume ($\rho\boldsymbol{g}\,\mathrm{d}V$). The generic law with $\varphi = \boldsymbol{u}$ and Gauss's theorem gives

$$\frac{\partial(\rho\boldsymbol{u})}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla\cdot\boldsymbol{\tau} + \rho\boldsymbol{g}, \tag{8.3.1}$$

with $\boldsymbol{\tau}$ the Newtonian stress of Chapter 7.

Symbols and units

$\rho\boldsymbol{u}$	momentum per unit volume	kg/(m^2 s)
$\boldsymbol{u}\boldsymbol{u}$	dyadic product: momentum flux tensor per unit density	m^2/s^2
p	pressure	Pa
$\boldsymbol{\tau}$	viscous stress tensor	Pa
$\boldsymbol{g}$	gravitational acceleration	m/s^2

8.4 Energy

Derivation: from total energy to temperature

Step 1: total energy $e_t = u + \frac{1}{2}|\boldsymbol{u}|^2$. Its sources are heat conduction $-\nabla\cdot\boldsymbol{q}$, work by pressure $-\nabla\cdot(p\boldsymbol{u})$, work by viscous stresses $\nabla\cdot(\boldsymbol{\tau}\cdot\boldsymbol{u})$ and gravity $\rho\boldsymbol{g}\cdot\boldsymbol{u}$:

$$\rho\frac{\mathrm{D}e_t}{\mathrm{D}t} = -\nabla\cdot\boldsymbol{q} - \nabla\cdot(p\boldsymbol{u}) + \nabla\cdot(\boldsymbol{\tau}\cdot\boldsymbol{u}) + \rho\boldsymbol{g}\cdot\boldsymbol{u}.$$

Step 2: remove kinetic energy. Dot (8.3.1) with $\boldsymbol{u}$ to get $\rho\,\mathrm{D}(\frac{1}{2}|\boldsymbol{u}|^2)/\mathrm{D}t = -\boldsymbol{u}\cdot\nabla p + \boldsymbol{u}\cdot(\nabla\cdot\boldsymbol{\tau}) + \rho\boldsymbol{g}\cdot\boldsymbol{u}$ and subtract. Using $h = u + p/\rho$ and mass conservation,

$$\rho\frac{\mathrm{D}h}{\mathrm{D}t} = \frac{\mathrm{D}p}{\mathrm{D}t} - \nabla\cdot\boldsymbol{q} + \boldsymbol{\tau} : \nabla\boldsymbol{u}. \tag{8.4.1}$$

Step 3: temperature. With $h = \sum Y_k h_k(T)$, $\rho\,\mathrm{D}h/\mathrm{D}t = \rho c_p\,\mathrm{D}T/\mathrm{D}t + \sum_k h_k \rho\,\mathrm{D}Y_k/\mathrm{D}t$. Replace $\rho\,\mathrm{D}Y_k/\mathrm{D}t$ by the right-hand side of (8.2.1) (in the form (2.5.2)). Then expand $\nabla\cdot\boldsymbol{q} = -\nabla\cdot(\lambda\nabla T) + \sum_k \nabla\cdot(h_k\boldsymbol{j}_k)$ with $\nabla h_k = c_{p,k}\nabla T$. The $\sum_k h_k\nabla\cdot\boldsymbol{j}_k$ terms cancel and what remains is

$$\boxed{\rho c_p\frac{\mathrm{D}T}{\mathrm{D}t} = \frac{\mathrm{D}p}{\mathrm{D}t} + \nabla\cdot(\lambda\nabla T) - \Big(\sum_k \rho Y_k c_{p,k}\boldsymbol{V}_k\Big)\cdot\nabla T - \sum_k h_k W_k\dot{\omega}_k + \boldsymbol{\tau} : \nabla\boldsymbol{u}} \tag{8.4.2}$$

Symbols and units

u (scalar)	specific internal energy (not velocity)	J/kg
$e_t = u + \frac{1}{2}\lvert\boldsymbol{u}\rvert^2$	specific total energy	J/kg
$h = u + p/\rho$	specific enthalpy	J/kg
$\boldsymbol{q}$	heat flux	W/m^2
$p\boldsymbol{u}$, $\boldsymbol{\tau}\cdot\boldsymbol{u}$	rate of work by pressure and viscous forces (per area)	W/m^2
$\boldsymbol{\tau} : \nabla\boldsymbol{u}$	viscous dissipation (heating)	W/m^3
c_p, $c_{p,k}$	specific heat capacity of mixture, of species k	J/(kg K)
λ	thermal conductivity	W/(m K)
$\boldsymbol{V}_k$, $\boldsymbol{j}_k$	diffusion velocity, diffusive mass flux of k	m/s, kg/(m^2 s)
$h_k W_k\dot{\omega}_k$	enthalpy carried by chemical production of k	W/m^3
$\mathrm{D}/\mathrm{D}t$	material derivative	unit of argument per s

Physical intuition: reading the temperature equation term by term

- $\rho c_p\,\mathrm{D}T/\mathrm{D}t$: heating of a moving fluid parcel.
- $\mathrm{D}p/\mathrm{D}t$: compression heating (important in engines, negligible in open flames).
- $\nabla\cdot(\lambda\nabla T)$: conduction from hot neighbours. In a flame this preheats the fresh gas.
- $-\sum \rho Y_k c_{p,k}\boldsymbol{V}_k\cdot\nabla T$: species with different heat capacities diffusing through a temperature gradient. Small, but kept by Cantera.
- $-\sum h_k W_k\dot{\omega}_k = \dot{q}$: **heat release** by chemistry, the same term as in the 0D reactor (6.1.2).
- $\boldsymbol{\tau} : \nabla\boldsymbol{u}$: viscous heating, negligible at low speed.

8.5 Closing the system

Unknowns: ρ, $\boldsymbol{u}$ (3), p, T, Y_k (K): $K+6$. Equations: mass (1), momentum (3), energy (1), species ($K-1$ independent, plus $\sum Y_k = 1$) and the ideal gas law (1): also $K+6$. The fluxes ($\boldsymbol{\tau}$, $\boldsymbol{q}$, $\boldsymbol{V}_k$) come from the transport laws, h_k and c_p from the NASA polynomials, and $\dot{\omega}_k$ from the mechanism. **Cantera supplies all these closures.** The flow solver only has to discretise derivatives.

8.6 The low-Mach-number approximation

A laminar methane flame moves at 0.4 m/s and the sound speed is 350 m/s: Mach number $\mathrm{Ma} \sim 10^{-3}$. Expanding $p = p_0(t) + \mathrm{Ma}^2 p_1(\boldsymbol{x}, t)$:

- The **thermodynamic pressure** p_0 is spatially uniform and is used in the ideal gas law and all property evaluations.
- The **hydrodynamic pressure** p_1 (tiny, a few Pa across a flame) appears only in the momentum equation to drive the flow.
- In an open domain p_0 is constant, so $\mathrm{D}p/\mathrm{D}t \approx 0$; viscous heating is dropped as well.

Sound waves are filtered out, which allows time steps set by the flow rather than by acoustics. Cantera's 1D flames use exactly this: `flame.P` is a single number.

Symbols and units

$\mathrm{Ma} = u/c$	Mach number	– (dimensionless)
c	speed of sound	m/s
$p_0(t)$	thermodynamic (uniform) pressure	Pa
$p_1(\boldsymbol{x}, t)$	hydrodynamic pressure perturbation	Pa

8.7 Why we do not start in 3D

For H_2/air that is already 16 coupled nonlinear PDEs. For methane with GRI-Mech 3.0 it is 59. The chemical source terms are stiff, flames are thinner than 1 mm while combustors are metres across, and in turbulent flows the flame is wrinkled at every scale. Before any of that, two 1D model problems isolate the physics:

- **Premixed flame** (Chapter 9): reactions and diffusion balancing convection. Output: flame speed and thickness.
- **Non-premixed counterflow flame** (Chapter 10): mixing controlled by strain. Output: flame structure in mixture-fraction space and extinction.

Both are also the building blocks ("flamelets") of turbulent combustion models.

Chapter 9

Laminar premixed flames

In a premixed flame fuel and oxidiser are mixed *before* they burn (gas stove, Bunsen burner, lean-premixed gas turbine). The flame is a thin front that propagates into the fresh mixture at a speed set by the mixture itself, the **laminar burning velocity** S_L. Project Part 3 is entirely about this object.

9.1 The physical picture

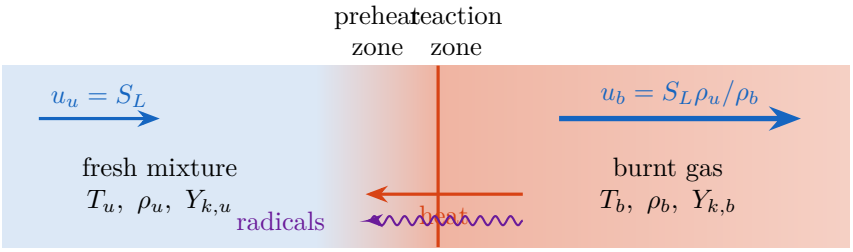


Figure 9.1: A planar premixed flame in its own reference frame. Fresh gas arrives at S_L. Heat and radicals diffuse *upstream* from the reaction zone and preheat the incoming mixture until it ignites. The burnt gas leaves faster because it has expanded.

Physical intuition

A premixed flame is a self-sustaining ignition wave. Each layer of fresh gas is heated by its already-burning neighbour, ignites and in turn heats the next layer. The speed of this relay depends on how fast heat diffuses (α) and how fast the hot gas reacts (τ_{chem}). That is why S_L is a property of the mixture and not of the burner.

9.2 Jump conditions across the flame

Derivation: mass and momentum

Steady and one-dimensional, (8.1.1) gives $\mathrm{d}(\rho u)/\mathrm{d}x = 0$:

$$\dot{m} \equiv \rho u = \rho_u S_L = \rho_b u_b = \text{const} \quad \Longrightarrow \quad \frac{u_b}{S_L} = \frac{\rho_u}{\rho_b} = \frac{T_b\, W_u}{T_u\, W_b} \approx 6\text{–}8. \tag{9.2.1}$$

Momentum: $\dot{m}\,\mathrm{d}u/\mathrm{d}x = -\mathrm{d}p/\mathrm{d}x$ (viscous terms cancel between the uniform ends), so

$$p_u - p_b = \dot{m}\,(u_b - u_u) = \rho_u S_L^2 \Big(\frac{\rho_u}{\rho_b} - 1\Big) \approx 1.2 \times 0.4^2 \times 6 \approx 1\,\mathrm{Pa}.$$

The pressure drop is negligible compared with 101 325 Pa, which justifies the low-Mach approximation (Section 8.6).

Symbols and units

$\dot{m} = \rho u$	mass flux through the flame	kg/(m² s)
ρ_u, ρ_b	density of unburnt, burnt gas	kg/m³
S_L	laminar flame speed (= u_u in the flame frame)	m/s
u_b	burnt-gas velocity	m/s
T_u, T_b	unburnt, burnt temperature	K
W_u, W_b	mean molar mass of unburnt, burnt gas	kg/kmol
p_u, p_b	pressure upstream, downstream	Pa

9.3 The 1D flame equations

Steady, 1D, low-Mach versions of (8.2.1) and (8.4.2), with $j_k = \rho Y_k V_k$:

$$\dot{m}\frac{\mathrm{d}Y_k}{\mathrm{d}x} = -\frac{\mathrm{d}j_k}{\mathrm{d}x} + W_k\dot{\omega}_k, \tag{9.3.1}$$

$$\dot{m}\,c_p\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\Big(\lambda\frac{\mathrm{d}T}{\mathrm{d}x}\Big) - \Big(\sum_k j_k c_{p,k}\Big)\frac{\mathrm{d}T}{\mathrm{d}x} - \sum_k h_k W_k\dot{\omega}_k. \tag{9.3.2}$$

Symbols and units

$\dot{m}$	constant mass flux ρu (eigenvalue)	kg/(m² s)
Y_k, j_k	mass fraction, diffusive mass flux of k	–, kg/(m² s)
$W_k\dot{\omega}_k$	chemical mass production rate	kg/(m³ s)
c_p, $c_{p,k}$	specific heat capacity of mixture, of k	J/(kg K)
λ	thermal conductivity	W/(m K)
h_k	specific enthalpy of k	J/kg
x	position	m

Boundary conditions: fresh state (T_u, $Y_{k,u}$) far upstream and zero gradients far downstream.

Key result: S_L is an eigenvalue

The equations are invariant under a shift $x \to x + c$, so the flame can sit anywhere. To get a unique solution one extra condition is imposed: Cantera fixes the temperature at one grid point. In exchange, $\dot{m}$ is no longer prescribed but *computed*. The flame "chooses"

its own mass flux, and $S_L = \dot{m}/\rho_u =$ `flame.velocity[0]`.

9.4 The burnt state is the HP equilibrium

Derivation: for any Lewis numbers

Write the enthalpy form (8.4.1) in 1D (low Mach, no viscous heating): $\dot{m}\,\mathrm{d}h/\mathrm{d}x = -\mathrm{d}q/\mathrm{d}x$ with $q = -\lambda\,\mathrm{d}T/\mathrm{d}x + \sum_k h_k j_k$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big[\dot{m}\,h + q\Big] = 0 \quad \Longrightarrow \quad \dot{m}\,h + q = \text{const.}$$

Far upstream and far downstream all gradients and diffusive fluxes vanish ($q = 0$), so $h_b = h_u$. The same argument applied to each *element* (chemistry does not create atoms) gives equal elemental composition at both ends. Far downstream the gas is also no longer changing, i.e. it is at chemical equilibrium. A state with the same h, p and elements that is at equilibrium is precisely `equilibrate("HP")` of the fresh gas.

Symbols and units

h, h_u, h_b	specific enthalpy (local, unburnt, burnt)	J/kg
q	total heat flux $-\lambda\,\mathrm{d}T/\mathrm{d}x + \sum_k h_k j_k$	W/m^2
$\dot{m}$	mass flux	kg/(m^2 s)
j_k	diffusive mass flux of k	kg/(m^2 s)

Pitfall: why the end of your domain is *not* that state

Recombination of radicals and, for hydrocarbons, oxidation of CO are slow. In a domain a few centimetres long, the gas leaving at `flame.T[-1]` may still be measurably short of equilibrium (Fig. 9.8). Also, a coarse grid gives an error in the burnt state. Hence the project's rule: **compare the end state with HP equilibrium; if the deviation is above 1%, refine; and quote T_{ad} from equilibrium, not from `flame.T[-1]`.**

9.5 Flame structure

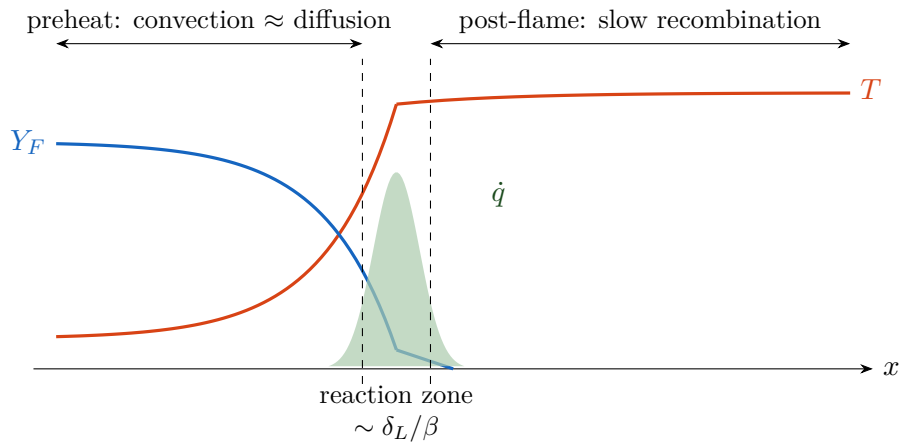


Figure 9.2: Classical three-zone structure. In the preheat zone chemistry is negligible and the profile is exponential. The thin reaction zone is where almost all heat is released. Downstream, slow recombination brings the gas to equilibrium.

Derivation: preheat zone profile

Upstream of the reaction zone $\dot{\omega}_k \approx 0$. With constant properties, (9.3.2) becomes $\dot{m} c_p T' = \lambda T''$, or $S_L T' = \alpha_u T''$. The solution that tends to T_u at $-\infty$ is

$$T - T_u = (T_r - T_u)\,\exp\!\Big(\frac{S_L\, x}{\alpha_u}\Big), \tag{9.5.1}$$

where T_r is the temperature where reaction starts. The e-folding length is the **diffusive flame thickness** $\delta_d = \alpha_u / S_L$.

Symbols and units

T_r	temperature at which reaction starts	K
α_u	thermal diffusivity of the unburnt gas	$\mathrm{m^2/s}$
S_L	flame speed	m/s
$\delta_d = \alpha_u / S_L$	diffusive flame thickness	m
x	position (reaction zone near $x = 0$)	m

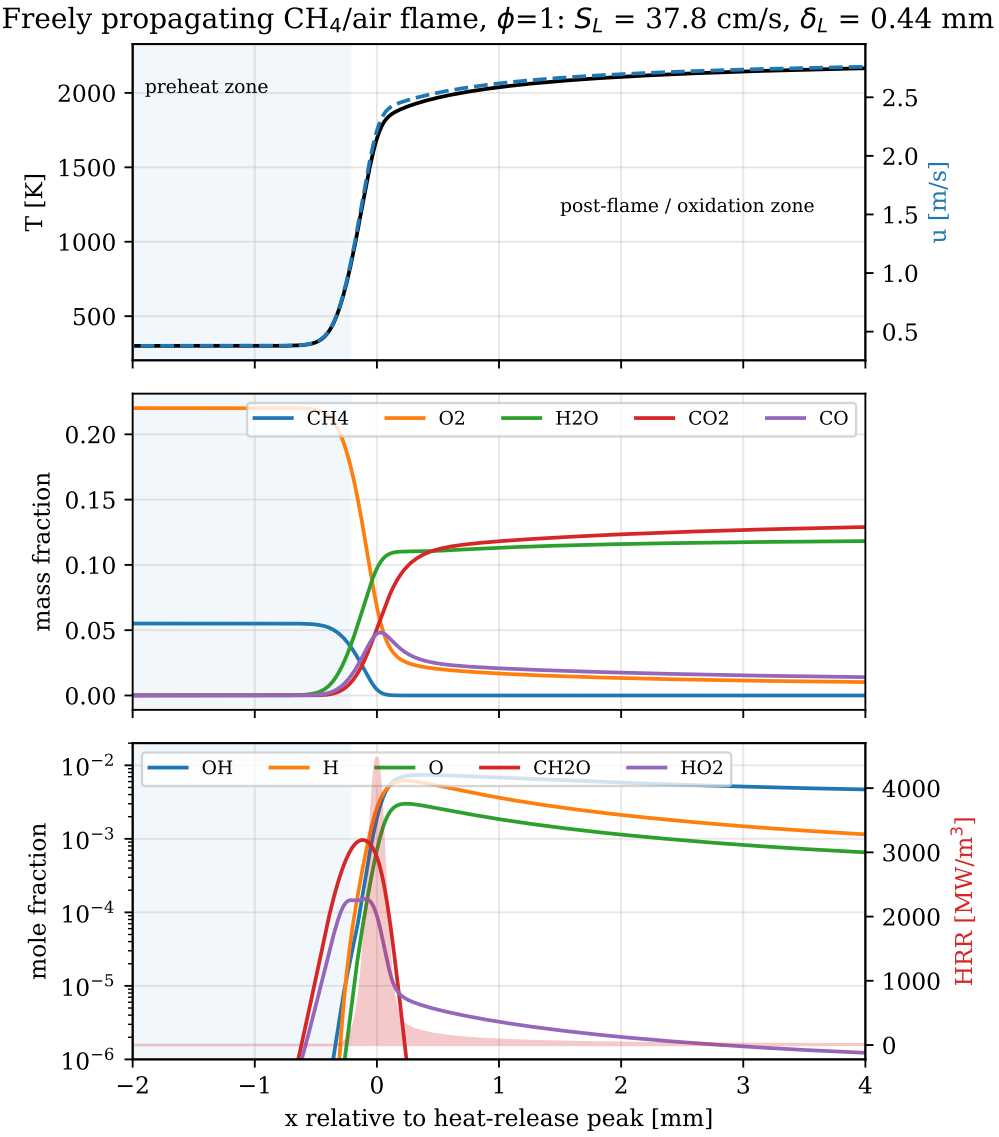


Figure 9.3: Computed stoichiometric CH_4/air flame (GRI-Mech 3.0, mixture-averaged). Top: T and u; the gas accelerates by ρ_u/ρ_b. Middle: major species. Bottom: radicals and the heat release rate (shaded). Note that HRR peaks where T is still rising. Intermediates such as CH_2O and HO_2 appear upstream of the HRR peak, and H, O, OH spill into the burnt gas and decay slowly.

9.6 How fast and how thick: scaling laws

Derivation: Mallard–Le Chatelier / Zeldovich scaling

Balance the three terms of (9.3.2) over a flame of thickness δ and temperature rise $\Delta T = T_b - T_u$:

$$\underbrace{\rho_u S_L c_p \frac{\Delta T}{\delta}}_{\text{convection}} \sim \underbrace{\lambda \frac{\Delta T}{\delta^2}}_{\text{diffusion}} \sim \underbrace{\dot{q}}_{\text{reaction}} .$$

Convection $\sim$ diffusion gives $\delta \sim \alpha / S_L$. Diffusion $\sim$ reaction gives $\delta^2 \sim \alpha \tau_c$ with chemical

time $\tau_c = \rho c_p \Delta T/\dot{q}$. Eliminating δ:

$$\boxed{S_L \sim \sqrt{\frac{\alpha}{\tau_c}}, \qquad \delta \sim \frac{\alpha}{S_L} \sim \sqrt{\alpha \tau_c}, \qquad \frac{S_L\,\delta}{\alpha} \sim 1.} \tag{9.6.1}$$

Symbols and units

δ	flame thickness	m
$\Delta T = T_b - T_u$	temperature rise across the flame	K
$\dot{q}$	typical heat release rate	$\mathrm{W/m^3}$
$\tau_c = \rho c_p \Delta T/\dot{q}$	chemical time scale	s
α, λ	thermal diffusivity, conductivity	$\mathrm{m^2/s}$, $\mathrm{W/(m\,K)}$
$S_L\delta/\alpha$	flame Péclet number	– (dimensionless)

Physical intuition

A flame is as thick as heat can diffuse in one chemical time, and travels one thickness per chemical time. Fast chemistry gives fast, thin flames. More diffusive mixtures (H_2) give faster, thicker ones. The product $S_L\delta$ is set by diffusivity alone: the **flame Péclet number** $S_L\delta/\alpha_u$ is a near-universal constant.

Temperature sensitivity. Reaction happens near T_b, where $\dot{q} \propto e^{-E/\mathcal{R}T_b}$, so

$$S_L \propto \exp\Big(-\frac{E}{2\mathcal{R}T_b}\Big). \tag{9.6.2}$$

The flame speed follows the burnt temperature, exponentially. Anything that raises T_b (stoichiometry near the T_{ad} peak, preheating) raises S_L strongly. The **Zeldovich number** $\beta = E(T_b - T_u)/(\mathcal{R}T_b^2) \approx 8$–$12$ measures this sensitivity. The reaction zone is thinner than the flame by a factor $\sim \beta$.

Symbols and units

E	global activation energy	J/kmol
T_b, T_u	burnt, unburnt temperature	K
β	Zeldovich number	– (dimensionless)

Pressure dependence.

Derivation: $S_L \propto p^{n/2-1}$

From Chapter 7, $\alpha \propto 1/p$. For a global reaction of overall order n (Section 5.10), $\dot{q} \propto p^n$, while $\rho c_p \Delta T \propto p$, so $\tau_c \propto p^{1-n}$. Then

$$S_L \propto \sqrt{p^{-1}\,p^{n-1}} = p^{n/2-1}, \qquad \delta \propto \frac{\alpha}{S_L} \propto p^{-n/2}.$$

For a bimolecular reaction ($n = 2$), S_L would be pressure-independent. Measured exponents are *negative* because three-body termination (R5) grows faster with pressure than branching, which lowers the effective order below 2. Note also that $S_L\delta \propto \alpha \propto 1/p$: the product is *not* constant with pressure, but the Péclet number $S_L\delta/\alpha_u$ is.

Symbols and units		
n	overall reaction order	– (dimensionless)
p	pressure	Pa
$\alpha \propto 1/p$	thermal diffusivity	m^2/s
$\tau_c \propto p^{1-n}$	chemical time	s
S_L, δ	flame speed, thickness	m/s, m

Preheat. Raising T_u raises T_b (Section 4.7) and α. Empirically $S_L \propto T_u^m$ with $m \approx 1.5$–2 for hydrocarbons. A common correlation is $S_L = S_L^0 (T_u/T_0)^m (p/p_0)^{n_p}$, fitted separately for each ϕ.

9.7 Thickness definitions

- **Thermal thickness** (used in the project):

$$\delta_L = \frac{T_b - T_u}{\max |\mathrm{d}T/\mathrm{d}x|}. \tag{9.7.1}$$

Symbols and units		
δ_L	thermal flame thickness	m
$\max \lvert \mathrm{d}T/\mathrm{d}x \rvert$	steepest temperature gradient	K/m
T_b, T_u	burnt, unburnt temperature	K

  Robust and easy to measure: draw the tangent at the steepest point.

- **Diffusive thickness** $\delta_d = \alpha_u / S_L$. It is much smaller (5–7×, Fig. 9.6), because α is evaluated in the cold gas while the gradient sits where the gas is hot and far more diffusive. Blint's correlation $\delta_L \approx 2\delta_d (T_b/T_u)^{0.7}$ accounts for this.

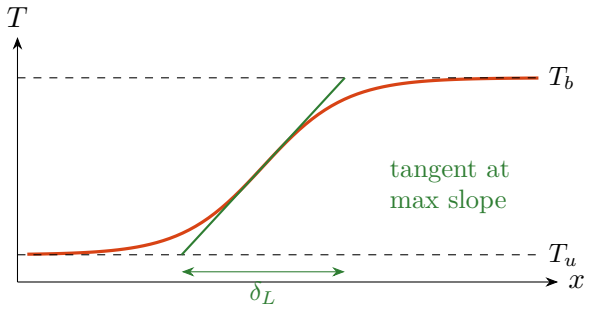


Figure 9.4: Thermal thickness: the width of the tangent at the inflexion point between T_u and T_b.

9.8 Two definitions of flame speed

Derivation: consumption speed

Integrate the fuel equation (9.3.1) over the whole domain. The diffusive flux vanishes at both ends:

$$\dot{m}\,(Y_{F,b} - Y_{F,u}) = \int_{-\infty}^{\infty} W_F \dot{\omega}_F \,\mathrm{d}x \quad \Longrightarrow \quad S_c = \frac{-\int W_F \dot{\omega}_F \,\mathrm{d}x}{\rho_u\,(Y_{F,u} - Y_{F,b})}.$$

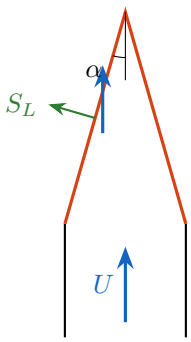
Symbols and units

S_c	consumption speed	m/s
$Y_{F,u}$, $Y_{F,b}$	fuel mass fraction in unburnt, burnt gas	– (dimensionless)
$W_F\dot{\omega}_F$	fuel mass production rate (negative: consumption)	kg/(m^3 s)
$\dot{m}$, ρ_u	mass flux, unburnt density	kg/(m^2 s), kg/m^3

- **Displacement speed** S_u: the velocity of the fresh gas relative to the flame front (`flame.velocity[0]` in a free flame). Depends on the frame and on stretch.
- **Consumption speed** S_c: how fast fuel is actually consumed per unit flame area. Robust in stretched or curved flames.

For a planar, unstretched, adiabatic flame they are identical, and a good check of your solution.

9.9 Measuring S_L: the Bunsen flame



Normal velocity component balances flame speed:
$S_L = U \sin\alpha$

Figure 9.5: The flame cone settles where the component of the jet velocity normal to the front equals S_L. Historically the main experimental method. Today counterflow and spherical-flame methods with stretch correction are preferred.

9.10 Dependence on equivalence ratio

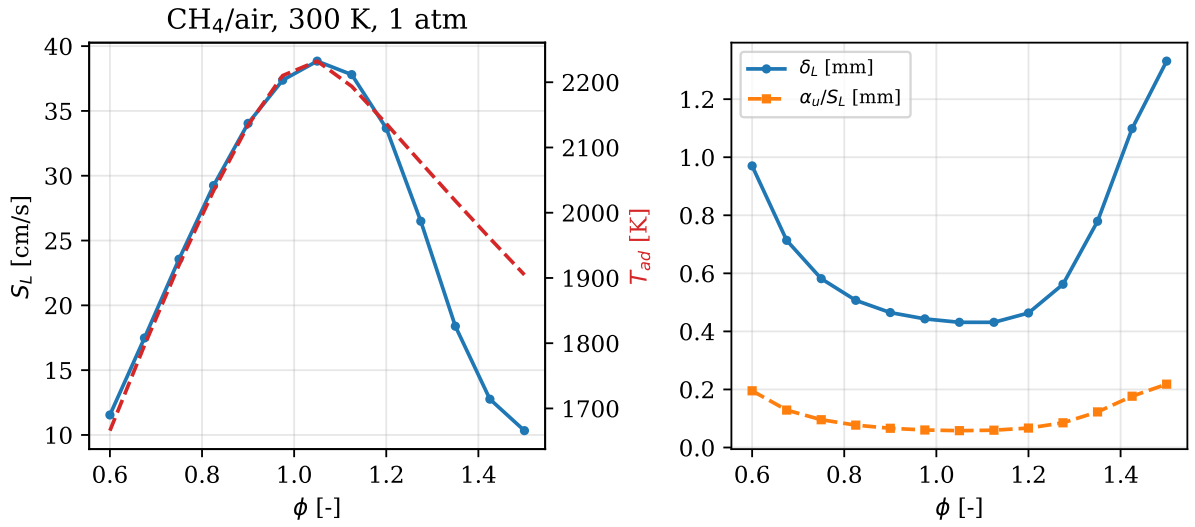


Figure 9.6: Left: computed CH_4/air flame speed (blue) and T_{ad} (red). S_L follows T_{ad} but falls much faster away from the peak: the exponential temperature sensitivity of Section 9.6. Right: thermal thickness δ_L and α_u/S_L. Slow flames are thick.

Physical intuition: what moves the S_L peak

For CH_4/air, the S_L peak sits near the T_{ad} peak. For other fuels the two can separate far apart, because $S_L \sim \sqrt{\alpha/\tau_c}$ depends not only on T_b but also on

- the *diffusivity* of the mixture, which grows with the fraction of a light fuel;
- the *radical pool*: how much H the reaction zone can produce and back-diffuse;
- which reactant is deficient and how fast *it* diffuses.

In Part 3, Task 4 you compare the $S_L(\phi)$ peak of H_2 with the $T_{ad}(\phi)$ peak from Part 1. Use these three items to explain the difference.

9.11 Lewis-number and preferential-diffusion effects

Under unity Lewis number (Section 7.5) enthalpy and element ratios are constant through the flame. Every point inside the flame has the fresh-gas ϕ and enthalpy. With real transport and a light fuel:

- H_2 **runs ahead.** In the preheat zone H_2 diffuses towards the reaction zone faster than heat diffuses out. The local element ratio changes, so a mixture fraction based on H atoms is *not* constant across the flame (Part 3, Task 3c).
- **H atoms run back.** The H radical (Le ≈ 0.2) diffuses far upstream into the cool preheat zone and starts the chain there. Forcing $D_H = \alpha$ suppresses this.
- **Local enthalpy deviates.** Inside the flame h can be above or below h_u even though both ends are equal (Section 9.4).

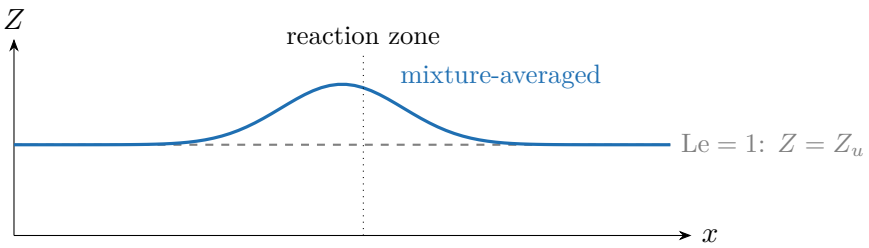


Figure 9.7: Qualitative sketch: with $\mathrm{Le}_{\mathrm{H_2}} < 1$, an H-element-based mixture fraction bulges inside the flame and returns to the fresh value downstream. Its shape, size and sign depend on ϕ. Measuring them is your task.

Stretch. Real flames are curved and strained. With Le < 1 a stretched lean flame burns *faster*, which leads to cellular instabilities in lean H_2 flames. The planar flame of the project is unstretched, the reference case to which stretch effects are compared through Markstein numbers.

9.12 Solving a free flame in Cantera

```
gas = ct.Solution("gri30.yaml")
gas.set_equivalence_ratio(1.0, "CH4", "O2:1, N2:3.76")
gas.TP = 300.0, ct.one_atm

flame = ct.FreeFlame(gas, width=0.03)            # domain [m]: ~50-100 flame
    thicknesses
flame.transport_model = "mixture-averaged"
flame.set_refine_criteria(ratio=2, slope=0.05, curve=0.1)
flame.set_max_grid_points(flame.flame, 10000)
flame.solve(loglevel=0, auto=True)               # auto: robust initial-guess strategy

S_L = flame.velocity[0]                          # [m/s]
dTdx = np.gradient(flame.T, flame.grid)
delta_L = (flame.T[-1] - flame.T[0]) / dTdx.max()
hrr = flame.heat_release_rate                    # [W/m^3]
```

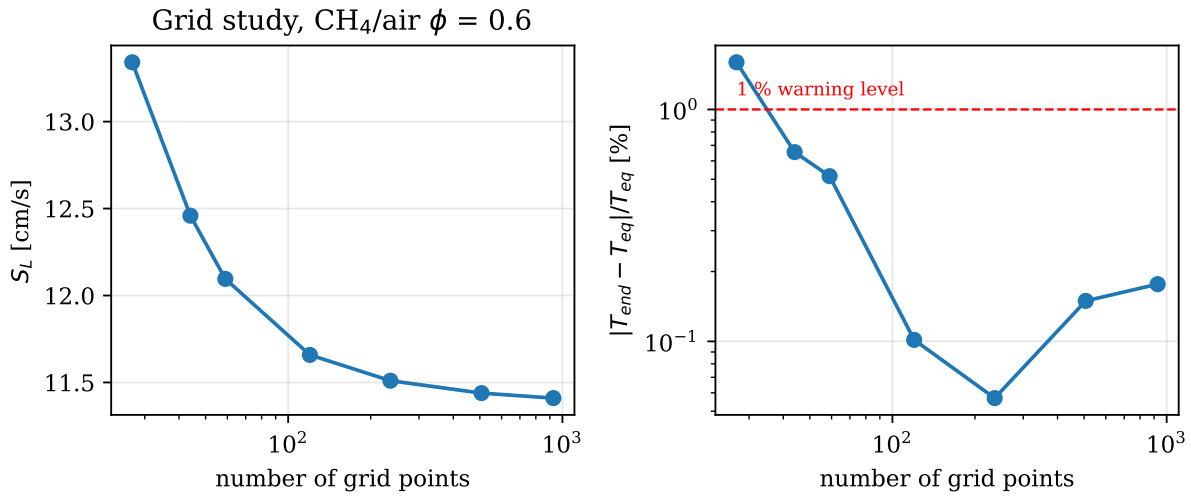


Figure 9.8: Grid study of a lean ($\phi = 0.6$) CH_4 flame. The coarsest grid (27 points, close to the default) over-predicts S_L by 17%. The burnt-temperature error falls with refinement until it reaches a floor set by the finite domain (slow CO oxidation), not by the grid. That floor is why T_{ad} must come from equilibrium.

Project link: Part 3

- Task 1: three panels against x in mm. Ask yourself why the HRR peak precedes the T maximum (Fig. 9.3), and where the radicals are made versus where they survive.
- Task 2: the equilibrium check is Section 9.4. Remember the warning above.
- Task 3: transport comparison, see Section 9.11. For a fair temperature comparison, shift each profile so that $T = (T_u + T_b)/2$ sits at $x = 0$.
- Task 4: sweep ϕ from a converged neighbour (Section 11.9). Check $S_L \delta_L / \alpha_u$ against (9.6.1).
- Task 5: fit $\ln S_L$ vs $\ln p$ and vs $\ln T_u$ by linear regression (`np.polyfit`). Compare your exponent with $n/2 - 1$ and decide what effective order n it implies.

Check your understanding

1. Why must `flame.velocity[-1]/flame.velocity[0]` equal ρ_u/ρ_b?
2. If the domain is too short, which boundary condition is violated and what happens to S_L?
3. A mechanism over-predicts T_{ad} by 30 K. Using $S_L \propto e^{-E/2\mathcal{R}T_b}$ with $E/\mathcal{R} = 20\,000$ K and $T_b = 2200$ K, by how much does it over-predict S_L?

Chapter 10

Non-premixed (diffusion) flames

In a non-premixed flame fuel and oxidiser arrive *separately* and must mix by diffusion before they can burn: candle, diesel spray, most industrial burners, rocket engines. The flame cannot propagate into either stream alone. It sits where the two meet in the right proportion. Project Part 2 is about this situation.

10.1 Configurations

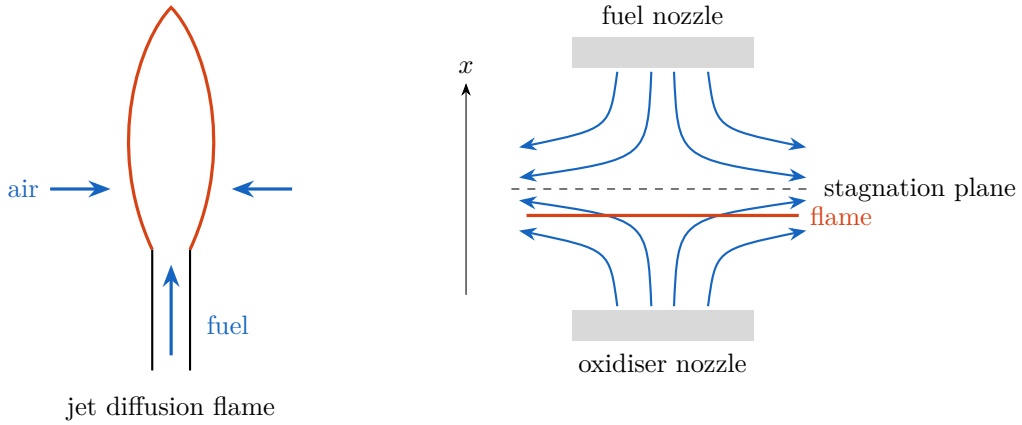


Figure 10.1: Left: jet flame (complex 2D geometry). Right: **counterflow** flame. Opposed jets create a stagnation plane. The flame sits on the oxidiser side (for most fuels Z_{st} is small) where the mixture is stoichiometric. Along the centreline the problem is exactly one-dimensional, which is why Cantera can solve it with detailed chemistry.

10.2 The mixture fraction

Derivation: a conserved scalar from two reacting ones

Model the chemistry as one step on a mass basis, $1\,\mathrm{kg\ F} + s\,\mathrm{kg\ O} \rightarrow (1+s)\,\mathrm{kg\ P}$, where s is the **stoichiometric oxygen-to-fuel mass ratio**. With equal diffusivities D and writing $\mathcal{L}(\cdot) = \partial_t(\rho\,\cdot) + \nabla\cdot(\rho\boldsymbol{u}\,\cdot) - \nabla\cdot(\rho D\nabla\,\cdot)$:

$$\mathcal{L}(Y_F) = W_F\dot{\omega}_F, \qquad \mathcal{L}(Y_O) = s\,W_F\dot{\omega}_F.$$

The combination $\beta = sY_F - Y_O$ satisfies $\mathcal{L}(\beta) = 0$: **no source**. Normalise it so that it is 1 in the fuel stream (1) and 0 in the oxidiser stream (2):

$$Z = \frac{sY_F - Y_O + Y_{O,2}}{sY_{F,1} + Y_{O,2}}. \tag{10.2.1}$$

Z is the mass fraction of material that originated in the fuel stream. It is conserved (convected and diffused but never created or destroyed), no matter how the chemistry proceeds.

Symbols and units

s	stoichiometric oxygen-to-fuel mass ratio (kg O per kg F)	– (dimensionless)
Y_F, Y_O	fuel, oxidiser mass fraction	– (dimensionless)
D	common diffusivity	$\mathrm{m^2/s}$
$\mathcal{L}(\cdot)$	transport operator (unsteady + convection − diffusion)	$\mathrm{kg/(m^3\,s)}$ × unit of argument
$W_F\dot{\omega}_F$	fuel mass production rate	$\mathrm{kg/(m^3\,s)}$
$\beta = sY_F - Y_O$	coupling function (conserved)	– (dimensionless)
$Y_{F,1}$, $Y_{O,2}$	fuel mass fraction in stream 1 (fuel), oxygen in stream 2 (oxidiser)	– (dimensionless)
Z	mixture fraction	– (dimensionless)

Derivation: stoichiometric mixture fraction

Stoichiometric means fuel and oxygen in exactly the ratio $1 : s$, i.e. $sY_F = Y_O$. Then $\beta = 0$ and

$$Z_{\mathrm{st}} = \frac{Y_{O,2}}{sY_{F,1} + Y_{O,2}} = \frac{1}{1 + sY_{F,1}/Y_{O,2}}. \tag{10.2.2}$$

Methane/air: $CH_4 + 2O_2$, so $s = 2 \cdot 32/16 = 4$; $Y_{F,1} = 1$, $Y_{O,2} = 0.233$: $Z_{\mathrm{st}} = 1/(1 + 4/0.233) = 0.055$. This matches Cantera's `gas.mixture_fraction(...)` (0.0552).

Symbols and units

Z_{st}	stoichiometric mixture fraction	– (dimensionless)
s	stoichiometric O-to-F mass ratio, $\nu_{O_2} W_{O_2}/W_F$	– (dimensionless)
$Y_{F,1}$, $Y_{O,2}$	fuel in fuel stream, oxygen in air (0.233)	– (dimensionless)
32, 16	molar masses of O_2 and CH_4	kg/kmol

Pitfall: Z_{st} formula with air

If you see $Z_{\mathrm{st}} = 1/(1 + \nu W_F/W_{O_2})$-type expressions, check what ν means and whether the oxidiser is pure O_2 or air. With air you *must* include $Y_{O_2,\mathrm{air}} = 0.233$ as in (10.2.2). A useful check: Z_{st} equals the fuel mass fraction of the stoichiometric *premixed* mixture (compare the H_2 example in Section 3.3).

Bilger's definition. With detailed chemistry "fuel" and "oxygen" are spread over many species. Bilger defines Z from *element* mass fractions, which only diffusion can change:

$$Z = \frac{\beta - \beta_2}{\beta_1 - \beta_2}, \qquad \beta = 2\frac{Y_C}{W_C} + \frac{1}{2}\frac{Y_H}{W_H} - \frac{Y_O}{W_O}, \tag{10.2.3}$$

Symbols and units

Y_C, Y_H, Y_O	elemental mass fractions of C, H, O	– (dimensionless)
W_C, W_H, W_O	atomic molar masses (12, 1.008, 16)	kg/kmol
β, β_1, β_2	Bilger coupling function (local, fuel stream, oxidiser stream)	kmol/kg

where the coefficients count how much oxygen each element needs. Cantera: `gas.mixture_fraction(fuel, oxidizer, element="Bilger")` or `flame.mixture_fraction("Bilger")`.

Derivation: $Z \leftrightarrow \phi$

Before reaction, a mixture with mixture fraction Z has $Y_F = ZY_{F,1}$ and $Y_O = (1-Z)Y_{O,2}$. Its equivalence ratio is $\phi = (Y_F/Y_O)/(Y_F/Y_O)_{\mathrm{st}}$, so

$$\phi = \frac{Z}{1-Z}\cdot\frac{1-Z_{\mathrm{st}}}{Z_{\mathrm{st}}}, \qquad Z = \frac{\phi Z_{\mathrm{st}}}{1-Z_{\mathrm{st}}+\phi Z_{\mathrm{st}}}. \tag{10.2.4}$$

Every point in a non-premixed flame is a premixed mixture of some local ϕ, from 0 (air) to ∞ (pure fuel).

Symbols and units

ϕ	local equivalence ratio of the unreacted mixture	– (dimensionless)
Z, Z_{st}	mixture fraction, stoichiometric value	– (dimensionless)
Y_F, Y_O	fuel, oxygen mass fraction before reaction	– (dimensionless)

10.3 Infinitely fast chemistry: Burke–Schumann

If chemistry is infinitely fast and complete, fuel and oxygen cannot coexist. Lean of Z_{st} there is no fuel; rich of it, no oxygen. Since Y_F, Y_O (and, with $\mathrm{Le} = 1$ and constant c_p, T) are then linear in Z on each side:

$$Z \le Z_{\mathrm{st}}: \quad Y_F = 0, \quad Y_O = Y_{O,2}\Big(1 - \frac{Z}{Z_{\mathrm{st}}}\Big), \quad T = T_2 + Z(T_1 - T_2) + \frac{QY_{F,1}}{c_p}Z, \tag{10.3.1}$$

$$Z \ge Z_{\mathrm{st}}: \quad Y_O = 0, \quad Y_F = Y_{F,1}\frac{Z - Z_{\mathrm{st}}}{1 - Z_{\mathrm{st}}}, \quad T = T_2 + Z(T_1 - T_2) + \frac{QY_{F,1}}{c_p}Z_{\mathrm{st}}\frac{1-Z}{1-Z_{\mathrm{st}}}. \tag{10.3.2}$$

Symbols and units

Q	heat of combustion per kg of fuel	J/kg
c_p	specific heat capacity (assumed constant)	J/(kg K)
T_1, T_2	fuel-stream, oxidiser-stream temperatures	K
Y_F, Y_O, Z	fuel and oxidiser mass fractions, mixture fraction	– (dimensionless)

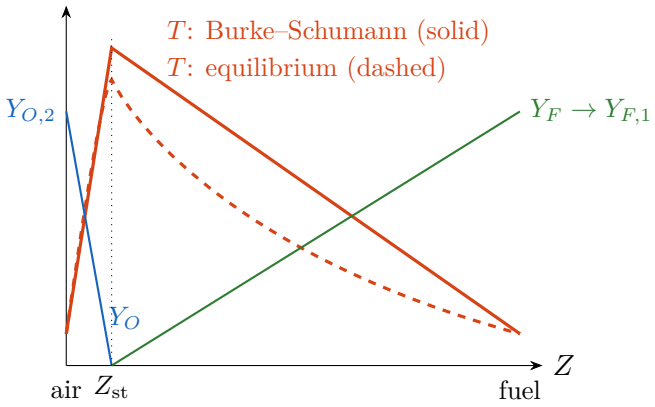


Figure 10.2: The Burke–Schumann limit in Z-space: a tent peaked at Z_{st}. Equilibrium (with dissociation) rounds the tip and lies below. Because Z_{st} is small, the lean branch is squeezed into a narrow interval and is therefore very steep.

Physical intuition: asymmetry of $T(Z)$

Two separate reasons make $T(Z)$ lopsided:

1. **Geometry:** the lean side occupies $[0, Z_{st}]$ and the rich side $[Z_{st}, 1]$. With $Z_{st} \ll 1$ the lean branch must climb as steeply as the rich branch descends gently.
2. **Heat capacity of the fuel stream:** on the rich side, the heat released is fixed by the available oxygen while each extra kg of fuel-stream gas adds $c_{p,F}$. If c_p varies, the rich branch of $T \approx T_2 + QY_{F,1}Z_{st}(1-Z)/[(1-Z_{st})\,c_p(Z)]$ with $c_p(Z) \approx Zc_{p,F} + (1-Z)c_{p,O}$ bends down faster when $c_{p,F} \gg c_{p,O}$. Compare c_p per kg of H_2, CH_4 and air before you explain your Part 2 plot.

On top of this, dissociation lowers the peak, and rich-side equilibrium can include endothermic reforming that real (finite-rate) flames do not have time to perform.

Equilibrium temperature profile $T_{eq}(Z)$. Mix the two streams adiabatically (under Le = 1 enthalpy is linear in Z), then equilibrate:

$$h(Z) = Zh_1 + (1-Z)h_2, \qquad Y_k(Z) = ZY_{k,1} + (1-Z)Y_{k,2}, \qquad \text{then } \texttt{equilibrate("HP")}. \tag{10.3.3}$$

```
gas.TPY = T_fuel, p, Y_fuel;  h1, Y1 = gas.enthalpy_mass, gas.Y.copy()
gas.TPY = T_ox,   p, Y_ox;    h2, Y2 = gas.enthalpy_mass, gas.Y.copy()
for Z in Z_grid:
    gas.HPY = Z*h1 + (1-Z)*h2, p, Z*Y1 + (1-Z)*Y2   # mix enthalpies, not temperatures
    !
    gas.equilibrate("HP")
    T_eq.append(gas.T)
```

Pitfall

Do *not* mix temperatures ($T = ZT_1 + (1-Z)T_2$). Enthalpy is the conserved quantity; temperature is not, because c_p differs between the streams. The code given on the project page for this step contains an unfinished placeholder (. . .). Use the pattern above.

10.4 The counterflow flame

Derivation: stagnation-point flow

Near the stagnation plane, the inviscid axisymmetric flow is $u = -ax$ (axial), $v = \frac{a}{2}r$ (radial). Check continuity in cylindrical coordinates: $\partial_x u + \frac{1}{r}\partial_r(rv) = -a + a = 0$. The constant a [1/s] is the **strain rate**: the rate at which fluid elements are stretched along r and compressed along x.

Symbols and units

u, v	axial, radial velocity	m/s
x, r	axial, radial coordinate	m
a	strain rate	1/s

Derivation: similarity reduction solved by Cantera

Assume $u = u(x)$, $v = r\,V(x)$, $T = T(x)$, $Y_k = Y_k(x)$ and $\partial p/\partial r = r\Lambda$ with Λ a constant. Substituting into the axisymmetric equations, all r-dependence cancels:

$$\frac{\mathrm{d}(\rho u)}{\mathrm{d}x} + 2\rho V = 0, \tag{10.4.1}$$

$$\rho u\frac{\mathrm{d}V}{\mathrm{d}x} + \rho V^2 = -\Lambda + \frac{\mathrm{d}}{\mathrm{d}x}\Big(\mu\frac{\mathrm{d}V}{\mathrm{d}x}\Big), \tag{10.4.2}$$

$$\rho u\frac{\mathrm{d}Y_k}{\mathrm{d}x} = -\frac{\mathrm{d}j_k}{\mathrm{d}x} + W_k\dot{\omega}_k, \qquad \rho c_p u\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\Big(\lambda\frac{\mathrm{d}T}{\mathrm{d}x}\Big) - \sum_k j_k c_{p,k}\frac{\mathrm{d}T}{\mathrm{d}x} - \sum_k h_k W_k\dot{\omega}_k. \tag{10.4.3}$$

These are Cantera's components `velocity` (u), `spreadRate` (V), `Lambda`, `T` and the species. The inlet mass fluxes $\dot{m}_F$, $\dot{m}_O$ are the controls; the eigenvalue Λ adjusts to fit both nozzles in the domain.

Symbols and units

$V = v/r$	radial velocity gradient (Cantera `spreadRate`)	1/s
Λ	radial pressure curvature $(1/r)\partial p/\partial r$, an eigenvalue	Pa/m^2
μ	dynamic viscosity	Pa s
ρu	axial mass flux	kg/(m^2 s)
j_k, $W_k\dot{\omega}_k$	diffusive mass flux, chemical mass source of k	kg/(m^2 s), kg/(m^3 s)
c_p, λ	heat capacity, conductivity	J/(kg K), W/(m K)

Derivation: choosing inlet flows for a target strain

For potential flow between nozzles a distance L apart, Seshadri & Williams give the strain rate on the oxidiser side as

$$a_O = \frac{2|u_O|}{L}\Big(1 + \frac{|u_F|}{|u_O|}\sqrt{\frac{\rho_F}{\rho_O}}\Big). \tag{10.4.4}$$

A common choice is to balance momentum fluxes, $\rho_F u_F^2 = \rho_O u_O^2$, which centres the

stagnation plane. Then $|u_F|/|u_O| = \sqrt{\rho_O/\rho_F}$, the bracket equals 2, and

$$u_O = \frac{aL}{4}, \qquad u_F = u_O\sqrt{\rho_O/\rho_F}, \qquad \dot{m}_O = \rho_O u_O, \quad \dot{m}_F = \rho_F u_F. \tag{10.4.5}$$

This is an estimate. Always report the strain the solver actually produced (`flame.strain_rate("max")`, or χ_{st} computed from the solution).

Symbols and units

a_O	strain rate on the oxidiser side	1/s
u_O, u_F	oxidiser, fuel inlet velocities	m/s
ρ_O, ρ_F	oxidiser, fuel inlet densities	kg/m^3
L	nozzle separation (Cantera `width`)	m
$\dot{m}_O$, $\dot{m}_F$	inlet mass fluxes (`mdot`)	kg/(m^2 s)

10.5 Scalar dissipation rate

$$\chi = 2D|\nabla Z|^2 \qquad [1/\mathrm{s}]. \tag{10.5.1}$$

Symbols and units

χ	scalar dissipation rate	1/s
D	diffusivity of the mixture fraction (often $\approx \alpha$)	m^2/s
∇Z	mixture-fraction gradient	1/m

Physical intuition

χ is the rate at which molecular diffusion destroys fluctuations of Z, i.e. how fast fuel and oxidiser are being mixed. $1/\chi$ is a *mixing time*. Large χ means steep gradients and fast diffusion: reactants are supplied quickly, but heat and radicals are also removed quickly. If $1/\chi_{\mathrm{st}}$ becomes shorter than the chemical time, the flame goes out.

Derivation: $\chi(Z)$ in a counterflow, and the erfc profile

Take constant density and diffusivity with $u = -ax$. The steady Z equation (no source) is

$$-ax\,Z' = D\,Z'' \quad\Longrightarrow\quad \frac{Z''}{Z'} = -\frac{ax}{D} \quad\Longrightarrow\quad Z' = C\,e^{-ax^2/2D}.$$

Integrating with $Z(-\infty) = 1$ (fuel side) and $Z(+\infty) = 0$:

$$Z = \tfrac{1}{2}\,\mathrm{erfc}\left(\frac{x}{\ell}\right), \qquad \ell = \sqrt{2D/a}. \tag{10.5.2}$$

Then $Z' = -\frac{1}{\sqrt{\pi}\,\ell}e^{-x^2/\ell^2}$ and $\chi = 2DZ'^2 = \frac{2D}{\pi\ell^2}e^{-2x^2/\ell^2} = \frac{a}{\pi}e^{-2\eta^2}$ with $\eta = x/\ell$. Eliminating $\eta = \mathrm{erfc}^{-1}(2Z) = \mathrm{erf}^{-1}(1-2Z)$:

$$\boxed{\chi(Z) = \frac{a}{\pi}\exp\Big(-2\big[\mathrm{erf}^{-1}(1-2Z)\big]^2\Big), \qquad \chi_{\mathrm{st}} = \chi(Z_{\mathrm{st}}) \propto a.} \tag{10.5.3}$$

Symbols and units

Z'	$\mathrm{d}Z/\mathrm{d}x$	1/m
C	integration constant	1/m
$\ell = \sqrt{2D/a}$	mixing-layer thickness	m
$\eta = x/\ell$	similarity variable	– (dimensionless)
χ, χ_{st}	scalar dissipation rate, its value at Z_{st}	1/s
a, D	strain rate, diffusivity	1/s, m^2/s
erf^{-1}	inverse error function	– (dimensionless)

Key result

χ_{st} is proportional to the strain rate. Strain $\rightarrow$ χ_{st} $\rightarrow$ flame structure $\rightarrow$ extinction is the causal chain of Part 2. In a computed flame, evaluate χ directly from the solution, $\chi = 2\,\frac{\lambda}{\rho c_p}\,(\mathrm{d}Z/\mathrm{d}x)^2$ (using $D \approx \alpha$), and read it at the grid point nearest Z_{st}. Then compare with (10.5.3).

10.6 Flamelets: the flame in Z-space

Derivation: sketch of the flamelet transformation

Use Z as a coordinate normal to the flame. By the chain rule $\partial_x = Z_x\partial_Z$ and $\partial_{xx} = Z_x^2\partial_{ZZ} + Z_{xx}\partial_Z$. Substitute into a species equation with $\mathrm{Le} = 1$. The terms containing ∂_Z multiply the Z-equation itself and vanish, which leaves

$$\rho\frac{\partial Y_k}{\partial t} = \rho\frac{\chi}{2}\frac{\partial^2 Y_k}{\partial Z^2} + W_k\dot{\omega}_k. \tag{10.6.1}$$

Symbols and units

Z_x, Z_{xx}	first and second x-derivatives of Z	1/m, $1/\mathrm{m}^2$
Y_k	mass fraction of k	– (dimensionless)
$\chi = 2DZ_x^2$	scalar dissipation rate	1/s
$W_k\dot{\omega}_k$	chemical mass source	$\mathrm{kg}/(\mathrm{m}^3\,\mathrm{s})$
ρ, t	density, time	kg/m^3, s

In Z-space, *all* flow information is in one parameter, χ: a diffusion–reaction balance with diffusivity $\chi/2$. Plotting T against Z removes the geometry and makes flames at different strain directly comparable (Fig. 10.3).

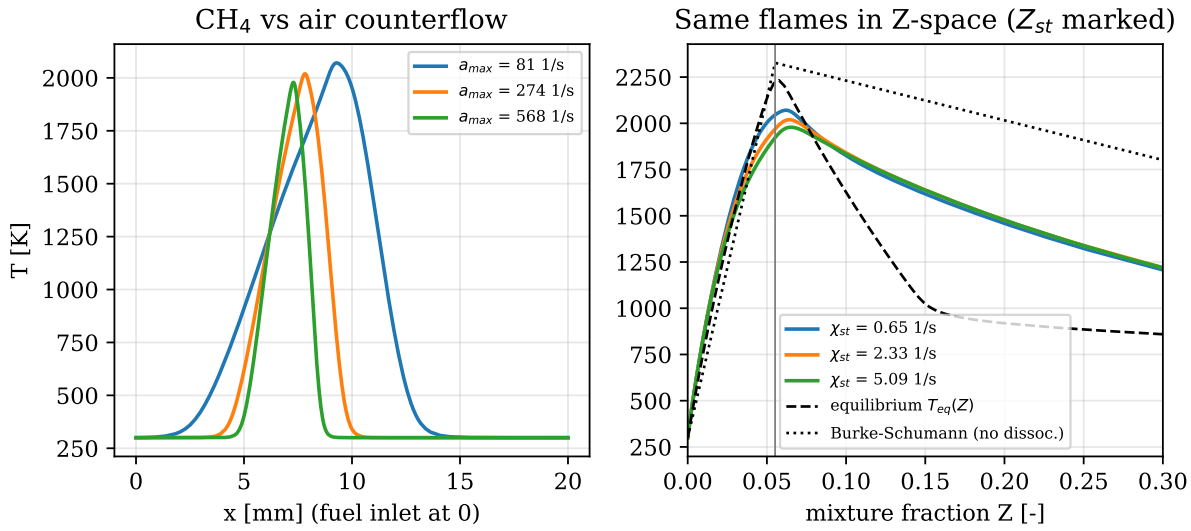


Figure 10.3: CH_4/air counterflow flames at three strain rates. Left: physical space; higher strain gives a thinner flame pushed towards the stagnation plane. Right: the same flames against Bilger Z, with $T_{\mathrm{eq}}(Z)$ and Burke–Schumann. The peak temperature falls with χ_{st}. On the rich side the flames are *hotter* than T_{eq}: equilibrium would reform CH_4+H_2O into CO+H_2 (endothermic), which the flame has no time to do.

10.7 Extinction and the S-curve

Define a Damköhler number $\mathrm{Da} = 1/(\chi_{\mathrm{st}}\tau_c)$. Plot the maximum temperature against Da (or against $1/\chi_{\mathrm{st}}$):

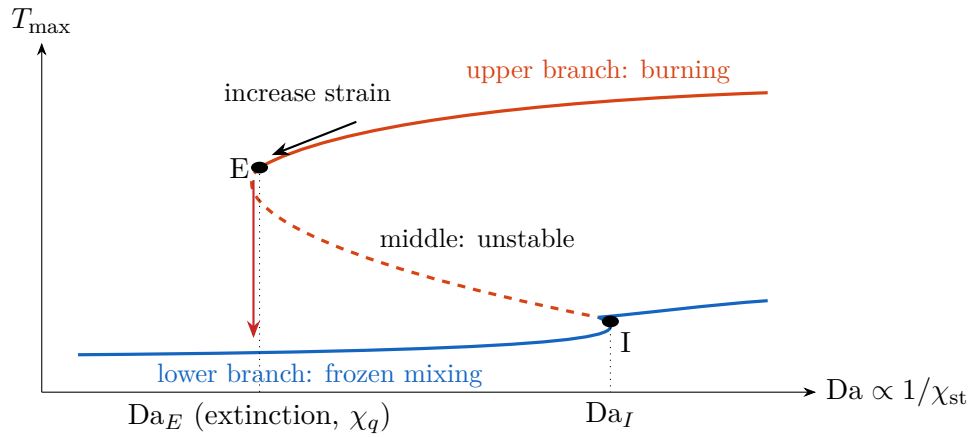


Figure 10.4: The S-curve. On the burning branch, increasing strain (decreasing Da) lowers $T_{\max}$ gradually until the turning point E. Beyond it no burning solution exists and the flame jumps to the frozen branch: **extinction**. Reigniting needs the much larger Da of point I: hysteresis.

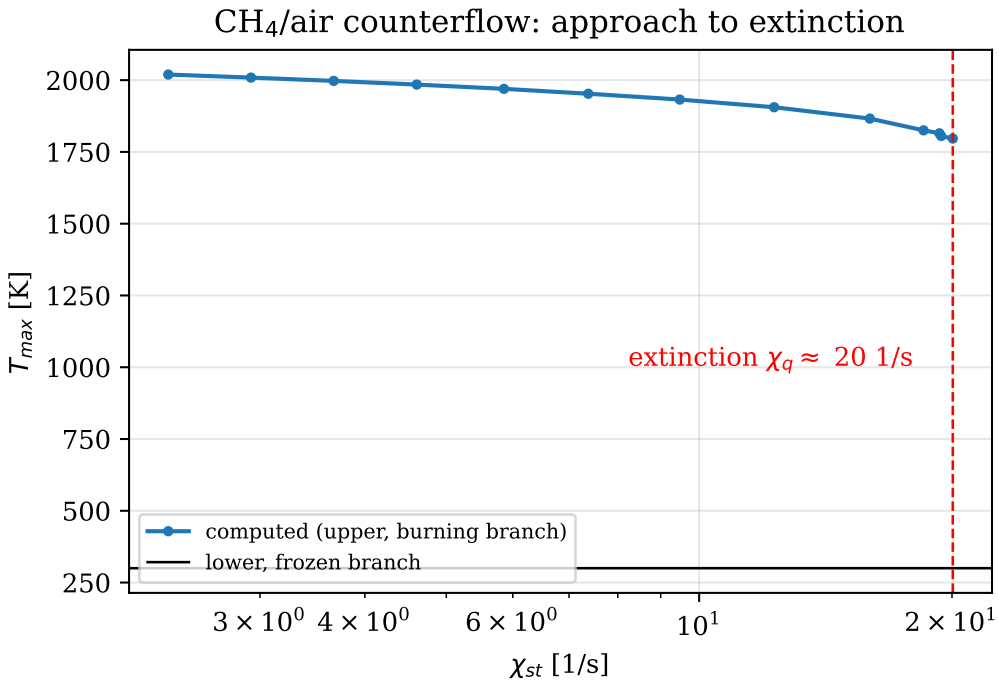


Figure 10.5: Upper branch computed with Cantera by scaling a converged CH_4/air solution to higher strain (and taking smaller steps whenever the next solution failed to burn). $T_{\max}$ falls slowly and then plunges near χ_q. Cantera cannot follow the unstable middle branch without special two-point control continuation.

Physical intuition: why extinction is sudden

As strain rises, heat leaves the reaction zone faster and $T_{\max}$ drops. Because of the Arrhenius sensitivity (and the crossover competition for H_2, Section 5.6), a modest drop in temperature causes a large drop in reaction rate. That reduces heat release, which lowers T further. Beyond a critical point this feedback has no stable balance and the flame collapses.

Detecting extinction in Cantera. After `solve`, check whether $\max T$ is close to the inlet temperatures (`flame.extinct()` does this test), or whether the heat release rate has vanished. A converged "solution" at 300 K is a valid solution of the equations: the frozen branch. It is not a burning flame.

10.8 Why even weakly strained flames are not at equilibrium

- Finite-rate chemistry: radical recombination needs time; at any finite χ some heat is still stored in radicals.
- Differential diffusion: with $\mathrm{Le}_{H_2} \neq 1$, elements separate and the local enthalpy is no longer $Zh_1 + (1-Z)h_2$. The local state is then not the one $T_{\mathrm{eq}}(Z)$ assumes, and different Z definitions disagree.
- The peak sits where reactants meet in a diffusion–reaction layer, not as a pre-mixed pocket given infinite time.

Project link: Part 2

- Put the flame solver and $T_{\mathrm{eq}}(Z)$ in `ct_utils`. Choose inlet $\dot{m}$ from target strains with the momentum-balanced formula above, then report the *actual* χ_{st}.
- Sweep strain from low to high, restarting each solve from the previous solution. When a solve lands on the frozen branch, you have bracketed extinction.
- Questions: asymmetry (the two reasons in the intuition box); peak shift with χ_{st} (look at where $T_{\max}$ sits relative to Z_{st} as χ grows); the area between $T(Z)$ and $T_{\mathrm{eq}}(Z)$ (energy not yet released: finite-rate plus differential diffusion); extinction detection (above).

Check your understanding

1. Compute Z_{st} for H_2/air by hand. Where, relative to the stagnation plane, will the flame sit?
2. If you double the strain rate, by what factor does χ_{st} change according to (10.5.3)? And the flame thickness ℓ?
3. Why does the frozen branch of the S-curve satisfy the steady counterflow equations?

Chapter 11

Cantera in practice

Cantera is an open-source C++ library with a Python interface that evaluates everything in Chapters 3–7 (thermodynamics, kinetics, transport) and solves the model problems of Chapters 6, 9 and 10. This repository uses Cantera 3.2.0 on Python 3.14.

11.1 Setup check

```
source .venv/bin/activate
python src/check_setup.py
# Cantera 3.2.0: adiabatic flame T = 2225.5 K
# Laminar flame speed = 0.419 m/s
```

11.2 The `Solution` object

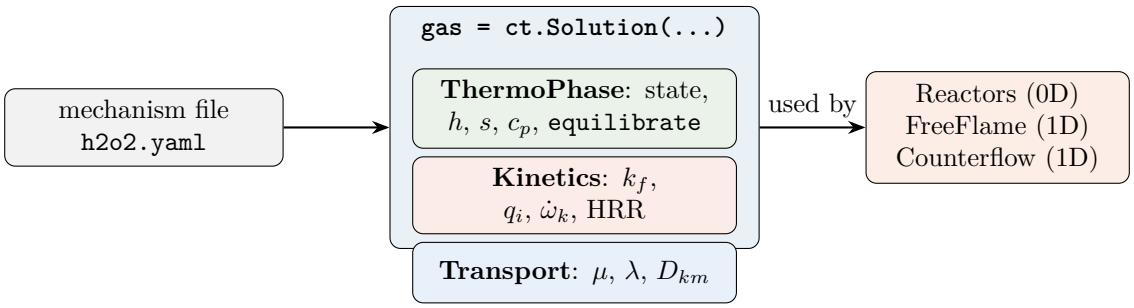


Figure 11.1: A `Solution` bundles three managers that all read the same thermodynamic state. Reactors and flames call them at every point and time step.

```
import cantera as ct
gas = ct.Solution("h2o2.yaml")                  # 10 species, 29 reactions
gas.TPX = 300.0, ct.one_atm, "H2:2, O2:1, N2:3.76"   # set state (T, p, composition)

gas.T, gas.P, gas.density                       # state
gas.X, gas.Y, gas.species_names                 # composition
gas.enthalpy_mass, gas.cp_mass, gas.entropy_mass       # J/kg, J/kg/K
gas.net_production_rates, gas.heat_release_rate        # kmol/m3/s, W/m3
gas.viscosity, gas.thermal_conductivity, gas.mix_diff_coeffs_mass
```

Setters come in pairs: `TP`, `TPX`, `TPY`, `HP`, `HPY`, `UV`, `SP`, … *All* properties update consistently with the state.

Pitfall: Python 3.14 / NumPy 2.5 quirk

In this environment `gas["OH"].X` raises `ValueError: cannot resize an array that may be referenced`. Use index access instead:

```
iOH = gas.species_index("OH")
x_OH = gas.X[iOH]
```

The project page uses `gas["OH"].X[0]`; translate it this way.

11.3 Anatomy of a YAML mechanism

Excerpts from `h2o2.yaml` (shipped with Cantera):

```
units: {length: cm, time: s, quantity: mol, activation-energy: cal/mol}

phases:
- name: ohmech
  thermo: ideal-gas
  elements: [O, H, Ar, N]
  species: [H2, H, O, O2, OH, H2O, HO2, H2O2, AR, N2]
  kinetics: gas
  transport: mixture-averaged

species:
- name: OH
  composition: {O: 1, H: 1}
  thermo:
    model: NASA7
    temperature-ranges: [200.0, 1000.0, 3500.0]
    data:                       # a1 ... a7, low range then high range
    - [3.99201543, -2.40131752e-03, 4.61793841e-06, -3.88113333e-09, 1.3641147e-12,
      3615.08056, -0.103925458]
    - [3.09288767, 5.48429716e-04, ...]
  transport: {model: gas, geometry: linear, well-depth: 80.0, diameter: 2.75}

reactions:
- equation: H + O2 <=> O + OH          # chain branching R2
  rate-constant: {A: 2.65e+16, b: -0.6707, Ea: 1.7041e+04}
- equation: H + O2 + M <=> HO2 + M     # termination R5
  type: three-body
  rate-constant: {A: 2.8e+18, b: -0.86, Ea: 0.0}
  efficiencies: {O2: 0.0, H2O: 0.0, N2: 0.0, AR: 0.0}
- equation: 2 OH (+M) <=> H2O2 (+M)
  type: falloff
  low-P-rate-constant: {A: 2.3e+18, b: -0.9, Ea: -1700.0}
  high-P-rate-constant: {A: 7.4e+13, b: -0.37, Ea: 0.0}
  Troe: {A: 0.7346, T3: 94.0, T1: 1756.0, T2: 5182.0}
```

- `units`: everything below is in cm, mol, cal; Cantera converts to SI.
- NASA7 `data`: the $a_1 \dots a_7$ of Section 3.7, for 200–1000 and 1000–3500 K. $a_6 = 3615$ encodes OH's positive heat of formation.

- `transport`: Lennard-Jones well depth ϵ/k_B [K] and diameter σ [Å], used by kinetic theory (Chapter 7).
- R2's $E_a = 17\,041$ cal/mol $= 71.3$ kJ/mol: the steep line of Fig. 5.3.
- In R5 the efficiencies of O_2, H_2O, N_2, Ar are zero because those colliders have their own explicit reactions (`H + O2 + N2 <=> HO2 + N2`, ...). Do not count them twice.

11.4 Equilibrium

```
gas.set_equivalence_ratio(phi, "H2", "O2:1, N2:3.76")
gas.TP = T_u, p
gas.equilibrate("HP")          # adiabatic, constant pressure
T_ad = gas.T
```

Key result

Compute the equilibrium state *before* any kinetic simulation. It is the value every reactor and flame must approach, and the sanity check for all of them.

11.5 `Solution` loops vs. `SolutionArray`

When states are independent (parameter sweeps, post-processing), a `SolutionArray` moves the loop into C++:

```
import numpy as np
phi = np.linspace(0.4, 3.0, 200)
states = ct.SolutionArray(gas, shape=phi.shape)
states.TP = 300.0, ct.one_atm
states.set_equivalence_ratio(phi, "H2", "O2:1, N2:3.76")
states.equilibrate("HP")
T_ad = states.T                          # numpy array, same shape as phi
```

Use a plain loop when each step depends on the previous one: time integration, or flame sweeps that reuse the previous solution. For 2D maps (Part 1, Task 4) use a 2D shape, `SolutionArray(gas, shape=(n1, n2))`, and `np.meshgrid` for the inputs.

11.6 Reactors

See Chapter 6. `IdealGasConstPressureReactor` (flames, open systems) or `IdealGasReactor` (closed bomb), wrapped in a `ReactorNet` and advanced with `net.step()` or `net.advance(t)`.

11.7 1D flames

class	physics	project
`FreeFlame`	freely propagating premixed flame, S_L eigenvalue	Part 3
`CounterflowDiffusionFlame`	opposed fuel/oxidiser jets	Part 2
`BurnerFlame`	burner-stabilised premixed (heat loss to burner)	–
`CounterflowPremixedFlame`	opposed premixed jets (stretch)	–

Useful attributes after `solve`: `grid` [m], `T`, `velocity`, `X`, `Y` ($K \times N$ arrays), `heat_release_rate`, `density`, `cp`, `thermal_conductivity`, `mixture_fraction("Bilger")` (counterflow), `strain_rate("max")`, `extinct()`.

11.8 Grid refinement: the most important setting

Cantera starts from a coarse grid and adds points where the solution is under-resolved, according to three criteria applied to every solution component φ:

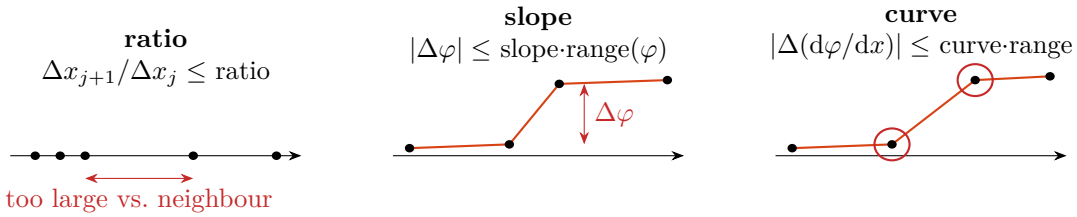


Figure 11.2: Refinement criteria. Smaller `slope` and `curve` mean more points in steep and strongly curved regions. `prune` removes points where they are no longer needed.

```
flame.set_refine_criteria(ratio=2.0, slope=0.005, curve=0.01, prune=0.0)  # research
    grade
flame.set_max_grid_points(flame.flame, 10000)
```

Pitfall: the defaults are not research grade

The ASEC tutorial page shows a lean H_2 flame ($\phi = 0.5$) where the default refinement gave only 27 points, with errors of $\sim 9\%$ in T_b, $\sim 11\%$ in Y_{H_2O} and $\sim 25\%$ in S_L, while the solver reported success. Fig. 9.8 shows the same lesson for lean methane. Always:

1. print `len(flame.grid)`;
2. check the end state against `equilibrate("HP")` ($< 1\%$);
3. tighten `slope`/`curve` until S_L changes by less than your tolerance;
4. **lighter fuels need tighter criteria**: hydrogen flames are thin in the reaction zone yet have long diffusive tails.

Do not use 1D flames for NO: thermal NO forms over milliseconds in the post-flame gas, far beyond the domain. Use a reactor or plug-flow model downstream.

A practical strategy is to solve with loose criteria first, then tighten and re-solve: the coarse solution is an excellent initial guess (Section 2.7).

```
for slope, curve in [(0.1, 0.2), (0.05, 0.1), (0.02, 0.04), (0.01, 0.02)]:
    flame.set_refine_criteria(ratio=2, slope=slope, curve=curve)
    flame.solve(loglevel=0, refine_grid=True, auto=False)
    print(len(flame.grid), flame.velocity[0])
```

11.9 Sweeps, continuation, saving

```
flame.solve(loglevel=0, auto=True)                # first point: robust
results = []
for phi in phi_values:                             # neighbours: start from previous
    solution
    gas.set_equivalence_ratio(phi, "H2", "O2:1, N2:3.76")
    gas.TP = T_u, p
    flame.inlet.T = T_u
    flame.inlet.X = gas.X
    flame.solve(loglevel=0, auto=False)
    results.append((phi, flame.velocity[0]))
    flame.save("flames.yaml", name=f"phi_{phi:.2f}", overwrite=True)

flame.restore("flames.yaml", name="phi_1.00")     # reload later without re-solving
```

- Sweep *outwards* from the easy case ($\phi \approx 1$) towards the limits; the solver struggles near flammability limits.
- If a solve fails, catch `ct.CanteraError`, go back to the last good solution and take a smaller parameter step (this is how Fig. 10.5 was made).
- Change the transport model on a converged solution and re-solve. This is faster than starting over: `flame.transport_model = "unity-Lewis-number"; flame.solve(auto=False)`.

11.10 API notes for Cantera 3.2

- Transport models: `"mixture-averaged"`, `"multicomponent"`, `"unity-Lewis-number"`.
- Reactors: pass `clone=True/False` explicitly; use `reactor.phase` (not `.thermo`).
- Flames: `flame.save/restore` write YAML (or HDF5 with `.h5`); `flame.to_pandas()` gives a DataFrame for plotting.
- Counterflow component names: `velocity`, `spreadRate`, `T`, `Lambda`, species.

Chapter 12

Project companion

This chapter walks through the three project parts task by task: what to compute, which theory applies, what the result should qualitatively look like, and how to sanity-check it. It does not give numerical answers. Producing and explaining them is the project.

12.1 Repository layout and workflow

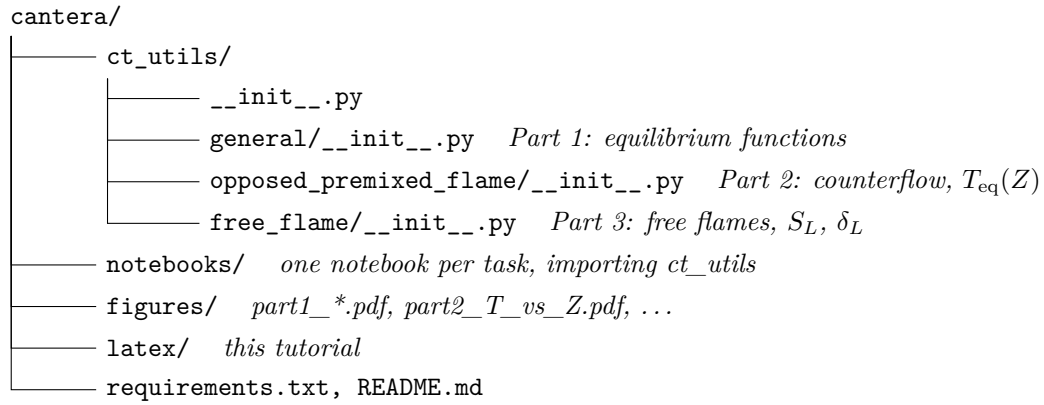


Figure 12.1: Suggested layout following the project page. The folder name `opposed_premixed_flame` is the one the page uses for the Part 2 (counterflow *diffusion* flame) code.

Good practice that will be reviewed.

- Small, focused commits with clear messages. The commit history is discussed.
- Library code in `ct_utils`, analysis in notebooks. No copy-paste between tasks: Task 2 of Part 1 explicitly asks you to reuse Task 1's code.
- A docstring on every function: arguments with units, return value, what it assumes.
- Tag each finished part: `git tag part-1-done && git push -tags`.

Every slide needs a title stating what was varied, axes with units, and at least one sentence of physical interpretation. Figures: vector PDF, readable fonts, legends that name the case.

12.2 Part 1: Equilibrium

Theory: Chapters 3, 4; Section 11.4, Section 11.5. **Mechanism:** `h2o2.yaml`. **Sanity check:** $\phi = 1$, 300 K, 1 atm gives $T_{\text{ad}} \approx 2382$ K (this Cantera version gives 2388 K).

API to implement in `ct_utils/general`

```
def equilibrium_state(phi, fuel="H2", T_u=300.0, p=101325.0,
                      mechanism="h2o2.yaml", oxidizer="O2:1, N2:3.76"):
    """Return a ct.Solution at the HP-equilibrium of the fresh mixture."""

def equilibrium_temperature(phi, fuel="H2", T_u=300.0, p=101325.0,
                            mechanism="h2o2.yaml", oxidizer="O2:1, N2:3.76"):
    """Adiabatic flame temperature [K] (scalar)."""          # -> uses
    equilibrium_state

def equilibrium_sweep(phi_range, **kw):        # np.ndarray, same shape as phi_range
def equilibrium_sweep_Tu(T_u_range, phi=1.0, **kw)
def equilibrium_map(phi_range, sweep_range, sweep_param="T_u",
                    fuel="H2", fixed_T_u=300.0, fixed_p=101325.0, ...):
    """Array of shape (len(phi_range), len(sweep_range))."""
```

Design tip: write `equilibrium_state` once; everything else calls it (or a vectorised `SolutionArray` version). Create the `ct.Solution` once per sweep, not once per point: loading the YAML is the slowest step.

Task 1: $T_{\text{ad}}(\phi)$, $\phi \in [0.4, 3]$

- Expect: a rise on the lean side, a peak slightly rich, and a slower decline on the rich side.
- Explain the peak position with dissociation and product c_p (Section 4.7, intuition box).
- Explain the asymmetry with the lean/rich model (Section 4.7, derivation box). For H_2, think about the heat capacity each additional kmol of H_2 brings compared with the heat it can release once O_2 is exhausted.
- 1000 K crossing: kinetics needs T above the crossover (Fig. 5.3) for branching to beat termination. Equilibrium alone does not decide flammability, but a mixture whose *maximum possible* temperature is below crossover cannot sustain a flame.

Task 2: $T_{\text{ad}}(T_u)$, $T_u \in [300, 800]$ K, for $\phi \in \{0.7, 1.0, 1.3\}$

- Expect nearly straight lines with slope below 1. Compute the slope with `np.polyfit` or `np.gradient`.
- Compare with $\bar{c}_{p,u}/\bar{c}_{p,b}$ evaluated by Cantera (Section 4.7, preheat derivation). Which ϕ dissociates most at the higher temperatures and therefore loses most of the extra enthalpy?

Task 3: $p \in [1, 20]$ atm; $T_{\text{ad}}(p)$ and $X_{\text{OH}}, X_{\text{H}}, X_{\text{O}}$

- Use a log x-axis for p; radicals on a log y-axis.
- Expect T_{ad} to rise modestly and the radicals to fall: Le Chatelier (Section 4.3). Which dissociation reaction has the largest $\Delta\nu$?

- NO_x: the thermal (Zeldovich) mechanism $O + N_2 \rightarrow NO + N$ has $E_a \approx 318\,\mathrm{kJ/mol}$. Weigh the higher temperature against fewer O atoms. Gas turbines run at high pressure for efficiency and lean to keep T low.

Task 4: maps $T_{\mathrm{ad}}(\phi, T_u)$ and $T_{\mathrm{ad}}(\phi, p)$

```
PHI, TU = np.meshgrid(phi_range, Tu_range, indexing="ij")    # shape (n_phi, n_Tu)
cs = plt.contourf(PHI, TU, T_map, levels=30, cmap="inferno")
plt.colorbar(cs, label=r"$T_{ad}$ [K]")
plt.contour(PHI, TU, T_map, levels=[1800], colors="w", linestyles="--")
```

- The 1800 K isoline is a design constraint: the combinations of ϕ and preheat that keep flame temperature below the level where thermal NO_x takes off and where turbine materials suffer.
- Which parameter moves T_{ad} more over its range? Compare the colour change along each axis.

12.3 Part 2: Non-premixed counterflow flames

Theory: Chapter 10; Lewis number effects Section 7.5. **Setup:** H_2 versus air (`O2:0.21, N2:0.79`), both at 300 K, 1 atm, width 0.02 m, refinement `ratio=3, slope=0.05, curve=0.1`.

Step 1: family of flames

```
def solve_counterflow(strain, fuel="H2:1", oxidizer="O2:0.21, N2:0.79", T=300.0,
                      p=101325.0, width=0.02, mechanism="h2o2.yaml", flame=None):
    """Solve a counterflow diffusion flame for a target strain rate [1/s].
    If `flame` is given, reuse it as initial guess (continuation)."""
    # rho_F, rho_O from ct.Solution at the inlet states
    # u_O = strain * width / 4 ;  u_F = u_O * sqrt(rho_O / rho_F)      (Sec. 9.4)
    # f.fuel_inlet.mdot = rho_F*u_F ; f.oxidizer_inlet.mdot = rho_O*u_O
    # f.set_refine_criteria(...) ; f.solve(loglevel=0, auto=flame is None)
```

Start with the lowest strain (`auto=True`) and increase strain step by step, reusing the solution. After each solve record `flame.strain_rate("max")`, $T_{\max}$, and `flame.extinct()`.

Step 2: $T(Z)$ and χ_{st}

```
Z = flame.mixture_fraction("Bilger")                  # check sign/limits: Z=1 at fuel
    inlet
alpha = flame.thermal_conductivity / (flame.density * flame.cp)
chi = 2 * alpha * np.gradient(Z, flame.grid)**2
i_st = np.argmin(abs(Z - Z_st));  chi_st = chi[i_st]
```

Compare your χ_{st} with (10.5.3) using the measured strain. Agreement within a factor of order one is expected (variable density, $D \neq \alpha$).

Pitfall

The project page's helper uses `flame.mixture_fraction_dissipation_rate`. That attribute does not exist in Cantera 3.2. Compute χ yourself as above.

Step 3: $T_{\mathrm{eq}}(Z)$

Enthalpy-weighted mixing then `equilibrate("HP")` (Section 10.3). Use a grid dense near Z_{st} (e.g. `np.concatenate([np.linspace(0, 3*Z_st, 80), np.linspace(3*Z_st, 1, 40)])`).

Step 4: the family plot

Strain rates 50, 200, 500, 1000, 2000 1/s (adjust if extinction comes earlier). Dashed $T_{\mathrm{eq}}(Z)$, vertical line at Z_{st}, x-limits $[0, 1]$. It is worth adding an inset or second panel zoomed into $[0, 0.2]$. Save to `figures/part2_T_vs_Z.pdf`.

Thinking about the questions.

1. *Asymmetry*: geometry ($Z_{\mathrm{st}} \ll 1$) plus heat capacity of the fuel stream per kg (Section 10.3).
2. *Peak shift*: track $Z(T_{\max})$ vs. χ_{st}. Differential diffusion of H_2 and H matters here.
3. *Area between curves*: chemical enthalpy not converted to sensible enthalpy: radicals not recombined, plus differential diffusion. It grows with χ_{st}, i.e. as Da shrinks.
4. *Closest to extinction*: highest χ_{st} that still burns; detect extinction via $T_{\max}$ or `extinct()`. A sharp drop in $T_{\max}$ between two strains brackets χ_q.
5. *Lowest strain ≠ equilibrium*: see the closing list of Chapter 10.

12.4 Part 3: Premixed free flames

Theory: Chapter 9, Section 11.8, Section 11.9.

API in `ct_utils/free_flame`

```
def solve_free_flame(phi, T_u=300.0, p=101325.0, transport="mixture-averaged",
                     fuel="H2", oxidizer="O2:1, N2:3.76",
                     mechanism="h2o2.yaml", width=0.03, flame=None):
    """Return a converged ct.FreeFlame (research-grade refinement)."""
def flame_speed(flame):     return flame.velocity[0]
def flame_thickness(flame): # (T_b - T_u) / max(dT/dx)
def flame_speed_sweep(phi_range, **kw)        # reuse previous solution as guess
def flame_thickness_sweep(phi_range, **kw)
```

Task 1: structure at $\phi = 1$

Three stacked panels against x [mm] with shared x-axis: (1) T with T_u, T_b and a bracket of width δ_L; (2) H_2, O_2, H_2O, OH, H; (3) HRR in MW/m^3. Centre x on the HRR peak and zoom to a few δ_L. Questions: why HRR precedes $T_{\max}$ (Fig. 9.2); where radicals are produced versus

where they survive; δ_L relative to the domain (`width` must be $\gg \delta_L$, otherwise the boundary conditions bite).

Task 2: equilibrium check

Error $|T_{\text{end}} - T_{\text{ad}}|/T_{\text{ad}}$ and a grouped bar chart of X_k (flame end vs. equilibrium) for H_2O, H_2, O_2, OH (log scale for minor species). Over 1%: tighten refinement or widen the domain (Section 9.4, Fig. 9.8).

Task 3: transport models

Solve once with `"mixture-averaged"`, then switch to `"unity-Lewis-number"` and re-solve.

```
x_mid = np.interp(0.5*(T_u + T_b), flame.T, flame.grid)     # T is monotonic in a free
    flame
x_shifted = (flame.grid - x_mid) * 1e3                       # mm
gas = flame.gas                                              # FreeFlame has no
    mixture_fraction()
Z = np.empty(len(flame.grid))
for j in range(len(flame.grid)):
    gas.TPY = flame.T[j], flame.P, flame.Y[:, j]
    Z[j] = gas.mixture_fraction("H2:1", "O2:1, N2:3.76", element="H")   # or element="
    Bilger"
```

(In Cantera 3.2 only `CounterflowDiffusionFlame` has a `mixture_fraction("Bilger")` method; `FreeFlame` and `SolutionArray` do not, hence the loop. Restore the flame's state afterwards if you keep using `gas`.) Interpretation: Section 9.11. For methane, look at Fig. 7.3: $\text{Le}_{\text{CH}_4} \approx 1$.

Task 4: $S_L(\phi)$, $\delta_L(\phi)$, $\phi \in [0.4, 2.5]$, both transport models

Sweep outwards from $\phi = 1$ in both directions, reusing solutions. Very lean H_2 needs a wider domain and tighter refinement. Plot $S_L\delta_L/\alpha_u$ as a check of (9.6.1). Compare the S_L peak with the T_{ad} peak from Part 1 using the three factors in the intuition box of Chapter 9.

Task 5: pressure and preheat

```
n, lnA = np.polyfit(np.log(p_atm), np.log(S_L), 1)        # S_L ~ p^n
m, lnB = np.polyfit(np.log(T_u / 300), np.log(S_L), 1)    # S_L ~ T_u^m
```

- Domain width: δ_L shrinks roughly as $1/p$, so a fixed 3 cm domain is wasteful at 10 atm, while the refinement must still resolve it. Check grid point counts.
- Compare the exponent with $n/2 - 1$ (Section 9.6), recalling the R2/R5 competition.
- On "$S_L\delta_L$ nearly constant": test it. Scaling says $S_L\delta_L \propto \alpha_u \propto 1/p$ while $\rho_u S_L \delta_L \approx \lambda/c_p$ is nearly pressure-independent. Decide which one your data supports and what that means physically.

12.5 Checklist before each tag

☐ Every number has units; every plot has labelled axes with units and a meaningful title.

- ☐ Equilibrium sanity values reproduced (2382 K-ish at $\phi = 1$).
- ☐ Grid convergence shown at least once (Part 3), with grid sizes reported.
- ☐ Flame end states within 1% of HP equilibrium.
- ☐ Functions importable: `from ct_utils.general import equilibrium_sweep`.
- ☐ Notebook runs top-to-bottom on a fresh kernel.
- ☐ Each slide has one sentence of physics, not just a description of the curve.

Appendix A

Nomenclature and data

Latin symbols

a	strain rate	1/s
A, β, E_a	Arrhenius pre-factor, temperature exponent, activation energy	–, –, J/kmol
$C_k = [X_k]$	molar concentration	$\mathrm{kmol/m^3}$
c_p, c_v	specific heat capacities	$\mathrm{J/(kg\,K)}$
D_{km}, $\mathcal{D}_{jk}$	mixture-averaged / binary diffusion coefficient	$\mathrm{m^2/s}$
g, h, s, u	specific Gibbs energy, enthalpy, entropy, internal energy	J/kg ($\mathrm{J/(kg\,K)}$)
$\boldsymbol{j}_k$	diffusive mass flux of species k	$\mathrm{kg/(m^2\,s)}$
K_p, K_c	equilibrium constants (pressure / concentration)	–
$\dot{m}$	mass flux	$\mathrm{kg/(m^2\,s)}$
p	pressure	Pa
q_i	rate of progress of reaction i	$\mathrm{kmol/(m^3\,s)}$
$\dot{q}$	heat release rate	$\mathrm{W/m^3}$
$\mathcal{R}$	universal gas constant, 8314.46	$\mathrm{J/(kmol\,K)}$
s	stoichiometric oxygen-to-fuel mass ratio (Ch. 10)	–
S_L, S_c	laminar flame speed, consumption speed	m/s
T	temperature	K
$\boldsymbol{u}$, $\boldsymbol{V}_k$	mass-averaged velocity, diffusion velocity	m/s
W_k, W	molar mass of species k, of mixture	kg/kmol
X_k, Y_k	mole / mass fraction	–
Z	mixture fraction	–

Greek symbols

$\alpha = \lambda/\rho c_p$	thermal diffusivity	$\mathrm{m^2/s}$
δ_L	thermal flame thickness	m
λ	thermal conductivity	$\mathrm{W/(m\,K)}$
μ, ν	dynamic / kinematic viscosity	$\mathrm{Pa\,s}$, $\mathrm{m^2/s}$
μ_k	chemical potential	J/kmol
$\nu'_{ki}, \nu''_{ki}, \nu_{ki}$	stoichiometric coefficients (reactant, product, net)	–
ξ	extent of reaction	kmol
ρ	density	$\mathrm{kg/m^3}$
ϕ	equivalence ratio	–
χ	scalar dissipation rate	1/s
$\dot{\omega}_k$	net molar production rate	$\mathrm{kmol/(m^3\,s)}$

Dimensionless numbers

$\mathrm{Re} = UL/\nu$, $\mathrm{Pr} = \nu/\alpha$, $\mathrm{Sc}_k = \nu/D_k$, $\mathrm{Le}_k = \alpha/D_k$, $\mathrm{Da} = \tau_{\mathrm{flow}}/\tau_{\mathrm{chem}}$, $\beta = E(T_b - T_u)/\mathcal{R}T_b^2$ (Zeldovich), $\mathrm{Ma} = u/c$.

Useful data

	H_2	O_2	N_2	H_2O	CH_4
W [kg/kmol]	2.016	32.00	28.01	18.02	16.04
$\Delta\hat{h}^\circ_f$ [MJ/kmol]	0	0	0	−241.8	−74.9
c_p at 300 K [kJ/kg/K]	14.3	0.92	1.04	1.86	2.23

Stoichiometric values in air (O_2:N_2 = 1:3.76): CH_4: $s = 4$, $Z_{\mathrm{st}} = 0.055$, $T_{\mathrm{ad}} = 2226\,\mathrm{K}$, $S_L \approx 0.38\,\mathrm{m/s}$.

Further reading

- S. R. Turns, *An Introduction to Combustion*, McGraw-Hill: the gentlest complete textbook.
- T. Poinsot & D. Veynante, *Theoretical and Numerical Combustion*: the governing equations, flame speeds and flamelets, for the CFD-minded.
- N. Peters, *Turbulent Combustion*, CUP: mixture fraction and flamelets.
- C. K. Law, *Combustion Physics*, CUP: stretch, Lewis-number effects, hydrogen.
- ANSYS Chemkin Theory Manual: exact forms of every property and rate expression.
- Cantera documentation and examples: https://cantera.org.
- ASEC onboarding: https://asec-laboratory.github.io/Onboarding/04_ReactiveFlows_and_Cantera/00_Introduction/.